

AIRCRAFT DESIGN - 2025

Design Project Report

For

A Modular Twin Hybrid Propeller Aircraft



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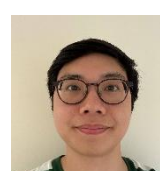
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FlowRider

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1. Company Structure

FlowRider was founded in 2025 with the objective of addressing the need for affordable and sustainable regional air transportation. The growing demand for regional passenger travel has

motivated our company to explore more affordable and environmentally friendly aircraft that can compete in the short-medium-haul markets. Besides just passenger travel, a more sustainable approach to travelling in the future involves aircraft that are modular. FlowRider introduces the FR25 program. The FR25 is a modular hybrid turboprop engine aircraft that will allow operators to easily shift their cabin configuration according to supply and demand of passengers and cargo. This report will provide details of the FR25 program.

The company is led by CEO and Systems Engineer, Florian, who oversees the operation cycle of its fleet production from conceptual design through to first delivery. Supporting him is Swee Woei, the structural lead, who is responsible for ensuring the structural development and safety of the aircraft through designing CAD models of the FR25

The engineering team includes Nikunj, materials lead, who manages material selection and innovation across the entire aircraft. Analysis Expert, Jason looks at how the aircraft dynamics during flight conditions through simulations and analysis using advanced tools such as Ansys.

Chief Technology Officer and wing design lead, Thaddeus, will design and optimize the airfoil and wing section. He is supported by mission lead, Jit Huat and propulsion lead, Heong Luck. Jit Huat oversees planning the mission profile to ensure that the aircraft can safely operate in various flight conditions. Propulsion Lead, Heong Luck, will look at state-of-the-art engines for the aircraft. The overall company structure can be seen in figure 1.

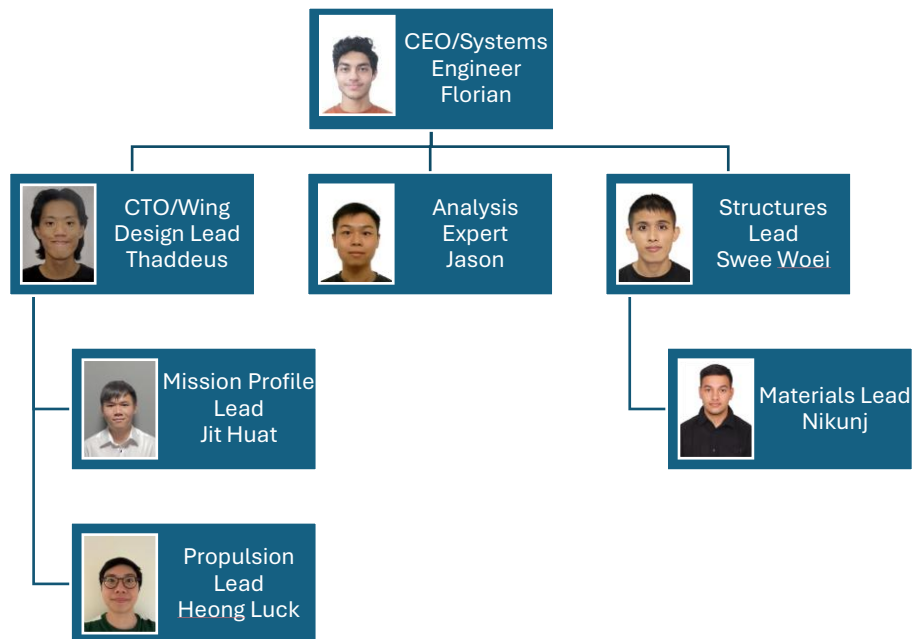


Figure 1: Group Organization

2. Introduction (Florian)

The aviation industry is undergoing a transformation due to a combination of technological, environmental, and socio-economic factors. According to the International Air Transport Association (IATA), the sector is expected to experience fundamental changes by 2035 and beyond. These drivers of change highlight the need for new aircraft concepts that are efficient, adaptable and resilient to uncertainties.

Alternative fuels and energy sources are one of the significant changes that will affect the aviation market. While oil continues to dominate energy, most countries will be getting more than 80% of their energy from renewables and nuclear by 2050 (IATA,2018). The aviation industry is expected to adopt non-traditional fuels such as bioenergy, hydrogen fuel cells and advanced storage systems for future designs.

As seen in figure 2, Climate change and extreme weather events are expected to increase in frequency and severity. Aircraft designs must adapt to more intermittent or persistent weather impacts such as heavy rainfall or convective weather (Burbidge,2018). New designs will need to be more robust and turbulent-resistant, capable of operating in hotter climates. In tandem, environmental regulations will evolve alongside climate change. Despite the aviation sector's low global CO_2 emission, international regulations on emissions and noise pollution are set to become more stringent (IATA,2018).

Climate risk	Impact	Actors	Key:								
 Precipitation change	• disruption to operations e.g. airfield flooding, ground subsidence • reduction in airport throughput • drainage system capacity • inundation of underground infrastructure (e.g. electrical) • inundation of ground transport access (passengers and staff) • loss of local utilities provision (e.g. power).	→ AO ANSP → AO ANSP AO AO → AO ANSP  → AO ANSP 	<table><tr><td>Aircraft operators</td><td>→</td></tr><tr><td>Airport operators</td><td>AO</td></tr><tr><td>ANSP</td><td>ANSP</td></tr><tr><td>External</td><td></td></tr></table>	Aircraft operators	→	Airport operators	AO	ANSP	ANSP	External	
Aircraft operators	→										
Airport operators	AO										
ANSP	ANSP										
External											
 Temperature change	• Changes in aircraft performance • Changes in noise impact due to changes in performance • heat damage to airport surface (runway, taxiway) • increased heating and cooling requirements • Increased pressure on local utilities e.g. water and power (for cooling).	→ ANSP → AO ANSP AO AO 									
 Sea-level rise	• loss of airport capacity • impacts on en-route capacity due to lack of ground capacity • loss of airport infrastructure • loss of ground transport access	→ AO ANSP → ANSP AO → AO ANSP 									
 Wind changes	• convective weather: disruption to operations • convective weather: route extensions • jet stream: increase in en-route turbulence • local wind patterns: disruption to operations and changes to distribution of noise impact	→ AO ANSP → ANSP → AO → AO									
 Extreme events*	• disruption to operations, route extensions • disruption to ground transport access • disruption to utilities supply	→ AO ANSP → AO  AO 									

Figure 2: Climate Change Impacts for Aviation Stakeholders (Eurocontrol,2014)

Another driver of change is the increased integration of global-air supply chain. Emerging markets in Africa and Asia-Pacific may boost global prices and make it trickier to configure supply chain assets. Beyond regulatory and environmental shifts, demographic and economic trends will reshape demand for travel. Trends in population growth, global ageing and middle-class growth in Africa and Asia-Pacific are expected to have a large impact on the airline industry (IATA, 2018). Although the potential of this new demographic is recognized, uncertainty remains about how well the industry will meet the needs of passengers as the consumer base diversifies.

The airline industry has been reactive to new technologies. The aviation industry must be prepared to involve technology at its nascent stages to disrupt the overall market. Over short to medium distances new modes of rapid transport (e.g. hyperloop) will influence the way people and goods travel.

3. Detailed Market Analysis (Florian)

FlowRider's strategy is to enter the emerging markets for future revenue. To maximize income revenue, the company will tap into countries with a growing aviation hub. According to IATA, the Chinese and African markets are set to rise in domestic travel due to a rising middle class. This provides FlowRider with an opportunity to capture a percentage of the overall market share. In the following sections, we investigate the Chinese and African Markets in more detail.

a. China Aviation Industry

China's Aviation market became the largest domestic market by seat capacity in 2020, surpassing the United States (CAPA, 2020). The country operates a fleet of 4,335 transport aircraft and manages 262 domestic airports with a total capacity of 1.6 billion passengers (Xinhua, 2024).

According to Airbus, China is set to become the largest market for aviation services by 2045. Its total market value will reach US\$61bn in 2045. Of the total aviation market cap, the civilian aviation market will capture around 25% of the overall market share as seen in figure 3. This means that the domestic Chinese aviation market will be valued at:

$$\text{Civillian Market Valuation} = \text{US\$61Bn} \times 25\% \approx \text{US\$15.25Bn}$$

In year 2025, China will have 11,160 aircrafts in which 9520 are new deliveries which accounts for 20% of the global demand (Airbus, 2024). This provides FlowRider with an ideal opportunity to tap in the Chinese markets.

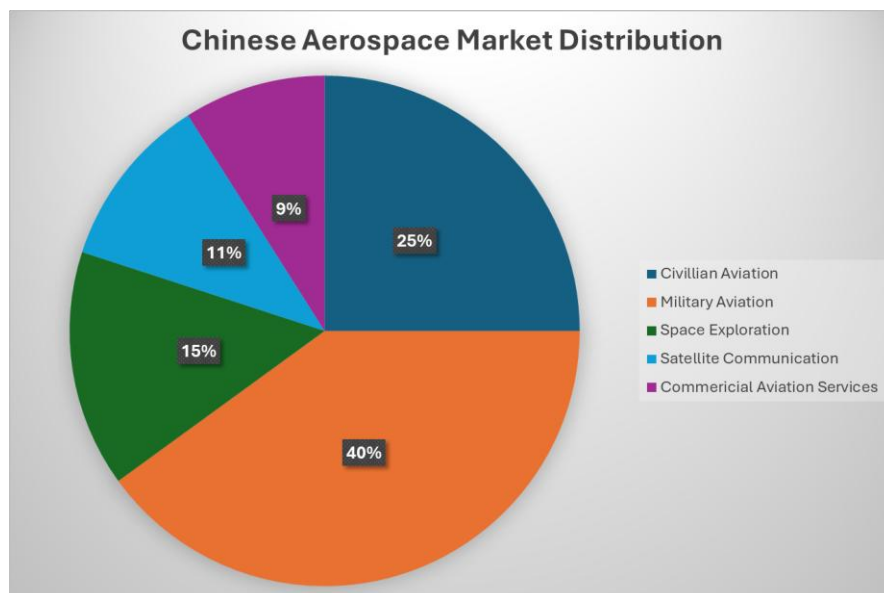


Figure 3: China Aerospace Market Segmentation (Ken Research, 2024)

The aviation sector spreads beyond just passenger transport. China, being the global manufacturer, has benefitted from a rise in e-commerce demand leading to the rise in air freight sector. Despite this, China's current cargo fleet is lagging international trade demand (Cirium, 2024).

i. Market Competition

The Chinese markets share for regional travel is currently divided between 3 major players, Boeing, Airbus and COMAC. In June 2025, reports suggest that China placed a major aircraft order between 200 and 500 units comprising a mix of single-aisle and twin-aisle models from Airbus with the majority comprising of single-aisle models. (Cirium, 2025). The same trend is expected to continue for the local manufacturer, COMAC. The trend suggests that COMAC's long term market could increase along with Airbus in China.

Currently, there are no competing aircraft models in China that follow our company's design of a hybrid-modular aircraft. However, for comparison, we will use the domestic aircrafts currently used in China as reference for our analysis.

Table 1: Aircraft Models in Chinese Domestic Markets in 2025

Aircraft Model	Manufacturer	Seat Capacity	Commerical Cargo Capacity	Market Share
C919	COMAC	156-168 Passengers	NA	0.5%
B737	Boeing	138-230 Passengers	NA	44.1%
A320	Airbus	140-170 Passengers	NA	55.1%
FR25	FlowRider	Passengers	0 Kg with 88 Passengers - 4800Kg with 48 Passengers	0%

As seen in Table 1, Airbus currently holds dominance in the Chinese market. With reports on COMAC's rise in market share, it would be practical if FlowRider enters this market to further divide the overall market share. FlowRider forecasts that it will be able to capture about 17 % of the Chinese markets as seen in table 2.

Table 2: Projected Aircraft Models in Chinese domestic markets in 2045

Aircraft Model(s)	Manufacturer	Seat Capacity	Commerical Cargo Capacity	Market Share
C919	COMAC	156-168 Passengers	NA	20.8%
B737	Boeing	138-230 Passengers	NA	32.0%

A320	Airbus	140-170 Passengers	NA	30.2%
FR25	FlowRider	48-88 Passengers	0 Kg with 88 Passengers - 4800Kg with 48 Passengers	17 %

With 17% of the Market Share captured by 2045 in China, this means that our company can approximately earn revenue of up to:

$$\text{Possible Revenue in 2047} = \text{US\$15.25 Bn} \times 17\% \approx \text{US\$2.59 Bn}$$

b. Africa Aviation Industry

Africa, as a continent, has experienced increasing demand for air travel due to the rising middle class. Countries such as South Africa, Kenya, Nigeria and Egypt lead the aviation sector due to higher GDPs (CAPA, 2025). Currently, the African aviation market is earning approximately US\$10.28bn with an annual growth rate of 9.91% (Statistica, 2025). At this rate, the African market will reach:

$$\begin{aligned}
 FV &= PV \times (1 + r)^t \\
 FV &= \text{US\$10.28Bn} \times (1 + 0.0991)^{20} \\
 &= \text{US\$ 68.03 Bn}
 \end{aligned}$$

where,

FV = Future Values

PV = Present Value

R = Growth Rate

T = years

Africa experiences an intra-regional travel of about 26% as seen in figure 4. Within the African continent, South Africa is the largest domestic market with 1.6 million seats per month in 2025 (OAG, 2025).

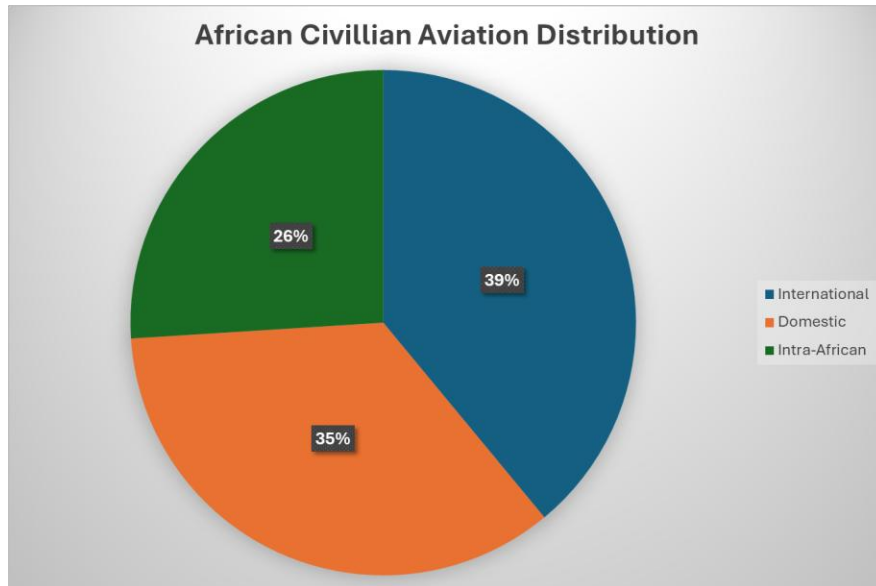


Figure 4: Africa Civilian Travel Market (Deutsche Aircraft,2025)

The African aviation market is poised for growth in the next 20 years. According to Boeing’s 2025 Commercial Outlook (CMO), the continent’s fleet is set to double by 2043. The continent can expect 1,680 new airplanes. Of the new airplanes, 70% will be single-aisle jets providing our company with an opportunity to enter this big market (Boeing, 2025). A key area of interest for our company is the rise of low-cost carriers (LCC) in the African Region. LCCs make air travel more accessible to a broader population that could expand domestic routes.

Apart from commercial travel, CMO estimates that the freighter fleet in Africa will triple by 2043. The expansion will support the region’s increasing e-commerce demand. This will increase the demand for freight aircrafts.

The given data suggests that our company should start its operation within South Africa with the FR25 to gain a foot inside the African market. Our design for a hybrid aircraft that balances passenger and cargo loading will provide us with the ideal opportunity in Africa.

i. Market Competition

The African aviation market is currently divided by major players from Airbus, COMAC, Boeing. By 2043, the African Aviation market is projected to connect 345 million passengers annually. Of the total passengers travel annually, domestic travel accounts for 37% of passenger travel. This is a substantial number as regional aircraft can demonstrate optimal performance. This number implies that there will be a strong demand for regional travel.

Besides the existing players, Deutsche Aircraft have announced their intentions on the Africa Market with their very own D328eco (Deutsche Aircraft, 2025) for the near to long term future. The D328eco is an ideal aircraft for intra African travel due to its compact size, fuel efficiency and multi-role capability. Taking this into account for our future projections, we can assume that the domestic market will be roughly divided by the 5 companies including ours as seen in table 3. This means that our company can capture around 20% of the African domestic market

Table 3: Projected Aircraft Models in African domestic markets in 2045

Aircraft Model(s)	Manufacturer	Seat Capacity	Commercial Cargo Capacity	Market Share
C919	COMAC	156-168 Passengers	NA	20%
B737	Boeing	138-230 Passengers	NA	20%
A320	Airbus	140-170 Passengers	NA	20%
D328eco	Deutsche Aircraft	40-50 Passengers	NA	20%
FR25	FlowRider	48-88 Passengers	0 Kg with 88 Passengers - 4800Kg with 48 Passengers	20%

A simple calculation shows that FlowRider will be able to capture US\$3.5 billion from the African markets.

FlowRider is projected to capture up to US\$6 billion in the China and African domestic aviation market alone. However, it is worth noting that FlowRider will expand into other domestic markets in the foreseeable future. The Chinese domestic markets can be more competitive due to the growth of local aircraft manufacturers such as COMAC. With governmental support for COMAC, it is probable that Chinese airliners will sign more contracts with COMAC. FlowRider will explore other domestic markets including those in the Northern/Southern American and European regions to ensure that the company can attract investors and customers at a global scale.

4. FR25 Program Overview (Florian)

The FR25 will be strategically designed to serve high-growth markets in Africa and China. While optimized for these regions, the FR25 is a versatile platform capable of meeting regional transportation needs worldwide.

According to the IATA outlook, several key factors will define the success of future aircraft. These are sustainability, resilience and adaptability. FR25 directly addresses these requirements through:

1. **Hybrid Propulsion:** The FR25 will be equipped with next generation hybrid-electric engines by RTX. The engines will combine biofuels and advanced battery systems to reduce emissions and operating costs while maintaining performance.
2. **Climate Resilience:** FR25 has been engineered to operate under higher temperatures, turbulence and weather conditions expected from climate change ensuring long-term reliability.
3. **Advanced Materials and Avionics:** Incorporates lightweight and durable materials with next generation avionics to enhance safety.

4. **Modular Cabin Design:** A dual passenger cargo configuration enables operators to maximise revenue. In full layout, the FR25 holds up to 88 passengers. Alternatively, the aft cabin can be reconfigured for cargo transport supporting the demand for e-commerce logistics holding up to 4800 kg in cargo.

The hybrid operational flexibility of the FR25 is uniquely positioned to maximise profit of airlines while meeting the demand for more affordable domestic passenger travel.

5. FR25 Development Program Schedule (Florian)

The overall program cost is estimated to be USD\$ 6.1 billion. The section covers the entire development program for the FR25 Model.

The development of the FR25 model is projected to take around 7-8 years to fully develop before first delivery to the intended market segments. The target for the company is to deliver its first aircraft in the year 2032. The overall structure of the development program is described in table 4. The first year of the program will focus on the trade study of existing aircrafts in the market. The company will look into future proof designs for an aircraft that is able to make profit in 2040. After a thorough examination of various aircraft designs, the company will select one of the designs for the preliminary design phase and then the detailed design.

Table 4: Aircraft Development program schedule

Year of Task	Development Phase
1	Conceptual Design
2	Preliminary Design Phase
3-4	Detailed Designs, Wind Tunnel Testing
4	Prototype development and load testing
4-7	Ground Vehicle Testing, Flight Testing, Tooling, Manual preparation, Training, Certifications
7 onwards	Delivery of 1 st Aircraft

After the detailed designs have been finalized, the company will conduct testing and make its first prototypes.

a. Key Performance Parameters (KPP)

The FR25 is designed to deliver optimal performance with a payload capacity of 9,200Kg. It will offer a range of 1,500 km making it suitable for short-medium haul regional operations. The aircraft's service ceiling is 7,630m. For operational efficiency in varied airport environments, the FR25 has a landing distance of 1,070 meters. It can achieve a cruise velocity of 667 km/h. The KPP values seen in table 5 are fundamental benchmarks that guide the overall design.

Table 5: FR25 KPP

Payload	9,200kg
Range	1,500km
Service Ceiling	7,630m
Landing Distance	1,070m
Cruise Velocity	667 km/h

b. Benchmark Analysis

A benchmark analysis was conducted to evaluate the FR25 competitiveness against existing and future aircraft in the turboprop segment with similar capacity and performance. This analysis aims to identify how the FR25 compares in terms of range, fuel efficiency, payload and cost with respect to the Maximum Take-Off Weight (MTOW). By assessing its position relative to market leaders, this comparison highlights the FR25's features that stand out from the rest.

Range is a critical performance metric that indicates how far the aircraft can operate. It is compared against the MTOW. When benchmarked against MTOW, the FR25 has a maximum range of 1500km. The graph in figure 5 shows that it is within the industry trend line range for aircraft of comparable size.

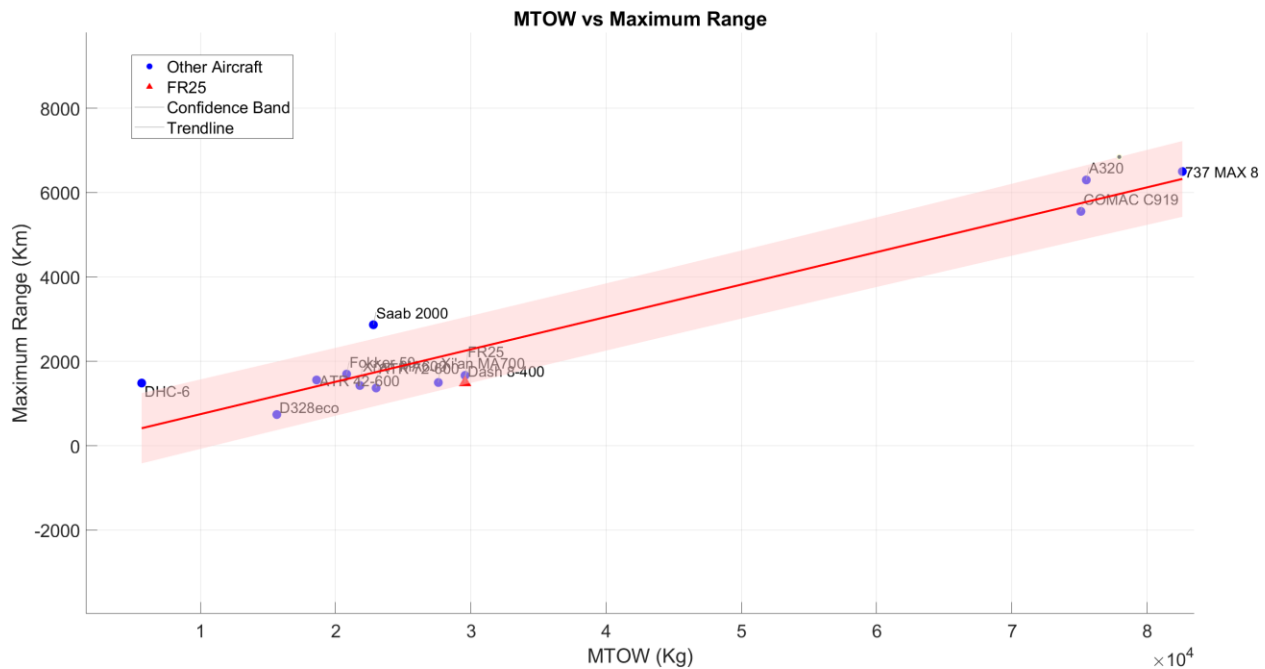


Figure 5: MTOW vs Maximum Range

Wing loading is a fundamental parameter that influences take-off and landing performance, climb capability and overall handling. The FR25 has a wing loading of 406 kg/m^2 . When compared to other models in figure 6, it is within the industry trendline.

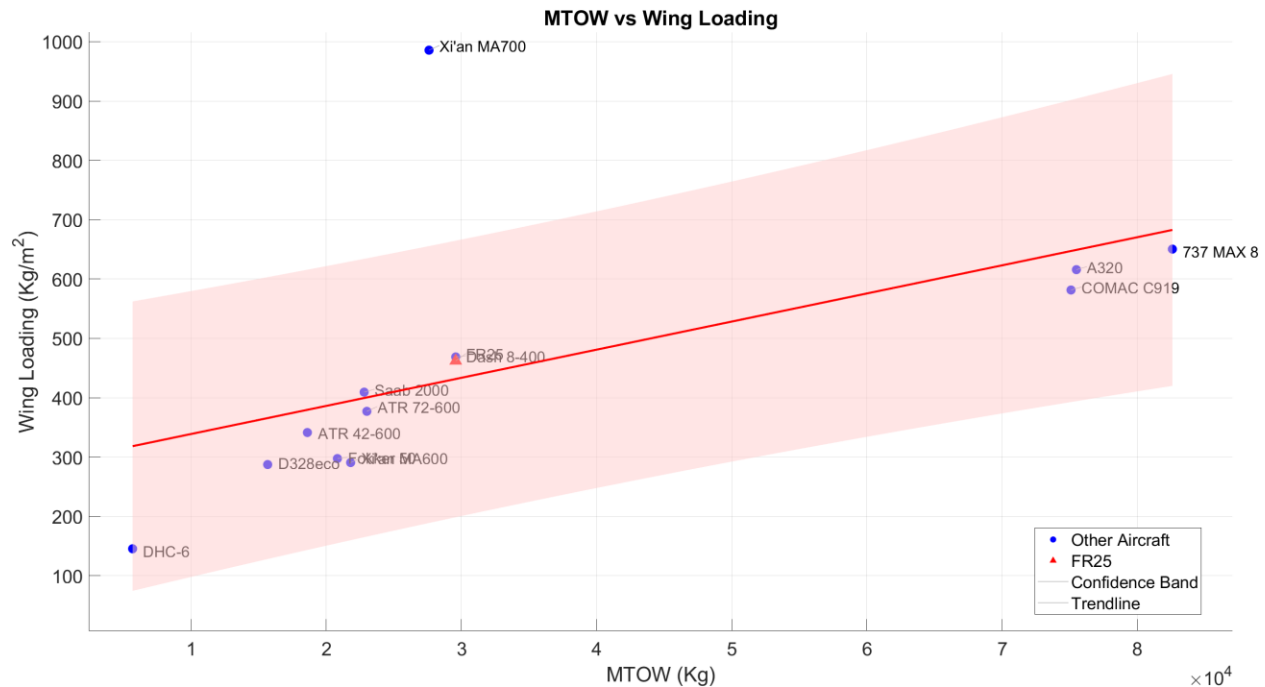


Figure 6: MTOW vs Wing Loading graph

A price-per-kg analysis shows that the FR25 is priced reasonably within its cluster. An analysis of the short-medium aircraft market seen in figure 7 indicates that the FR25 is competitively priced at USD\$1314.30 per empty weight. Comparing other aircrafts of similar capabilities shows that the FR25 is priced below the centre of its cluster. This indicates that the FR25 offers a lower price point as compared to its competitors.

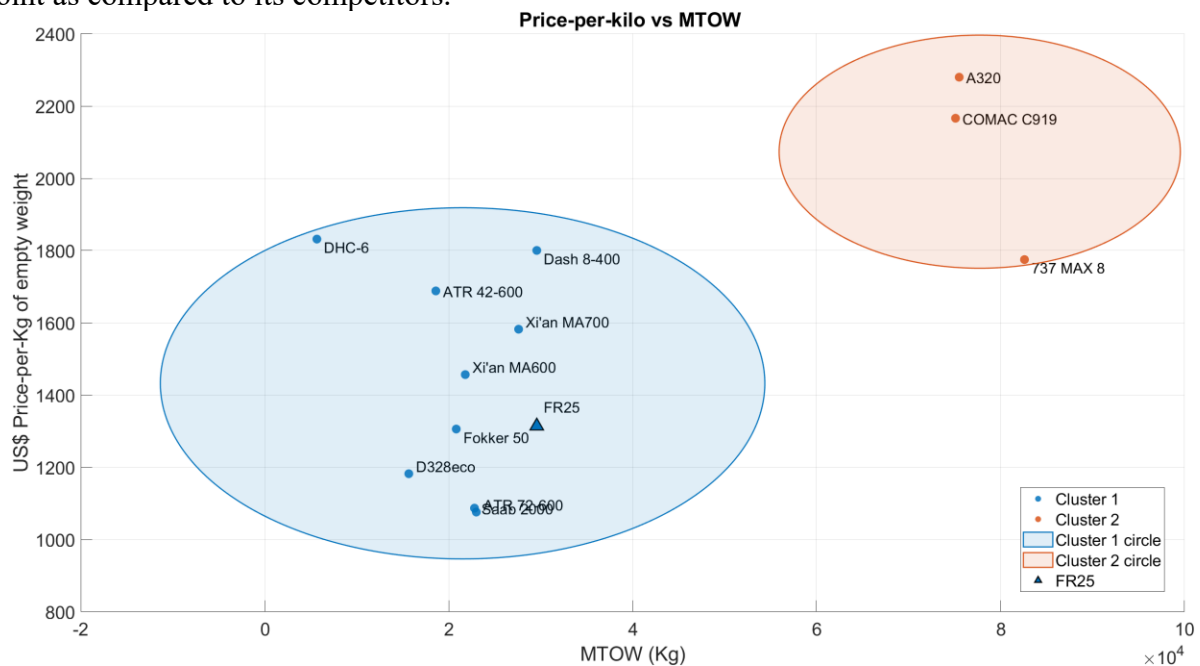


Figure 7: Price-per-kilo selling price of empty weight

c. Breakeven Analysis

FlowRider projects that the FR25 program will achieve profitability by 2035, 10 years after the start of conceptual design in 2025. Table 6 summarizes the financial trajectory of the FR25 program. The program will cost an estimated USD\$6.1B in development and certification costs before attaining its first delivery in 2032 as seen in table 7. Production will begin modestly with 10 aircraft delivered in the first year and 50 in the following. The data is summarized into a bar chart with a trendline in figure 8. The company will eventually scale up annually as demand in the African and Chinese domestic markets grow.

Table 6: Incurred Cost Summary Table

	Values
Development Cost (Entire Program)	USD 6.1 Billion
Production Cost (Per Unit)	USD\$ 15 million
Aircraft Selling price	USD\$ 20 million
Number of Aircraft to Break Even	210

Table 7: Breakeven Analysis for the FR25 line

Year	Phase	Development Cost (USD\$B)	Units Delivered	Production Cost (B\$)	Revenue from Sales (B\$)	Profit / Debt (B\$)
2025	Conceptual Design	1.2	0	0	0	-1.2
2026	Preliminary Design	1.2	0	0	0	-2.4
2027	Detailed Designs	1	0	0	0	-3.4
2028	Detailed Designs	1.1	0	0	0	-4.5
2029	Prototyping and Fight Testing	0.5	0	0	0	-5
2030	Prototyping and Fight Testing	0.5	0	0	0	-5.5
2031	Prototyping/Production and Fight Testing	0.5	0	0.1	0	-6.1
2032	First Delivery	0	10	1.5	2	-5.6
2033	Delivery	0	50	7.5	10	-3.1
2034	Delivery	0	50	7.5	10	-0.6
2035	Delivery	0	100	15	20	4.4
2036	Delivery	0	100	15	20	9.4
2037	Delivery	0	100	15	20	14.4
2038	Delivery	0	150	22.5	30	21.9
2039	Delivery	0	150	22.5	30	29.4
2040	Delivery	0	200	30	40	39.4

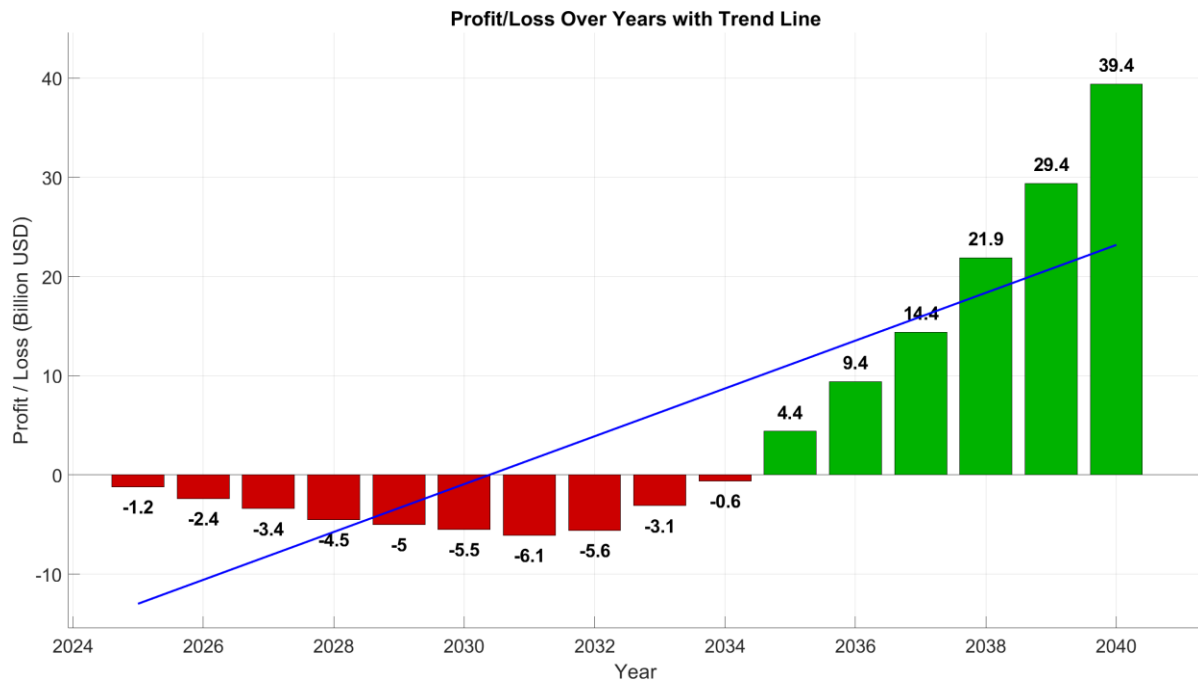


Figure 8: Profit/Loss trajectory for the overall FR25 program until 2040

6. Aircraft Specifications (Jit Huat)

Table 8 provides a general overview of the specifications of the FR25. The values obtained in the table have been derived in the following sections.

Table 8: Specifications of FR25

Specifications (Unit)	Data	Remarks
General Specifications		
Payload (kg)	9200	
Number of Passengers (pax)	88	In full passenger seating configuration
Range (max payload, km)	1,500	
Empty Weight (Kg)	19,615.08	
Maximum Take Off Weight (Kg)	30,693.07	
Maximum Fuel Capacity (Kg)	1859.3	
Cruise Speed (knots)	360 knots	667 km/h
Cruise Altitude (ft)	25,000 ft (FL250)	7,630 m
Take Off Distance (m)	1,700	
Landing Distance (m)	1,070	
Fuselage Specifications		
Length of Fuselage (m)	33.04	
Diameter of Fuselage (m)	3.09	
Wing Specifications		
S_{REF} (Wing Area / m ²)	64	
b (Wingspan / m)	27	
A (Aspect Ratio)	11.4	
λ (Taper Ratio)	0.5	Tip Chord / Root Chord
Tip Chord (m)	1.5	
Root Chord (m)	3	
Quarter-Chord Sweep Angle (degrees)	12	
Engine Specifications		
Engine Model	Hybrid-Electric Turboprop	
Number of Engines	2	
Maximum Power per Engine	4,535 SHP	3,382 kW
Fan Diameter (m)	3.93	

a. Overview (Jit Huat & Jason)

With the aforementioned factors taken into consideration, there are a few implications that need to be considered when designing a new aircraft for successful operation in 20 years' time. Firstly, the aircraft will be dedicated for domestic travel within African and Asia Pacific region. The aircraft accommodates up to 88 passengers in an all-passenger configuration, or 48 passengers with additional cargo, and offers a maximum range of approximately 1,500 km.

Furthermore, given the highly unpredictable weather conditions, the aircraft should also be designed using materials that are strong and stiff to better able to withstand turbulence and not contribute to the fatigue cycle. The avionics system should be able to counter the effects of extreme turbulence.

On top of that, the aircraft will be designed with alternative fuel technology to make travelling more sustainable. The engine will be dependent Sustainable aviation fuel (SAF) and batteries. Since it is an aircraft designed for regional travel, the aircraft will be suited for take-off and landing at smaller airfields. By utilizing a hybrid-electric turboprop engine, the aircraft will be able to take-off from shorter runways, as well as reducing noise pollution surrounding the airport.

Lastly, with fluctuating passenger demand, coupled with rising demand for cargo, the future of regional aircraft will be modular, meaning that the seats and can be removed and be converted to cargo space when passenger demand is not met. The modular aircraft makes it convenient to reconfigure such that it can target airports with limited MRO infrastructure. The aircraft specifications can be summarized in table 9.

Table 9 : Aircraft Design Parameters

Design Aspect	Requirement/Feature
Mission Profile	Range of 800-1500 km
Market Profile	Regional and domestic routes in Africa and Asia-Pacific
Take-Off Performance	Capable of Short Runway Operations to access smaller regional airports
Propulsion	Hybrid Turboprop engines that can integrate hydrogen/biofuel or battery-assisted technology
Structural Design	Hybrid Material Structure using composites and advanced allows for weight reduction and durability
Environmental	Designed for hotter climates and irregular weather
Avionics	Turbulent management capability
Flexibility	Modular cabin and cargo configuration to handle unpredictable passenger loads

With the current design aspects, the proposed model will take some inspiration from the ATR 72-600 and the Dash 8-400 seen in figure 9. Although they are dominating the regional turboprop market today, they rely on conventional engines with fixed cabin layouts. Current models are not adaptable to the shift in passenger and cargo demand. While there are cargo variants for the ATR 72-600 and Dash 8-400, they lack the flexibility to be converted to combi or passenger aircraft.



Figure 9: ATR 72-600 and Dash 8-400 respectively

b. Mission Profile (Jit Huat)

The mission profile of the aircraft follows that of a commercial aircraft and is segmented into seven distinct phases as seen in figure 10: Taxi Out, Take-off, Climb, Cruise, Descent, Landing and Taxi In. For missed approach scenarios, an additional Loiter segment is incorporated.

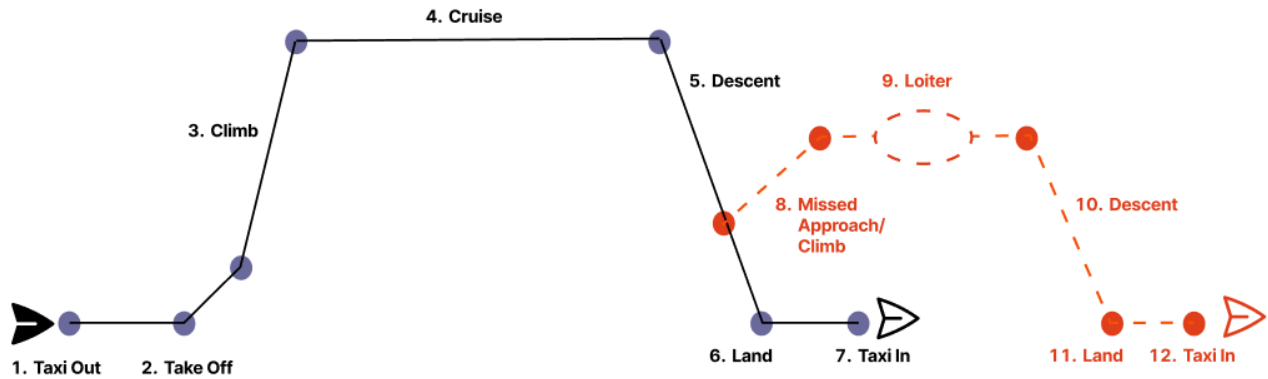


Figure 10: Primary Mission Profile

The aircraft is designed for a maximum range of 1500 km. Key specifications include a maximum fuel weight of 1859.3 kg and a MTOW of 30,693.07kg. The mission has a cruise speed of 667 km/h (360 knots) and cruise altitude of 7630 m (25,000 ft).

For engine performance estimation of the mission, the total Shaft Horsepower (SHP) is 9071 SHP (6764 kW), approximately 4535.5 SHP (3382 kW) per engine, considering all the power losses a combination of both electrical and mechanical power, and propeller efficiency.

The Power Specific Fuel Consumption (PSFC) will be calculated for ground and cruise conditions. Considering the hybrid design of the aircraft, about 73% of the total SHP will be considered for calculations related to fuel, which is a SHP of 6684 (4948 kW), while the rest of the SHP will be powered by electricity and not fuel.

$$PSFC = \frac{\dot{m}_f}{P_{net}}$$

$$PSFC_{ground} = 0.000066 \frac{kg/s}{kW}$$

$$PSFC_{cruise} = 0.0000302 \frac{kg/s}{kW}$$

General Formulas:

$$SHP_{shaft, mech} = \frac{T \cdot \omega}{745.7}$$

$$SHP_{propulsive} = SHP_{shaft, mech} + SHP_{motor, electric}$$

$$T = \text{Torque (Nm)}$$

$$\omega = \text{Angular Velocity} \left(\frac{rad}{s} \right)$$

$$Fuel\ Burn_{phase} = PSFC \left(\frac{kg}{kW} \right) * Propulsive\ Power\ (kW) * Time\ (s)$$

$$Fuel\ Consumption_{phase} = \left(\frac{Fuel\ Burn}{Total\ Fuel} \right) * 100$$

$$W_{after\ phase} = 1 - \frac{Fuel\ Burn}{Take - off\ Weight}$$

c. Primary Mission

The aircraft has been designed to operate in compliance with the regulatory framework established by the Federal Aviation Administration (FAA). Specifically, its certification and design follow the requirements of **14 CFR Part 25**, which specifies the airworthiness standards for transport category airplanes and ensures compliance with structural, performance, and safety requirements (FAA Part 25).

Furthermore, its intended mission profile is aligned with the operational requirements outlined in **14 CFR Part 121**, which defines the requirements for domestic, flag, and supplemental air carrier operations (FAA Part 121). By adhering to these regulations, the aircraft's mission profile supports safe and reliable commercial passenger service while meeting FAA certification and operational standards.

a. Taxi Out

The taxi-out phase begins when the aircraft leaves the gate and ends when it lines up on the runway for take-off. During this segment, the aircraft moves under its own power at low speeds, usually between 10–20 knots.

Duration: 15 minutes (900 seconds)

Taxi Speed: 35-50 km/h (19-27 knots).

Shaft Horsepower (SHP) during taxi: Approximately 6% of 9071.

$$SHP_{Taxi\ Out} = 0.06 \cdot 9071 = 544.26$$

$$Fuel\ Burn_{Taxi\ Out} = 0.000066 * (0.06 \times 4948) * 900 = 17.63\ kg$$

$$Fuel\ Consumption_{Taxi\ Out} = \left(\frac{17.63}{1859.3} \right) * 100 = 0.948\%$$

$$\frac{W_{after\ Taxi\ Out}}{W_{Before\ Taxi\ Out}} = 1 - \left(\frac{17.63}{29277.31} \right) = 0.99939$$

b. Take Off

The take-off segment covers a short period from the start of the take-off roll until the aircraft is airborne and climbing away from the runway. Take-off thrust will be maintained until 250 m above ground level.

Duration: 2 minutes (120 seconds)

Take-off Distance: 1700 m.

Flaps Setting: 15 degrees.

Take-off Speed (V_r): 110-125 KIAS / 203-231 km/h (110-125 knots).

Shaft Horsepower (SHP) during Take-off (Both Engine Maximum Output): 9071 SHP.

$$\begin{aligned} SHP_{Take-off} &= 9071 \\ Fuel\ Burn_{Take\ Off} &= 0.000066 * (4948) * 120 = 39.19\ kg \\ Fuel\ Consumption_{Take\ Off} &= \left(\frac{39.19}{1859.3} \right) * 100 = 2.11\% \\ \frac{W_{after\ Take\ Off}}{W_{Before\ Take\ Off}} &= 1 - \left(\frac{39.19}{29277.31} \right) = 0.99867 \end{aligned}$$

c. Climb

The climb segment follows take-off phase at 250 m above ground level and continues until the aircraft reaches its cruising altitude of 7630 m (25,000 ft).

Duration: 15 minutes (900 seconds)

Climb to FL250 (7630m / 25,000 ft).

Angle of attack decreases while maintaining a climb rate.

Begin flap retraction and reduce thrust to around 90% of max thrust.

Climb Speed: 296-342 km/h (160-185 knots)

Climb Rate: 457-609 m/min (1500-2000 ft/min)

Shaft Horsepower (SHP) during Climb: Approximately 80% of 9071.

$$\begin{aligned} SHP_{climb} &= 0.8 * 9071 = 7256.8 \\ Fuel\ Burn_{climb} &= 0.000066 * (0.8 * 4948) * 900 = 235.13\ kg \\ Fuel\ Consumption_{climb} &= \left(\frac{235.13}{1859.3} \right) * 100 = 12.65\% \\ \frac{W_{after\ Climb}}{W_{Before\ Climb}} &= 1 - \left(\frac{235.13}{29277.31} \right) = 0.99196 \end{aligned}$$

d. Cruise

The cruise segment is the longest and most fuel-efficient segment, where the aircraft maintains constant altitude and speed while covering most of the planned route.

Duration: 116.5 minutes (6990 seconds)
 Cruise Altitude: FL250 (7630m / 25,000 ft).
 Cruise Speed: 667 km/h (360 knots).
 Range: 800 to 1500 km.
 Shaft Horsepower (SHP): Approximately 75% of maximum SHP.

$$\begin{aligned}
 SHP_{cruise} &= 0.75 \cdot 9071 = 6803.25 \\
 Fuel\ Burn_{cruise} &= 0.0000302 * (0.75 \times 4948) * 6990 = 783.38\ kg \\
 Fuel\ Consumption_{cruise} &= \left(\frac{783.38}{1859.3} \right) * 100 = 42.13\% \\
 \frac{W_{after\ Cruise}}{W_{Before\ Cruise}} &= 1 - \left(\frac{783.38}{29277.31} \right) = 0.97324
 \end{aligned}$$

e. Descent

The descent segment starts at the top of descent and continues until the aircraft reaches traffic pattern altitude near the destination airport.

Duration: 15 minutes (900 seconds)
 Descend Speed: 667 km/h (360 knots) initially, reducing to 463–500 km/h (250-270 knots) below 3000 m (9842 ft).
 Descend Rate: 460–610 m/min.
 Shaft Horsepower (SHP): Approximately 10% of maximum SHP.

$$\begin{aligned}
 SHP_{descend} &= 0.1 \cdot 9071 = 907.1 \\
 Fuel\ Burn_{Descent} &= 0.000066 * (0.1 \times 4948) * 900 = 29.39\ kg \\
 Fuel\ Consumption_{Descent} &= \left(\frac{29.39}{1859.3} \right) * 100 = 1.58\% \\
 \frac{W_{after\ Descent}}{W_{Before\ Descent}} &= 1 - \left(\frac{29.39}{29277.31} \right) = 0.9989
 \end{aligned}$$

f. Landing

The landing segment covers the final approach after descent segment, flare, touchdown, and rollout until the aircraft slows to taxi speed.

Duration: 5 minutes (300 seconds)
 Approach Speed: 210 km/h (113 knots).
 Estimated Maximum Landing Weight (MLW): 28,000 kg.
 Flaps Setting (Maximum): 35 degrees.
 Landing Distance: 1070 m (ISA, SL, MLW, Flaps 35).
 Touchdown Speed: 200 km/h (108 knots).
 Shaft Horsepower (SHP): Approximately 10% of maximum SHP.

$$\begin{aligned}
 SHP_{Land} &= 0.1 \cdot 9071 = 907.1 \\
 Fuel\ Burn_{Land} &= 0.000066 * (0.1 \times 4948) * 300 = 9.8\ kg
 \end{aligned}$$

$$Fuel\ Consumption_{Land} = \left(\frac{9.8}{1859.3} \right) * 100 = 0.527\%$$

$$\frac{W_{after\ Land}}{W_{Before\ Land}} = 1 - \left(\frac{9.8}{29277.31} \right) = 0.99967$$

g. Taxi In

The taxi-in segment begins once the aircraft exits the landing runway and continues until it reaches the assigned gate. Engine power is set at idle or slightly above, and ground speed remains low, usually under 20 knots.

Duration: 10 minutes (600 seconds)

Taxi Speed: 35-50 km/h (19-27 knots).

Shaft Horsepower (SHP) during taxi: Approximately 6% of 9071 SHP.

$$SHP_{Taxi\ In} = 0.06 \cdot 9071 = 544.26$$

$$Fuel\ Burn_{Taxi\ In} = 0.000066 * (0.06 \times 4948) * 600 = 11.76\ kg$$

$$Fuel\ Consumption_{Taxi\ In} = \left(\frac{11.76}{1859.3} \right) * 100 = 0.632\%$$

$$\frac{W_{after\ Taxi\ In}}{W_{Before\ Taxi\ In}} = 1 - \left(\frac{11.76}{29277.31} \right) = 0.99959$$

$$Total\ Remaining\ Weight = \frac{W_{final}}{W_{Initial}}$$

$$= 0.99939 * 0.99867 * 0.99196 * 0.97324 * 0.9989 * 0.99967 * 0.99959$$

$$= 0.9618$$

h. Mission Profile Plot

Figures 11 and 12 show the mission profile in a graphical form. In figure 11, it shows the fuel properties during the difference mission phases. Figure 12 shows the altitudes of the FR25 during each mission phase.

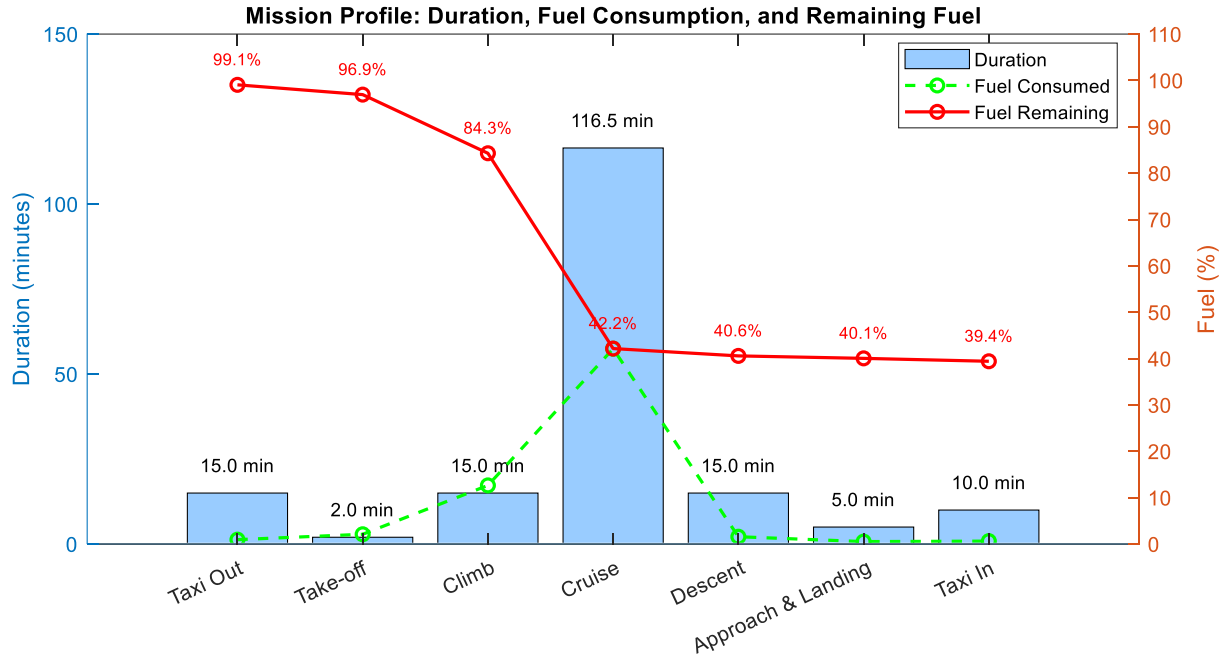


Figure 11: Mission Plot: Duration, Fuel Consumption and Remaining Fuel

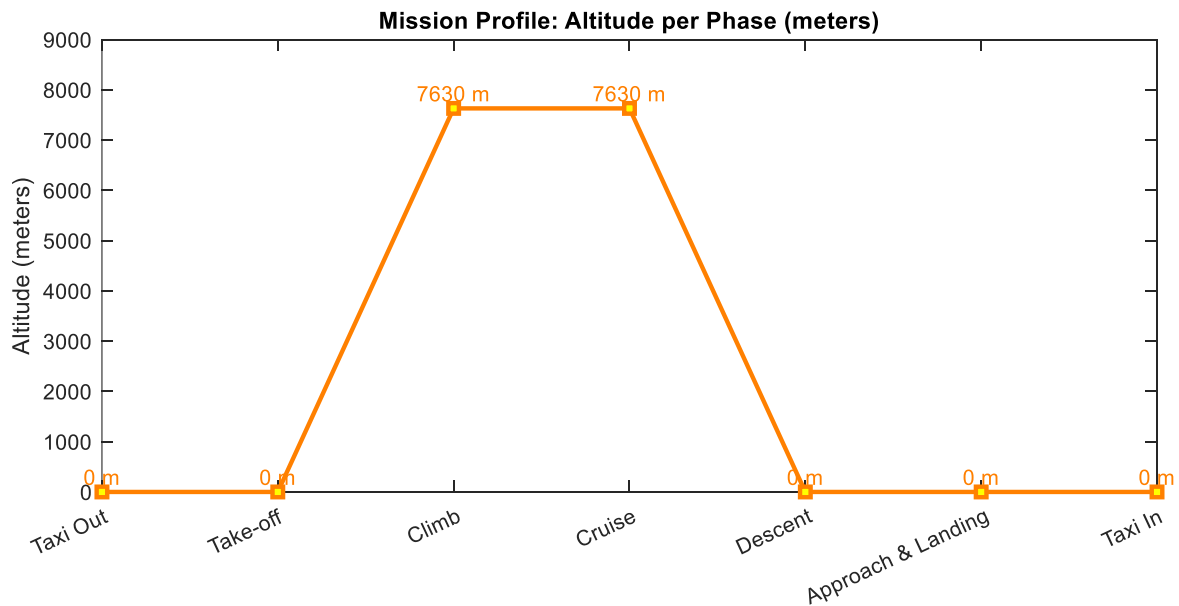


Figure 12: Mission Profile: Altitude

d. Payload Range Diagram (Jit Huat)

The payload range diagram illustrates the trade-off relationship between the payload and the range of an aircraft. The structure of an aircraft is designed to be able to sustain certain number of loads, these loads depend on the amount of weight the aircraft carries. Thus, there is a payload value for the aircraft MTOW that cannot be exceeded, or the aircraft frame would collapse. When an aircraft is carrying maximum payload, the amount of fuel that can be embarked to fly a certain range is then limited by the MTOW. After arriving at this point, any further range can only be achieved by reducing the payload, which follows a linear relationship. The amount of fuel an aircraft can carry is also limited, hence at a sudden point this linear relationship changed drastically.

The range of the FR25 hybrid-electric was calculated using the modified Breguet range equation for hybrid-electric aircraft with constant power-split. This equation is developed by de Vries, Hoogreef and Vos (De Vries, 2024). This equation accounts for the energy characteristics of both fuel and battery sources operating in parallel. The hybrid range equation is expressed as:

$$R = \eta_3 \frac{e_f}{g} \frac{L}{D} \left(\eta_1 + \eta_2 \frac{\Phi}{1 - \Phi} \right) * \ln \left(\frac{W_{OE} + W_{PL} + \left(\frac{g}{e_{bat}} \right) E_{0;tot} \left(\Phi + \left(\frac{e_{bat}}{e_f} \right) (1 - \Phi) \right)}{W_{OE} + W_{PL} + \left(\frac{g}{e_{bat}} \right) \Phi E_{0;tot}} \right)$$

Where:

R = range

η_1, η_2, η_3 = transmission and propulsive efficiencies

e_f, e_{bat} = specific energy of fuel and battery

g = gravitational acceleration

$\frac{L}{D}$ = lift to drag ratio

Φ = supplied power ratio

W_{OE}, W_{PL} = operating empty weight and payload weight

$E_{0;tot}$ = total energy at start of mission

Table 10 : Values used for range calculation

Parameter	Value	Unit
η_1	0.90	-
η_2	0.90	-
η_3	0.90	-
e_f	43,000	-
e_{bat}	900	KJ/kg
$\frac{L}{D}$ (Cruise)	15	KJ/kg
Φ	0.40	40% battery/turboprop

W_{OE}	192423	N
W_{PL}	90252	N
$E_{0,tot}$	300	MJ

Using the parameters in table 10, the FR25 achieves a cruise range of:

$$R = 1531 \text{ km} = 827 \text{ nm}$$

This range represents the maximum distance achievable at constant cruise altitude and speed of 185 m/s with simultaneous power contributions from both the turboprop engine and battery system. The respective range values are calculated using the Hybrid Range Equation formula by varying the payload weight. The calculations are done by utilizing MATLAB software. (Refer to Appendix for calculations).

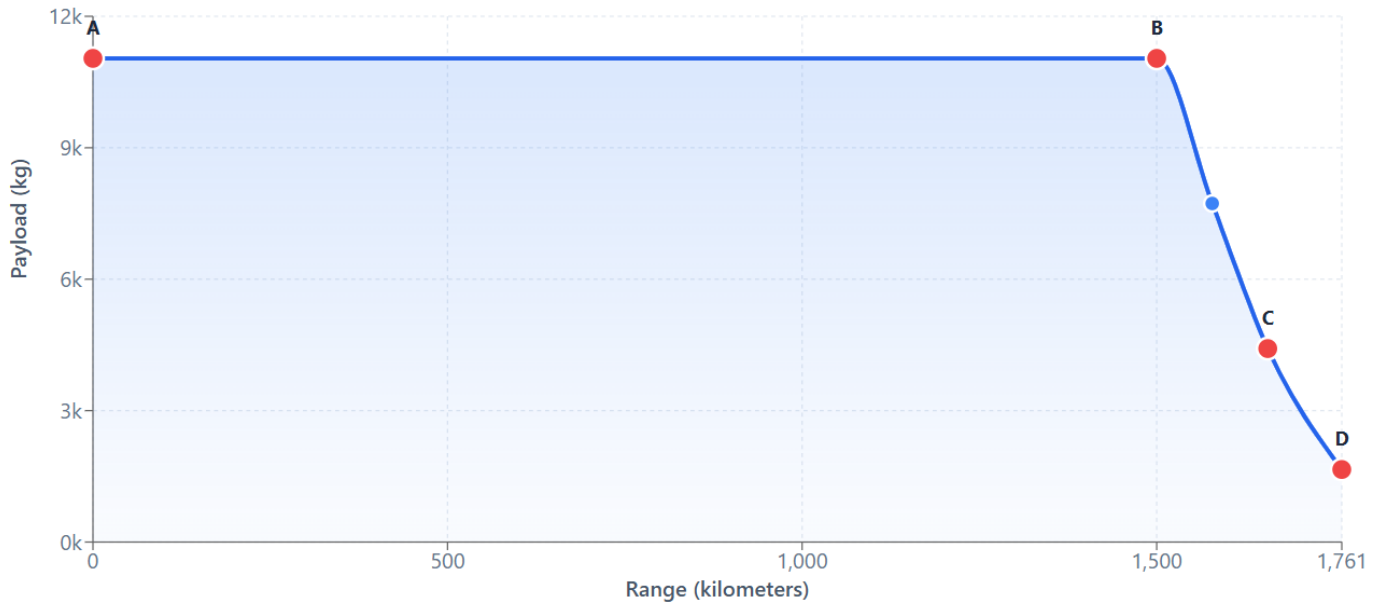


Figure 13: Payload Range Diagram

The graph in figure 13 can be depicted as such:

- Point A: Maximum structural payload with minimum fuel - **shortest range**
Fuel Weight, $W_{fuel} = 0 \text{ kg}$ (Minimum)
Payload Weight, $W_{payload} = 19252 \text{ kg}$ (Maximum)
Range, $R = 0 \text{ km}$
- Point B: Maximum payload with maximum fuel - **optimum payload range**
Fuel Weight, $W_{fuel} = 1859.3 \text{ kg}$ (Maximum)
Payload Weight, $W_{payload} = 19252 \text{ kg}$ (Maximum)
Range, $R = 1531 \text{ km}$
- Point C: Reduced payload to accommodate full fuel - **extended range**

$$\begin{aligned} \text{Fuel Weight, } W_{fuel} &= 1859.3 \text{ kg (Maximum)} \\ \text{Payload Weight, } W_{payload} &= 6000 \text{ kg (Reduced)} \\ \text{Range, } R &= 1610 \text{ km} \end{aligned}$$

- Point D: Maximum fuel with minimum payload - **maximum ferry range**

$$\begin{aligned} \text{Fuel Weight, } W_{fuel} &= 1859.3 \text{ kg (Maximum)} \\ \text{Payload Weight, } W_{payload} &= 0 \text{ kg (Minimum)} \\ \text{Range, } R &= 1761 \text{ km (Maximum)} \end{aligned}$$

7. FR25 CAD Model (Swee Woei)

Regional and/or business aircraft are known to be smaller in size, to contain material costs, and for aerodynamic efficiency, promoting cost savings. The selection of the fuselage size therefore demonstrates the need to balance cost effectiveness and spaciousness.

FlowRider have incorporated the ability for the FR25 to be modular in the cabin layout. Under general demand circumstances, the aircraft will be able to hold a maximum of 88 passengers. This maximizes the available space within the fuselage while ensuring adequate spacing between each seat.

If the demand for cargo exceeds that of personnel transportation, the cabin layout of the aircraft can be retrofitted easily with a cargo compartment. The aircraft's modularity therefore ensures that every operational cycle is maximized both in terms of profitability and efficiency in transportation.

The retrofitting of the rear seatings to fit cargo, and vice versa, is done via a cargo door at the rear of the fuselage.

A high tail design is likely to be incorporated, considering the high-wing turboprop nature of our design; the horizontal stabilizer will be placed outside of the slipstream of the propellers.

a. Passenger Configuration

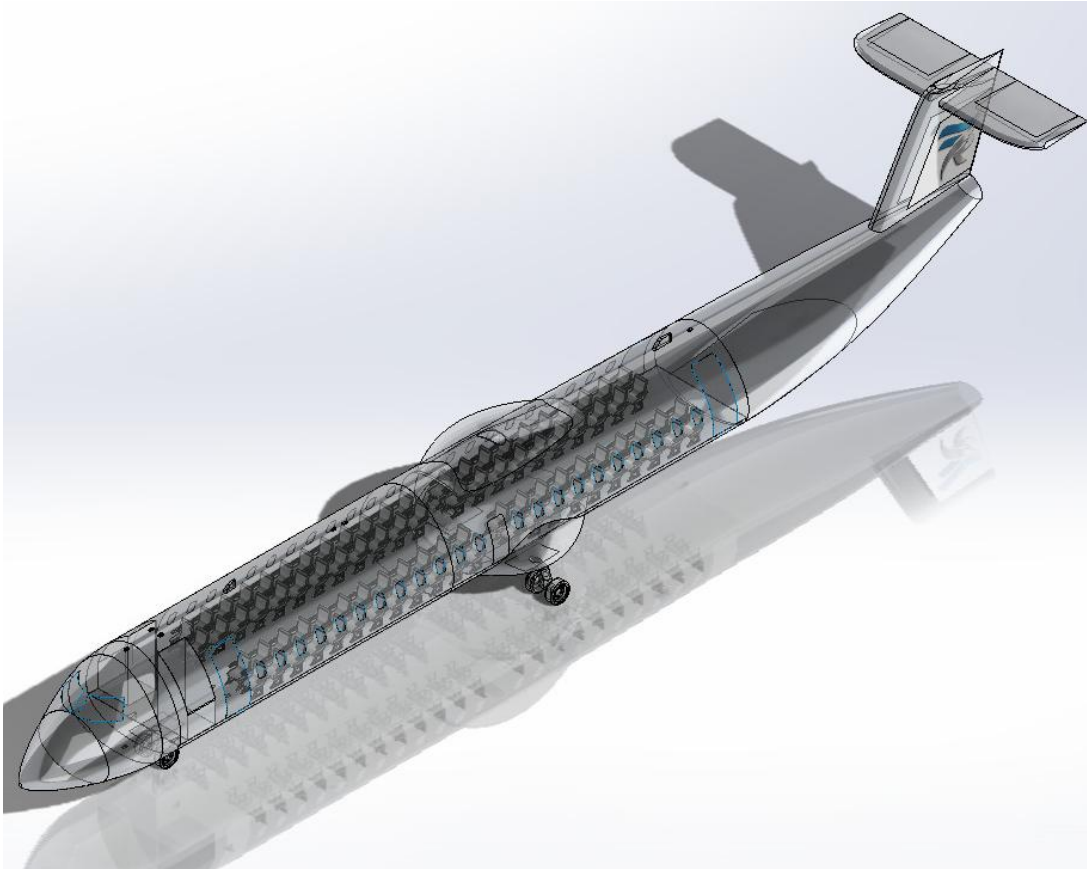


Figure 14: Fuselage Passenger Configuration

The aircraft is fitted with a 2 by 2 configuration, featuring 48 seats along the front, and 40 seats along the rear as seen in figure 14. The lavatory is designed to be at the front of the aircraft, and the 48 seats at the front are meant to be permanent.

b. Passenger and Cargo Configuration

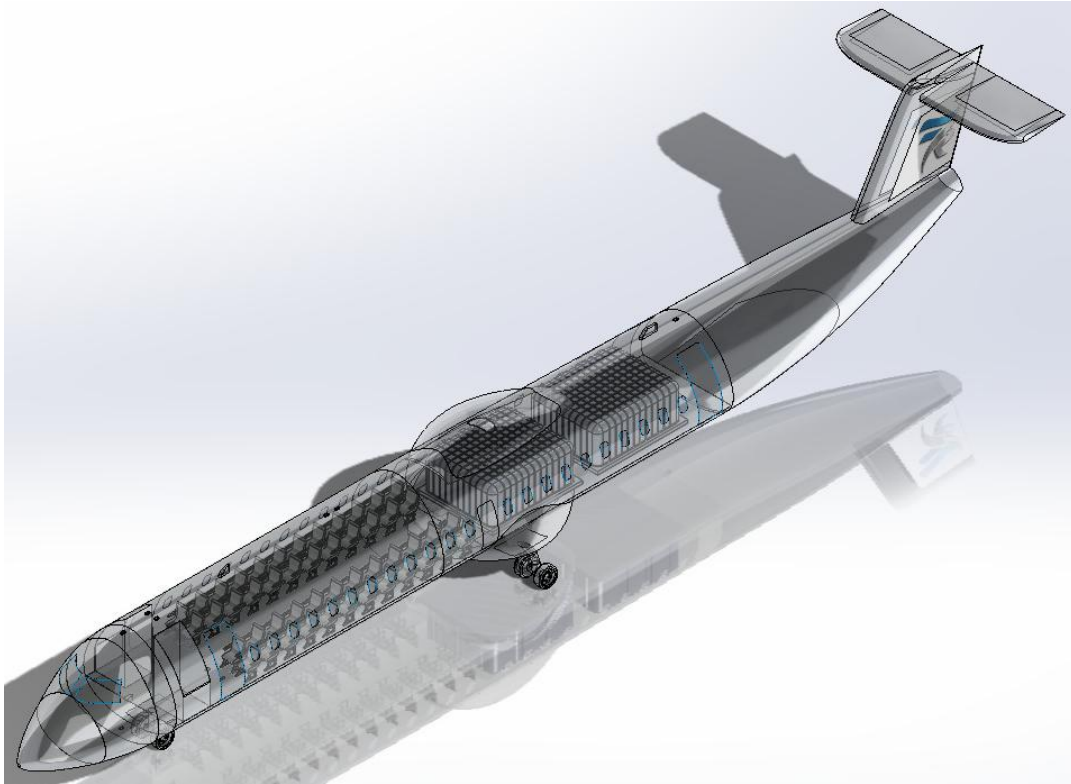


Figure 15: Fuselage Passenger and Cargo Configuration

When the rear 40 seats are removed, the aircraft will have the capacity to house upwards of two cargo pallets, subject to prevailing dimensions as seen in Figure 15.

Structural requirements are necessary and mandated under FAR 25. These requirements work to ensure (1) no permanent deformation under normal conditions, and (2) sufficient structural overhead in the event of adverse conditions.

The requirements for the FR25, is laid out by the FAA in FAR title 14, Chapter I (Federal Aviation Administration, Department of Transportation), Subchapter C (Aircraft), Part 25 (Airworthiness Standards: Transport Category Airplanes).

In general, the chapters are grouped into the following categories (Cornell Law School, n.d.):

- § 25.1 to § 25.5 General
- § 25.21 to § 25.255 Flight
- § 25.301 to § 25.581 Structures
- § 25.601 to § 25.899 Design and Construction
- § 25.901 to § 25.1207 Powerplant
- § 25.1301 to § 25.1461 Equipment
- § 25.1501 to § 25.1587 Operating Limitations and Information
- § 25.1701 to § 25.1733 Electrical Wiring Interconnection Systems
- § 25.1801 Special Federal Aviation Regulations

The relevant requirements for FR25 includes the following non-exhaustive list:

- § 25.33 Propeller Pitch and Pitch Limits
- § 25.251 Vibration and Buffeting
- § 25.301 Loads
- § 25.303 Factor of Safety
- § 25.321 General Flight Loads
- § 25.341 Gust and Turbulence Loads
- § 25.361 Engine and Auxiliary Power Unit Torque
- § 25.362 Engine Failure Loads
- § 25.363 Side Load on Engine and Auxiliary Power Unit Mounts
- § 25.365 Pressurized Compartment Loads
- § 25.367 Unsymmetrical Loads Due to Engine Failure
- § 25.483 One-gear Landing Conditions
- § 25.561 General Emergency Landing Conditions
- § 25.562 Emergency Landing Dynamic Conditions
- § 25.563 Structural Ditching Provisions
- § 25.571 Damage-tolerance and Fatigue Evaluation of Structure
- § 25.581 Lighting Protection
- § 25.631 Bird Strike Damage
- § 25.651 Proof of Strength of Control Surfaces
- § 25.775 Windshields and Windows
- § 25.905 Propellers
- § 25.907 Propeller Vibration and Fatigue
- § 25.925 Propeller Clearance

To that end, the team will work with the Federal Aviation Administration (FAA), and Civil Aviation Administration of Singapore (CAAS) to fulfil the necessary requirements.

c. Fuselage Exterior

i. Radome

The radome works to house important antennas and radars and is thus generally made of composite materials to not interfere with the signals sent out by the antennas and radars.

ii. Fuselage Body

The fuselage body works as the primary structure carrying all relevant payload – both living things and cargo. To that end, both static and dynamic loads likely to be experienced by the aircraft must be accounted for, and the structure must be overdesigned by a factor of safety to ensure (1) safe operation under normal operating conditions, and (2) continued compromised operation until the aircraft can land at a safe area.

iii. Aircraft Wings

In being the primary load carrying structure on the aircraft, the aircraft wings likewise must be designed and built to sustain and withstand sufficient forces acting on it, in ways not detrimental

or permanent under normal operating conditions. Flight controls like slats, and ailerons must be proven to be able to withstand limit loads.

iv. Aircraft Engines

Particularly in FR25's case, the engines are mounted on the wings and are turboprop engines. The vibrations from the normal operating range of the engines must not coincide with the natural frequency of the wing, and the engine must be demonstrated to not fail destructively in the event of an adverse incident.

v. Horizontal Stabilisers

The horizontal stabilisers house the elevators which enables the pitching motion of the aircraft. In FR25, the horizontal stabilisers and elevators are designed to be slightly oversized to (1) provide greater stability along the pitching axis, and (2) ensure effective elevator controls during low airspeed operations.

vi. Vertical Stabiliser

The vertical stabiliser works as the primary structure stabilising the aircraft's yaw axis. The vertical stabiliser also houses the rudder, both of which are designed to be slightly oversized to (1) provide greater stability along the yaw axis, and (2) ensure effective rudder control during low airspeed operations.

vii. Landing Gears

While the number of wheels is not mandated, the landing gear must be designed to be of proper size and fit, ensuring that each tire is able to sustain loads under critical operating conditions.

viii. Cargo Door

The cargo door in the FR25 serves to allow for the quick conversion of the aft passenger seats to a cargo section. The legislations governing the cargo door includes a mixture from § 25.783 (Fuselage Doors), § 25.787 (Stowage Compartments) and others.

d. Fuselage Interior

The fuselage interior is designed to maximise passenger comfort and safety. It is also designed in such a way where the rear passenger seats can be easily removed to slide in the cargo modules.

While the design and implementation of a hybrid passenger and cargo configuration is a challenge on its own, the key challenge is the need for both safety and regulatory compliance when both passenger and cargo are to be packed within the same space.

i. Passenger Seats

The seats onboard the aircraft seen in figure 16 will feature a balance between comfort and profitability, ensuring adequate dimensions and spacing. The placement, spacing and dimensions of the seats are not lined out within the FAR 25. The seats must however be designed to withstand up to 16g in dynamic loadings, as per FAR § 25.562.



Figure 16: Passenger Seat

ii. Cargo Pallets

While the exact dimensions are not mandated in FAR 25, the figure below illustrates a cargo pallet of fixed dimensions. The retrofitted rear compartment can realistically fit any form of cargo that can be fitted within the fuselage.

In satisfying safety and legislative requirements, the cargo must be secured by approved means, in ways that prevent shifting during both flight and ground movement. Especially when near passengers, injuries must be prevented, and access to emergency exits must not be obstructed.

To that end, mounting points will be designed along the floorboard at the rear, and the team will work closely with the authorities to ensure both safety and regulatory compliance. Best practices and recommendations like the FAA Advisory Circular 120-85B which outlines the recommended procedures for cargo operations is an example of the few documents that will be referred to.

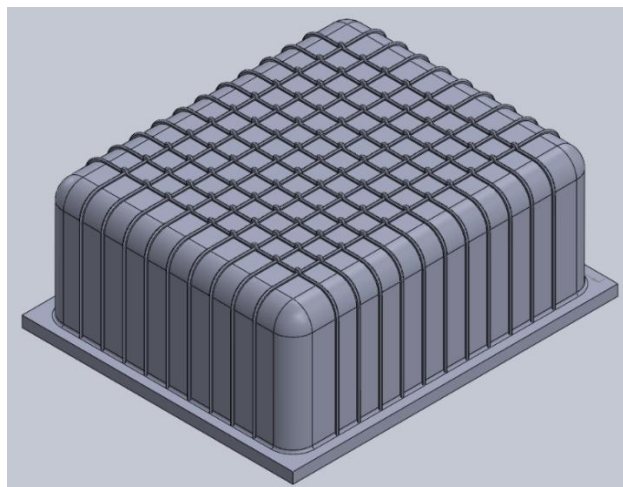


Figure 17: Cargo Pallet

iii. Lavatory

While there is no specified number of lavatories necessary to be onboard transport aircrafts, the U.S. Department of Transportation (DOT) has recently passed a rule in 2023 concerning the accessibility of lavatories on single-aisle aircrafts. Specifically, at least one lavatory onboard single-aisle aircrafts must be able to be accessed by personnel with relevant disabilities. This ruling is mandatory for single-aisle aircrafts with upwards of 125 seating capacity (U.S. Department of Transportation, 2023).

In our design, we feature a single-aisle design, with a maximum of 88 seating capacity, while featuring two lavatories at the front of the aircraft.

iv. Fuselage Doors

Pursuant to FAR § 25.807, the number and types of doors necessary onboard aircrafts is scaled to the seating capacity. In our design, the specific clause that applies to us states " For a passenger seating configuration of 41 to 110 seats, there must be at least two exits, one of which must be a Type I or larger exit, in each side of the fuselage."

Type I of which, are defined as "...a floor-level exit with a rectangular opening of not less than 24 inches wide by 48 inches high, with corner radii not greater than eight inches". Type II are then defined as "...not less than 20 inches wide by 44 inches high, with corner radii not greater than seven inches. Type II exits must be floor-level exits..."

In metric terms, type I doors then translate to minimum dimensions of 609.6 mm wide by 1219.2 mm high, with corner radii not greater than 203.2 mm. Type II doors translate to minimum dimensions of 508 mm wide by 1117.6 mm, with corner radii not greater than 177.8 mm.

The FR25 is fitted with four (4) type 1 doors, and two (2) type 2 doors.

Type 1 doors measure 780 mm wide by 1650 mm high, with corner radii of 60 mm, and are in direct fulfilment of type I doors. Type 2 doors measure 510 mm wide by 1120 mm high, with corner radii of 80 mm, and are in direct fulfilment of type II doors.

Type 2 doors are designated emergency doors located on both sides of the fuselage, just aft of the engines.

When in full passenger configuration, the FR25 will have all four (4) type 1 doors, and two (2) type 2 doors operational.

When in passenger and cargo configuration, while the rear two (2) type 1 doors will be inaccessible, there will still be two (2) type 1 doors at the front of the aircraft, sufficiently fulfilling FAR § 25.807's requirement of minimally two (2) type I or larger exit doors.

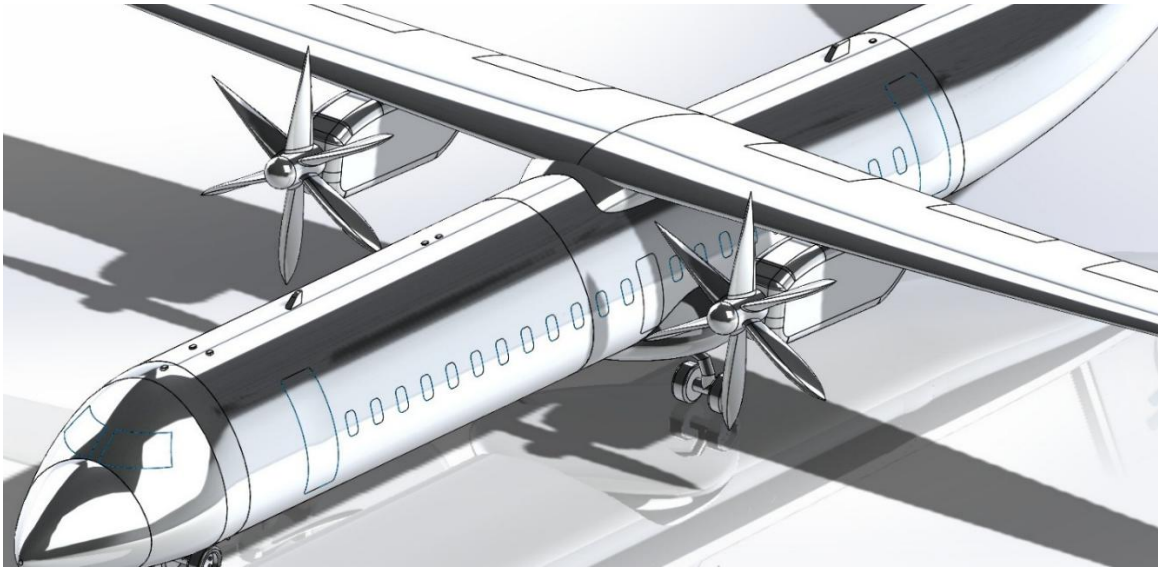


Figure 18: Close up of Fuselage Doors

v. Battery

In reference to § 25.1351 through § 25.1365, the design and management of battery safety must be factored in as the FR25 materializes. As we aim to leverage on emerging technologies in both battery, and hybrid electric engines, care must be taken to incorporate these emerging technologies while maintaining high safety standards, and regulatory compliance.

e. Component Sourcing

The manufacturing of an aircraft requires the expertise of market experts in various fields to gather and synergize the best possible setups. Figure 19 illustrates a sample of FlowRiders prospective business partners to be engaged with as the company progresses down the design timeline.

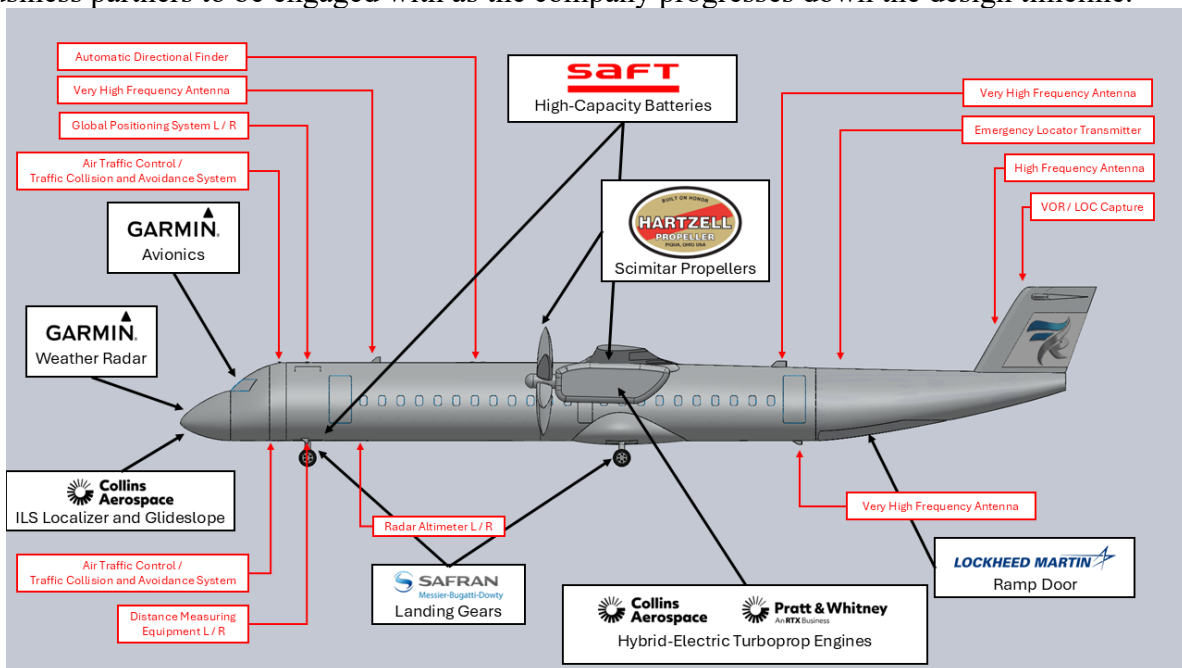


Figure 19: Aircraft Components and Partners

f. Assembled Design

Subsequent chapters will explore the details about the design methodology, calculations, choices for the wings, and engines respectively. The chapter here will contain the fully assembled design of the aircraft.

The areas where the wing joins the fuselage presents areas of sharp edges and drastic changes. These areas as we come to know, are not ideal both in terms of drag, and stress concentration. To that end, a fairing is designed to smoothen the area of intersection.

Figure 20 serves to illustrate the exterior of the assembled aircraft. Figure 21 shows the fuselage interior with passenger seats configuration. Figure 22 shows the fuselage interior with both passenger seats and cargo configuration.

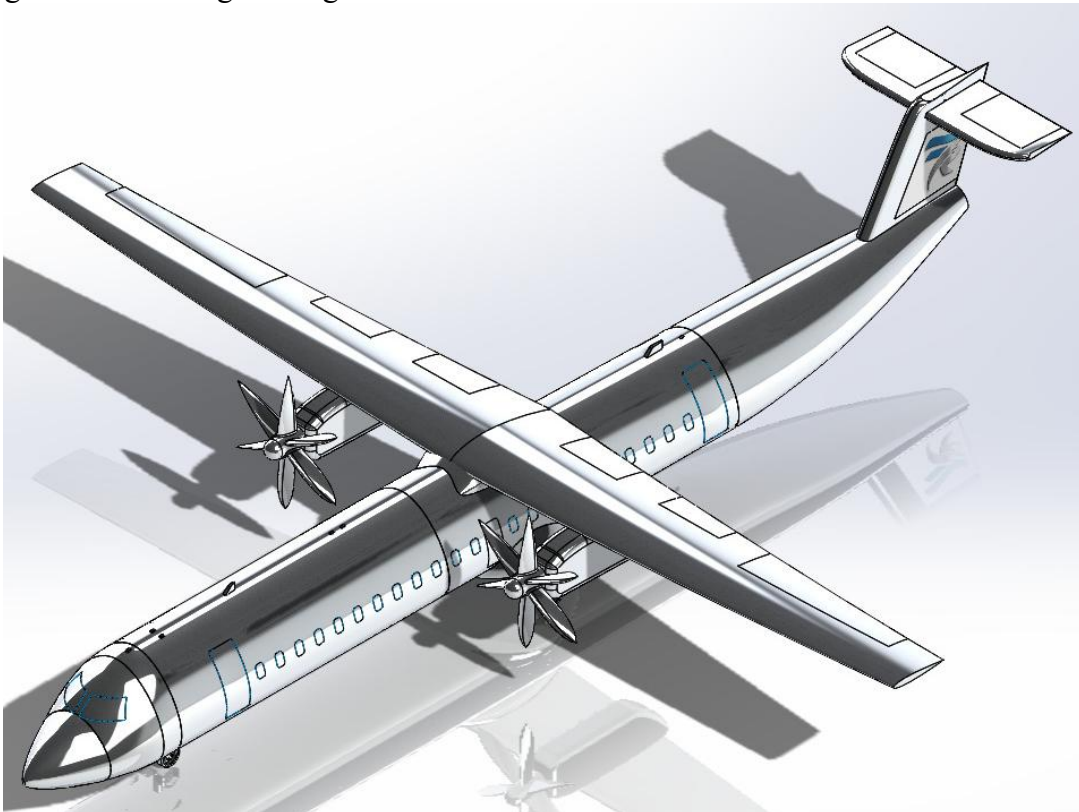


Figure 20: Aircraft assembled (Exterior)

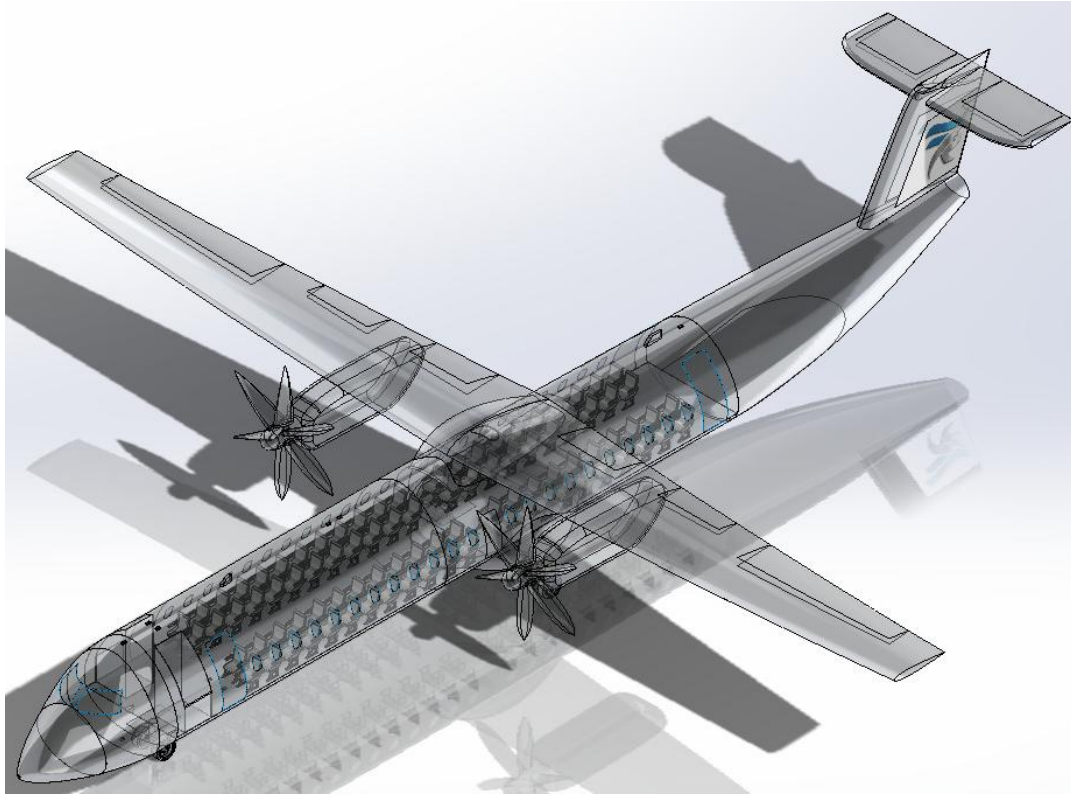


Figure 21: Aircraft assembled (Seats)

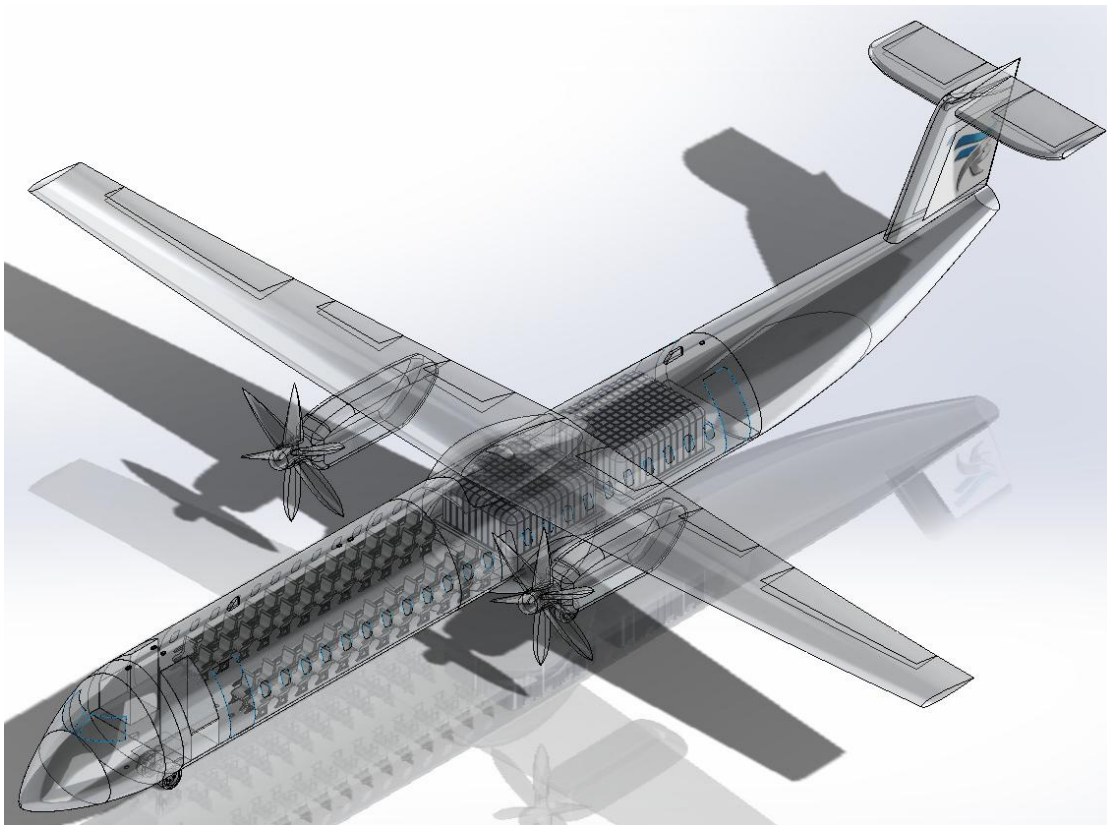


Figure 22: Aircraft assembled (Seats and Cargo)

i. Front View

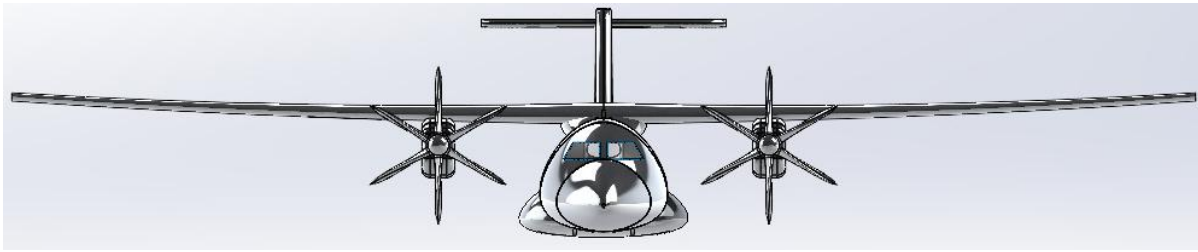


Figure 23: Front View (Gear Up)



Figure 24: Front View (Gear Down)

ii. Left View

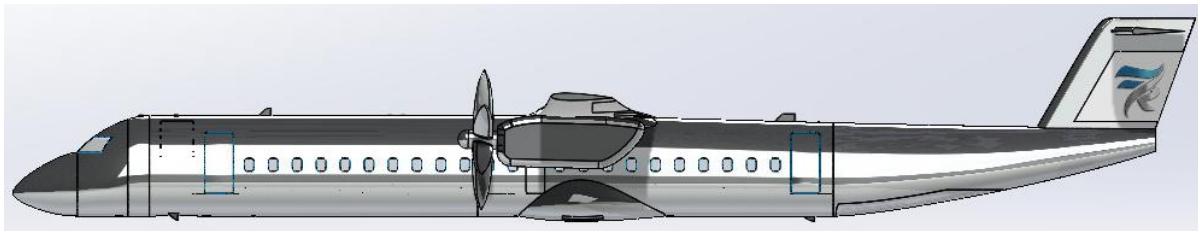


Figure 25: Left View (Gear Up)

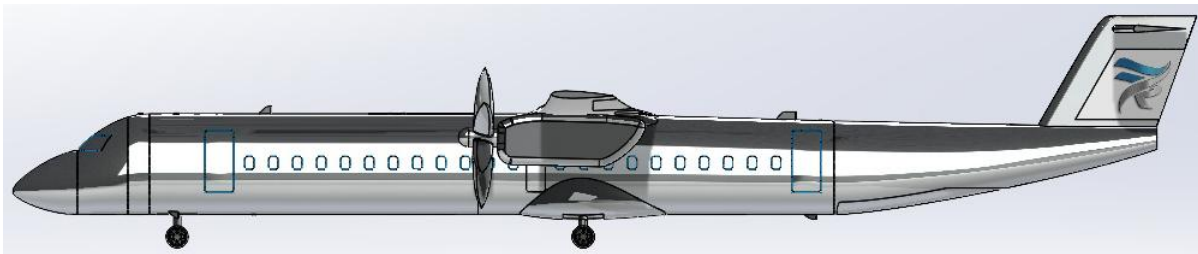


Figure 26: Left View (Gear Down)

iii. Rear View

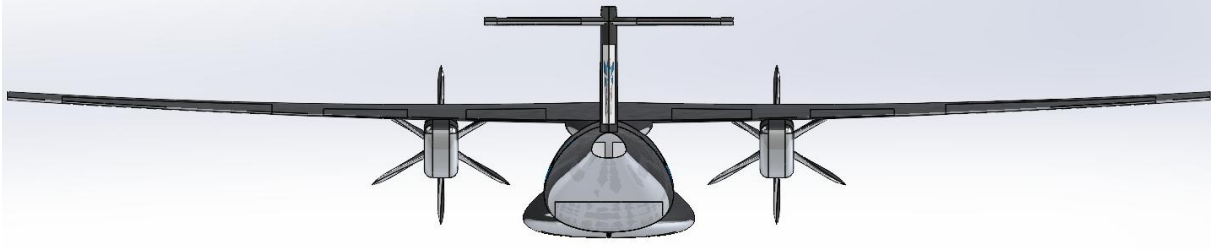


Figure 27: Rear View (Gear Up)

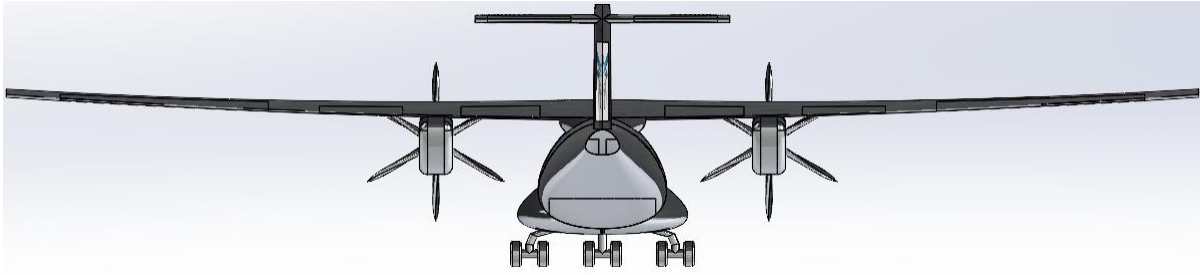


Figure 28: Rear View (Gear Down)

iv. Right View

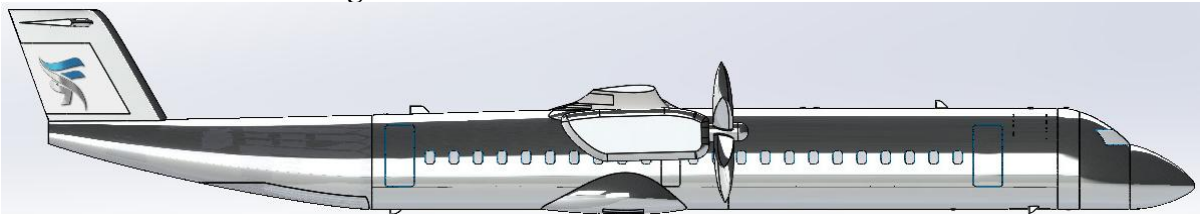


Figure 29: Right View (Gear Up)

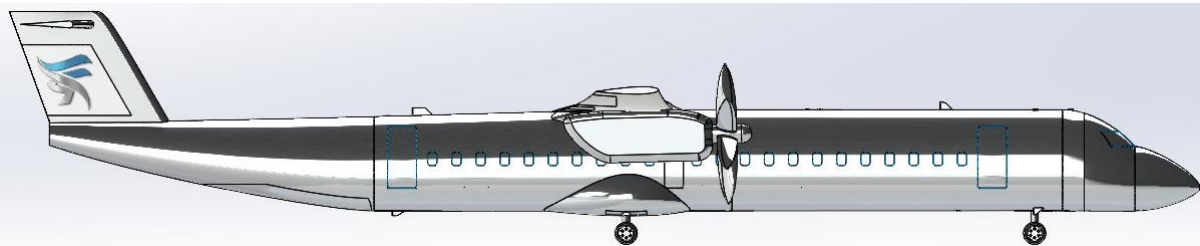


Figure 30: Right View (Gear Down)

v. Plane View

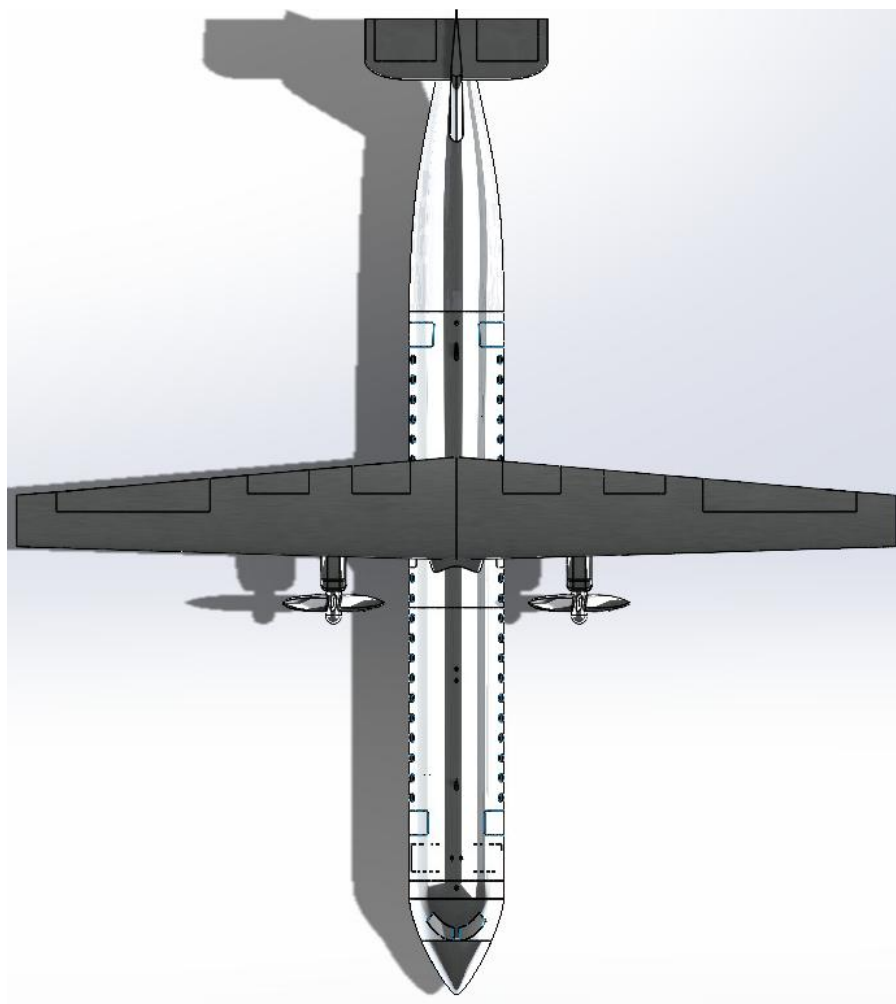


Figure 31: Plane View

vi. **Bottom View**

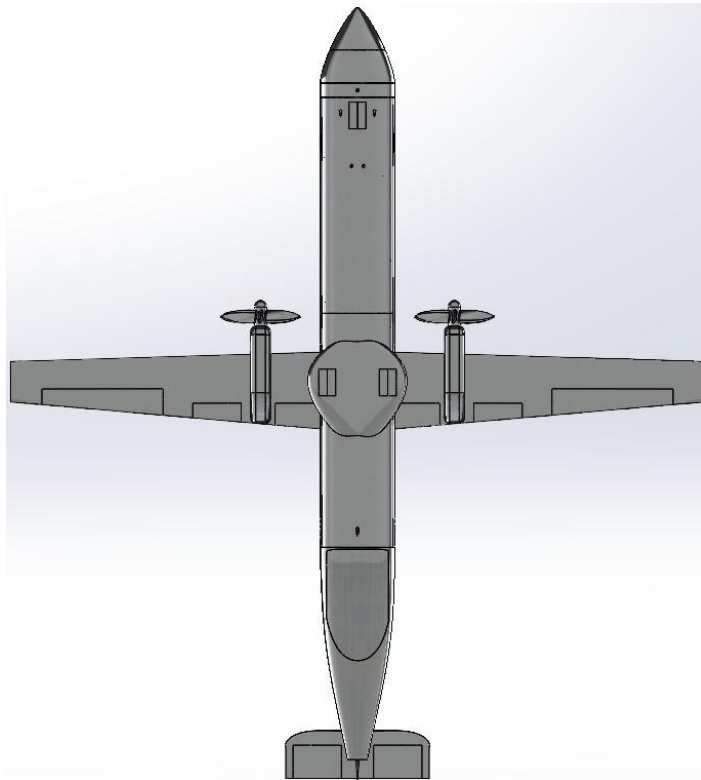


Figure 32: Bottom View (Gear Up)

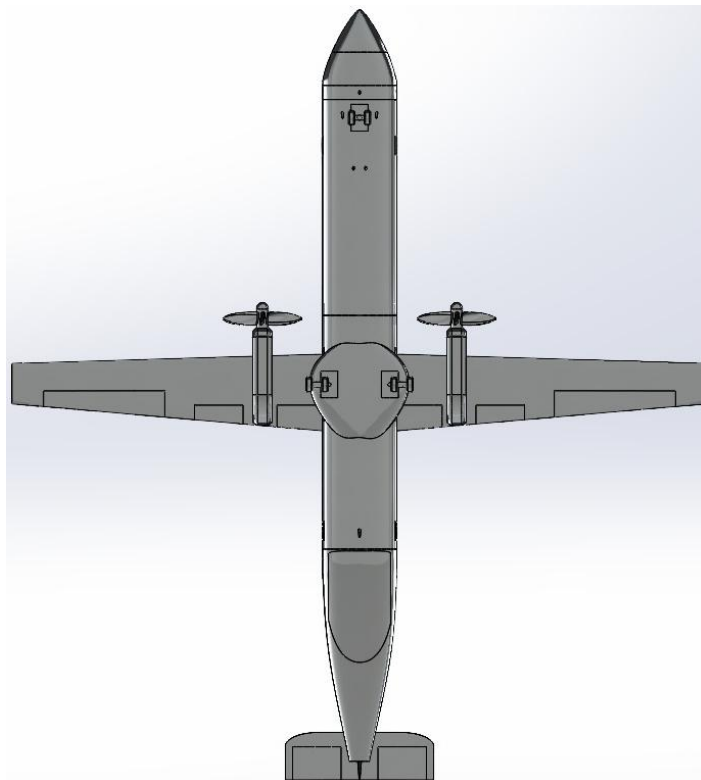


Figure 33: Bottom View (Gear Down)

8. Technical Details of FR25 (Heong Luck, Jason, Jit, Florian)

FR25 Sizing (Heong Luck)

The following sizing process determines the appropriate wing, horizontal stabilizer, and the vertical stabilizer to ensure sufficient lift, control, and stability for the FR25 design. The calculations in this section will be based on key parameters such as lift coefficient, tail volume coefficient and many other aerodynamic factors to effectively determine the right sizing for these surfaces to ensure that the FR25 can achieve the optimal lift generation, control effectiveness and overall aerodynamic balance and stability of the aircraft.

i. Wing Sizing

The wing is the main component on the aircraft that will generate the lift, which will balance out with the weight of the aircraft during all phases of flight. Furthermore, it also houses many key control surfaces that will help control the aircraft's motion and flight stability. The ailerons, located near the wing tip on the trailing edge of the wing, will help to provide roll control, allowing the aircraft to roll left and right during manoeuvre.

The flaps located on the trailing edge of the wing nearer to the wing root, will increase the lift and drag during take-off and landing. More lift and drag during landing will improve control of sink rate and help slow down the aircraft before touchdown.

The wing sizing process is therefore important and is to validate and justify that the proposed wing geometry of the FR25 can provide sufficient lift for the aircraft's operating condition according to its defined weight. Since the FR25 is designed for short to medium haul hybrid turboprop aircraft, the wing geometry must be able to provide high efficiency for regional operations.

From the lift equation:

$$L = W = \frac{1}{2} \rho V^2 S C_L$$

By rearranging the lift equation and making the wing area, S, the subject, we get:

$$S = \frac{2W}{\rho V^2 C_L}$$

The wing area is defined to be 72m². The equation for the wing area is as follows:

$$S = \frac{2 \times 30693.072 \times 9.81}{0.549 \times 185.2^2 \times C_L} = 72m^2$$

Where:

$$\rho = 0.549kg/m^3$$

$$V = 185.2m/s$$

Making C_L the subject:

$$C_L = 0.4442$$

With the wing area, S, now known, wing loading can be calculated using:

$$\frac{W}{S} = \frac{30693.072 \times 9.81}{72} = 4182 \text{ N/m}^2$$

Based on the lift equation and some aerodynamic parameters, the proposed wing area of 72 m² produces a practical cruise lift coefficient of C_L of 0.4442 and a wing loading of 4182 N/m². These values are aligned closely with the performance envelopes of the competitors' aircraft such as the ATR 72-600 and the Dash 8-400.

ii. Horizontal Stabilizer Sizing

The horizontal stabilizer is a fixed aerodynamic surface and is located at the tail of the aircraft, controlling the aircraft's pitching motion. The horizontal stabilizers pitches the nose of the aircraft upwards or downwards. The horizontal stabilizer is a major contributor to the longitudinal static stability of the aircraft. On top of that, the elevator, which is attached to the trailing edge of the horizontal stabilizer, allows the pilot to control the pitching motion.

Therefore, sizing the horizontal stabilizer becomes very important to ensure that the aircraft can maintain pitch stability and balance out the pitching moment generated by the main wing and the fuselage.

The governing equation to calculate the horizontal tail volume ratio is as follows:

$$V_H = \frac{l_t S_H}{\bar{c} S}$$

Where:

V_H is the horizontal tail volume coefficient of 0.9 for a typical value for turboprop

l_t is the distance from wing aerodynamic centre to tail aerodynamic centre of 10 m

$\bar{c}$ is the MAC of 2.606m

S_H is the horizontal stabilizer area

S is the main wing area

Substituting in the values and making S_H the subject:

$$S_H = \frac{0.9 \times 72 \times 2.606}{10} = 16.889 \text{ m}^2$$

iii. Vertical Stabilizer Sizing

The purpose of the vertical stabilizer is to provide directional stability, provide yawing motion to the aircraft and maintaining a straight flight. It counters the yawing moment due to crosswind to maintain proper orientation. The rudder, mounted on the trailing edge of the vertical stabilizer, provides yawing control to coordinate turns. In the event of asymmetric power, the rudder can be used to counter yaw the moment created.

Therefore, it is essential for the vertical tail to be appropriately sized to ensure effective yaw control, maintain stability during crosswind operation and recover from a spin that is caused by a stall.

The governing equation for vertical stabiliser sizing is as follows:

$$V_v = \frac{S_v l_v}{S b}$$

Where:

V_v is vertical tail volume coefficient, which is about 0.08 for a turboprop aircraft

S_v is vertical stabilizer area to be determined

l_v is distance from centre of gravity of main wing to the vertical stabilizer aerodynamic centre of 10m

S is 72m²

b is wingspan of 27m

Rearranging the formula and making S_v the subject:

$$S_v = \frac{V_v S b}{l_v}$$

$$S_v = \frac{0.08 \times 72 \times 28.65}{10} = 16.502m^2$$

b. FR25 Flight Dynamics (Jason)

A good aircraft design should have good control and handling qualities and efficient performance characteristics that suit their specific requirements, all while ensuring stability and safety. An aircraft in flight is often subjected to disturbances like wind gusts or turbulence, and thus it is crucial for it to be stable so that the aircraft can recover without pilot intervention. There are two types of stability concepts, namely static stability and dynamic stability.

Static stability refers to initial tendency of the aircraft to return to its original equilibrium state after a small disturbance, while dynamic stability refers to how the aircraft behaves over time after the disturbance. For FR25, despite its modular configuration, it is still a passenger aircraft. Hence, passengers' comfort and safety are of utmost importance in the design of the aircraft. The analysis for stability of the FR25 along the three rotational axes is provided below:

- Lateral axis (Pitching): Longitudinal stability
- Longitudinal axis (Rolling): Lateral stability
- Vertical axis (Yawing): Directional stability

i. Longitudinal Stability

To ensure the FR25's longitudinal static stability, the location of the aircraft centre of gravity (CG) and its neutral point (NP) must be derived. The CG is the point where the total weight of the aircraft acts, while the NP is the aerodynamic centre (AC) of the entire aircraft. The location of the CG and NP is crucial in ensuring that the aircraft remains statically stable. The CG is typically located ahead of the NP to ensure a positive static margin (SM). If the CG is positioned too far ahead of the NP, the aircraft will produce a very strong restoring pitching moment when it encounters any disturbances. Despite being very stable, strong pitching moments will result in reduced control and manoeuvrability,

which is not ideal. If the CG is located aft of the NP, the aircraft becomes statically unstable. Any nose-up disturbance will result in a further increase in the angle of attack, increasing the risk of a deep stall. Hence, the location of the CG and NP is crucial in ensuring the safety of the passengers.

To calculate the CG of the aircraft, the masses of the individual components and the distance between the CG of the component and a reference datum point is required. Using the equation below, the weighted average location of all the aircraft masses is:

$$x_{CG} = \frac{\sum(w_i x_i)}{W_{total}}$$

Where:

x_{CG} is the location of the aircraft's centre of gravity from the datum

w_i is the mass of each component

x_i is the distance of each item from the datum

W_{total} is the total aircraft mass

1. Estimation of Cg at Take Off

With the ability to change the configuration of the cabin, the change in CG of the aircraft when it is in full passenger configuration or modular configuration must be taken into account. However, the CG of the aircraft will not experience much change between both configurations. Assuming maximum passenger capacity, the weight of the passengers is evenly distributed along the fuselage of the aircraft. The moment can be calculated by assuming a point load located in the middle of the fuselage.

$$\text{Max payload: } (88 + 4) \text{ passengers} \times 120 \text{ kg} = 11040 \text{ kg}$$

Similarly, in modular configuration, assuming full passenger capacity of 48 passengers at the front of the aircraft, the weight of the cargo is limited to the remaining balance of the maximum payload. Based on the equation below, the maximum cargo weight should not exceed:

$$\text{Max cargo weight: } (88 - 48) \text{ passengers} \times 120 \text{ kg} = 4800 \text{ kg}$$

Using the nose of the aircraft as datum, the weight and distance between datum and the individual components is tabulated in Table 11. Assuming maximum weight prior to take off with maximum fuel, the aircraft CG is calculated by dividing the sum of individual moments by the total mass.

Table 11 : Weight and distance of FR25 Components during Take Off

Component	Mass (kg)	Distance from nose (m)	Moment (Mass * Distance)
Wing Structure	2104.29	17.02	35815.02
Fuselage	3992.39	15.49	61842.12
Empennage Structure	464.49	32.51	15100.57

Landing Gear	1176.1	16.62	19546.78
Engine and Pylons	2974.2	15.92	47349.26
Operating Systems	3123.81	8.21	25646.48
Furnishing	1606.92	16.62	26707.01
Operational Items	3372.88	15.49	52245.91
Batteries	800	5.98	4784.00
Fuel	1675.84	16.32	27349.71
Max Payload	9200	15.49	142508.00
Sum	30490.92		458894.86
CG from nose (m)	15.05		

The location of the NP can be estimated based on the location of the wing's AC. The AC typically lies near 25% of the mean aerodynamic chord (MAC). Assuming the AC is $\frac{1}{4}$ of the MAC, which is calculated to be 2.61 m, the AC of the wing from the nose is 15.24 m. The NP is generally aft of the wing's AC. For FR25, the NP of the entire aircraft from the nose is estimated to be around 15.35 m. Based on the estimated CG of the entire aircraft, we observe that the CG is located ahead of the NP. This results in a positive SM (NP - CG), making the aircraft statically stable in the longitudinal axis. The values are summarized in table 12.

Table 12 : Summary of Longitudinal stability during take off

Stability Criterion	Distance (m) from Nose	Stability
Neutral point	15.35	-
Aerodynamic Centre	15.24	-
CG location	15.05	-
Static Margin	Positive	Stable

The values for static margin for commercial aircraft, as Raymer suggests, are typically 5 – 15%. This gives the aircraft a good balance between sufficient stability and adequate manoeuvrability. For FR25, the static margin is calculated to be at 11.49% of the MAC.

2. Estimation of CG before landing

We will also have to consider the change in CG prior to landing, when the fuel is depleted. From Section (Weight Analysis Section):

For Range of 1500 km:

$$\text{Fuel Burn Rate} = \frac{0.18}{0.185277} = 0.9715 \frac{kg}{km}$$

$$\text{Trip Fuel} = (0.9715)(1500) = 1457.25 \text{ kg}$$

$$\text{Reserve Fuel (15\% of Trip Fuel)} = (0.15)(1457.25) = 218.59 \text{ kg}$$

The new change in CG is tabulated in Table 13 below.

Table 13 : Weight and distance of FR25 Components during landing

Component	Mass (kg)	Distance from nose (m)	Moment (Mass * Distance)
Wing Structure	2104.29	17.02	35815.02
Fuselage	3992.39	15.49	61842.12
Empennage Structure	464.49	32.51	15100.57
Landing Gear	1176.1	16.62	19546.78
Engine and Pylons	2974.2	15.92	47349.26
Operating Systems	3123.81	8.21	25646.48
Furnishing	1606.92	16.62	26707.01
Operational Items	3372.88	15.49	52245.91
Batteries	800	5.98	4784.00
Fuel (15% remaining)	218.59	16.32	3567.39
Max Payload	9200	15.49	142508.00
Sum	29033.67		435112.54
CG from nose (m)	14.99		

The new CG of the aircraft is 14.99 m, which is ahead of the NP, hence making the aircraft more longitudinal statically stable as seen in table 14.

Table 14 : Summary of Longitudinal stability during take off

Stability Criterion	Distance (m) from Nose	Stability
Neutral point	15.35	-
Aerodynamic Centre	15.24	-
CG location	14.87	-
Static Margin	Positive	Stable

As the position of the CG moves forward, the SM increased to 13.93%, which makes the aircraft more longitudinally stable. This may also result in reduced control effectiveness and reduced manoeuvrability. However, the change in SM is small and it is still within the typical SM value of 5 – 15%. More importantly, the CG location and SM calculated is just an approximate and the values are overestimated.

ii. Directional and Lateral Stability

During flight, an aircraft is prone to sideslip, which may arise due to a sideways wind gust or other disturbances. Sideslip increases drag due to the increase in frontal area of fuselage, which reduces the overall efficiency. Aircraft that have positive directional static stability will generate restoring yawing moment to counter the sideslip. Directional static stability is the ability for the aircraft to yaw into wind to maintain directional equilibrium (Cook, M. V., 2013). Similarly, aircraft that is lateral statically stable will generate restoring rolling moment after a slight disturbance, without pilot input. Since the directional and

lateral stability of the aircraft is interdependent, the design of FR25 is analysed to ensure static stability in both the directional and lateral directions.

1. Wing Configuration

The wing configuration plays an important role to the aircraft's lateral static stability. The FR25 has a slight dihedral with high wing configuration, which generates a rolling moment that nullifies any rolling moment disturbances.

When the aircraft experiences a bank angle disturbance, the imbalance in lateral forces will result in sideslip and rolling moment disturbances. When the aircraft experiences positive disturbing rolling moment, the sideslip velocity of starboard wing will be larger than the port wing. This difference in velocity will result in a difference in dynamic pressure and a lift differential between the wings. The starboard wing will experience increased lift and drag, which produces a stabilising rolling moment in the negative direction. This will oppose the sideslip, making the aircraft more lateral statically stable.

FR25 has a high wing configuration, which also creates a lift differential when the aircraft experiences positive sideslip and rolling moments. The increased lift and drag on the starboard wing due to the high wing will yield a stabilising negative rolling moment, which opposes the sideslip and makes the aircraft more lateral statically stable.

2. Fuselage Configuration

The fuselage will often contribute a destabilising moment. When the aircraft experiences sideslip, it will produce fuselage side force, which is located ahead of the CG. This side force, when multiplied by the moment arm, will generate a destabilising moment.

3. Vertical Tail Configuration

The vertical tail is the primary contributor to directional static stability. As the aircraft experiences positive sideslip, the vertical fin experiences a side force acting to the left due to the incoming relative wind, generating a positive yawing moment. Hence, it is important to size the vertical tail properly as it will affect the directional stability of the entire aircraft. The sizing of the vertical tail is shown in Section 8i.

Each individual component will play a crucial role in affecting the directional and lateral static stability of our aircraft. With the above factors in mind, the design of various components in FR25 has been optimised to ensure that the aircraft is both directional and lateral statically stable.

Flight control refers to the ability of the pilot to manoeuvre the aircraft to a desired attitude. The control surfaces are vital elements that enable the pilot to manoeuvre the aircraft along 3 primary axes:

- Lateral axis (Pitching): Controlled by the elevators
- Longitudinal axis (Rolling): Controlled by the ailerons
- Vertical axis (Yawing): Controlled by the rudder

To understand the effect of the control surfaces on the aircraft, the moments are first calculated. There are three main moments generated by the control surfaces:

- Pitching Moments M
- Yawing Moments N
- Rolling Moments L

iii. Pitching Moment (M)

The pitching moment controls the rotation of the aircraft about the lateral axis. The elevator, located on the vertical tail, is the primary directional control surface. The total aircraft pitching moment is the combination of the individual components' pitching moment of the CG. The pitching moment equation is shown below:

$$M = \frac{1}{2} \rho V^2 C_m S \bar{c}$$

Where:

ρ : air density

V: velocity of aircraft

C_m : pitching moment coefficient

S: planform wing area

$\bar{c}$: mean aerodynamic chord (MAC)

Based on the equation above, we can identify several factors that will affect the total aircraft pitching moment. It is important to optimise these parameters to achieve a balance between the aerodynamic efficiency and control responsiveness.

1. Pitching Moment Coefficient (C_m)

The pitching moment coefficient is a dimensionless quantity that describes the pitching moment of an aircraft. It is derived from several factors, such as the incident angle of attack but largely contributed by the angle of elevator deflection. Below are some factors to optimise C_m :

- Elevator control power

- As the elevator is the primary control surface for pitching moments, optimising the design of the elevators is crucial in enhancing its effectiveness. The elevator control power is affected by the horizontal tail volume and the elevator size. Larger tails and elevators can create more significant pitching moment coefficient, which increases the pitching moments.

- Elevator deflection angles (δ_e)

- A change in elevator deflection angles directly affects the pitching moment of an aircraft. A large deflection angle will result in more drastic pitching moments, while a smaller deflection angle will provide a smoother response.

2. Change in Wing Design

The planform wing area, S , directly affects the pitching moments as it is the main contributor to the amount of lift generated by the aircraft. The FR25 utilises a positive cambered airfoil, hence the pitching moment coefficient about the aerodynamic centre of the aircraft is negative. An increase in wing area will produce more lift, which also increases the nose-down pitching moment, enabling the aircraft to oppose pitch disturbances more effectively.

A change in wing geometry will affect the MAC, which directly affects the pitching moments. The MAC represents the moment arm in the pitching moment; thus a larger MAC will produce larger pitching moments, and the tail must provide more counter-moment to counteract these disturbances.

Changes in the wing design have its disadvantages too. A larger wing area will also result in an increase in induced drag and weight, which will affect the range and fuel efficiency of the aircraft. With these factors in mind, we must weigh the pros and cons to maximise the aircraft's stability and control while minimising its effects on the performance.

3. Tail Design

A high-winged propeller aircraft like the FR25 will produce a large region of downwash and propeller wash behind the wing. Airflow over the tail might be affected by the turbulent airflow due to the downwash. To ensure better control effectiveness, a T-tail design was utilised so that the tail can operate in cleaner air, allowing the elevators to respond more consistently. By keeping the tail out of the propwash, it will also reduce the stress and structural fatigue loading on the tail.

Additionally, FR25 is designed as an aircraft that has a modular cabin configuration. A rear loading ramp is included for loading and unloading cargo, which will be located at the rear half of the aircraft. Having a T-tail will allow better ground clearance and prevent the tail from obstructing during the loading and unloading process.

However, a T-tail will also lead to increased vulnerability to deep stalls. At high angles of attack, the turbulent airflow over the wing will blanket the T-tail, rendering the pitching control surfaces ineffective. This will result in a loss of control and affect the safety of our passengers. It is important for a tail design that will enable the FR25 to have more predictable and consistent pitching control without hindering the safety of passengers.

iv. Yawing moment (M)

The yawing moment controls the rotation of the aircraft about the vertical axis. The rudder, located on the vertical tail, is the primary directional control surface. The yawing moment equation is shown below:

$$N = \frac{1}{2} \rho V^2 C_n S b$$

where:

ρ : air density

V : velocity of aircraft

C_n : yawing moment coefficient
 S : planform wing area
 b : wingspan

Based on the equation above, we can identify several factors that will affect the total aircraft yawing moment. It is important to optimise these parameters to achieve a balance between the aerodynamic efficiency and control responsiveness.

1. Yawing Moment Coefficient (C_n)

The yawing moment coefficient describes the yawing moment of an aircraft, and it is derived from the size of the vertical tail and the rudder deflection angles as observed in the equation below:

$$\Delta C_n = -\eta_{vt} V_{vt} \frac{dC_{L_{vt}}}{d\delta_r} \delta_r$$

where:

η_{vt} : vertical tail efficiency

V_{vt} : vertical tail volume

δ_r : rudder deflection angle

- Size of vertical tail

- Since the rudder is the primary control surface for yawing moments, optimising the design of the rudder is crucial in enhancing its effectiveness. Changing the size of the vertical tail will change the efficiency and its volume, which will affect the airflow around tail. A larger vertical tail can achieve the desired yaw moments with lesser rudder input, improving fuel efficiency and reducing drag.

- Rudder deflection angles (δ_r)

- A change in rudder deflection angles will result in a change in the vertical tail side force (Y_{vt}). A large deflection angle will result in a more significant yawing moment.
- For higher directional static stability, a larger rudder deflection angle is needed. However, if the rudder deflection angle is too large, it will lead to more sluggish control.

2. Vertical Tail Design

As the vertical tail directly affects the stability and control on the vertical axis, the goal is to design a tail that provides enough control authority across all flight conditions, while minimising weight and drag. The size of the vertical tail is optimised to improve directional stability.

v. Rolling Moment (L)

The rolling moment controls the rotation of the aircraft about the longitudinal axis. The ailerons, located on the wing, are the primary directional control surface. Spoilers can also be used as an alternate directional control surface. The rolling moment equation is shown below:

$$L = \frac{1}{2} \rho V^2 C_l S b$$

where:

ρ : air density

V : velocity of aircraft

C_l : rolling moment coefficient

S : planform wing area

b : wingspan

Based on the equation above, we can identify several factors that will affect the aircraft rolling moment. The basic principle is to alter the flow over the wing at the ailerons to modify the spanwise lift distribution. The difference in lift distribution will create a rolling moment.

1. Rolling Moment Coefficient (C_l)

The rolling moment coefficient describes the rolling moment of an aircraft, and it is derived from the characteristics of the ailerons and the aileron effectiveness as observed in the equation below:

$$C_l = \frac{2C_{L_{\alpha w}} \tau \delta_a}{S b} S_a$$

where:

$C_{L_{\alpha w}}$: lift coefficient due to wing's angle of attack

τ : flap effectiveness

δ_a : aileron deflection angle

S : planform wing area

b : wingspan

S_a : planform aileron area

- Characteristics of ailerons

- A larger aileron area will produce a larger lift differential, which will increase the rolling moment coefficient and rolling moments. This will enhance the aircraft's ability to roll quickly.
- The location of the ailerons matters as it will change the moment arm, which directly affects the rolling moment. Positioning the ailerons further away from the wing root will increase the moment arm, making the ailerons more effective in producing rolling moment.

- Aileron deflection angles (δ_a)

- A change in aileron deflection angles will affect the airflow over the wing, specifically at the ailerons. This will result in lift differential will create rolling moment. Larger deflection angles will result in greater rolling moments, allowing the aircraft to roll faster.
- However, if the aileron deflection angle is too large, it will lead to increased drag, affecting the aircraft's efficiency.

2. Spoilers

The main control surfaces that affect rolling moments are the ailerons and spoilers, which are both located on the wing. Spoilers on each side of the wing can be independently deployed to intentionally alter the airflow over the wing. If the spoiler on one wing is deployed, airflow over that wing is no longer laminar. The wing lift is reduced, resulting in a lift difference. This will result in a rolling moment where the aircraft will roll towards the direction of the wing with lesser lift. The rolling moment can be controlled by changing the spoilers' deflection angles. A larger deflection angle will result in larger lift difference, hence increasing the rolling moment, enabling the aircraft to roll at a faster rate.

c. FR25 Weight Analysis (Jit Huat)

For the initial design phase, the MTOW of the aircraft will be estimated. The MTOW (W_{TO}) will consist of three major components, namely the empty weight (W_{empty}), payload (W_{pay}) and fuel (W_{fuel}):

$$W_{TO} = W_{empty} + W_{fuel} + W_{pay}$$

i. Empty Weight (W_{empty})

To obtain the W_{empty} , an estimated W_{TO} from a similar aircraft will be used. The weight fraction will be taken from figure 34 which shows the historical data for aircraft weight and fuel fractions from 1980-2012.

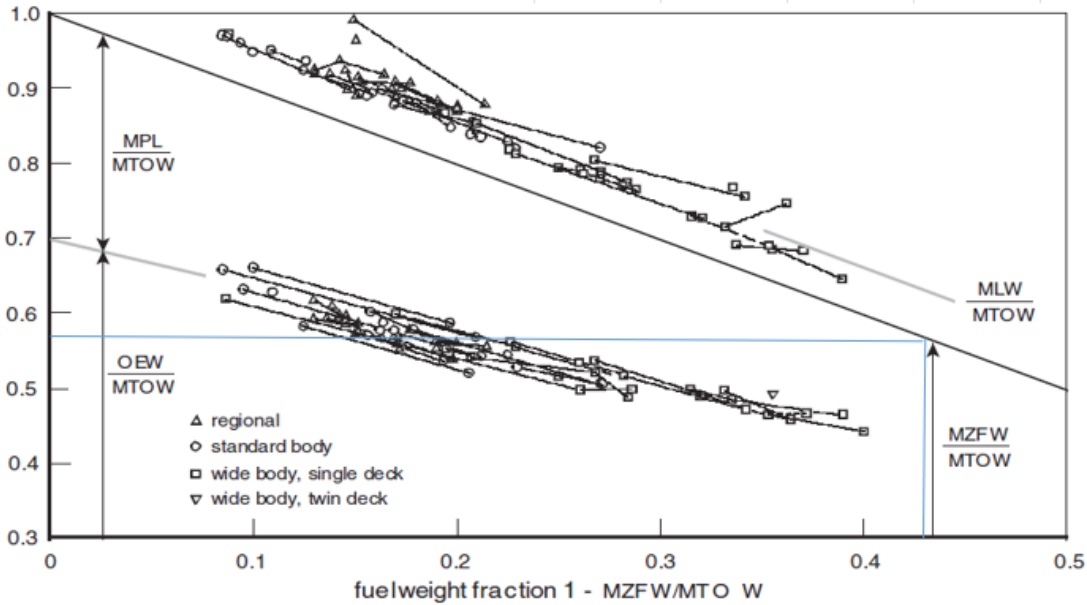


Figure 34: Aircraft weight and fuel fractions

The W_{TO} is calculated following the relation:

$$\text{Estimated Max Takeoff Weight}(W_{TO}) = 29574 \text{ kg}$$

$$\text{Empty Weight Fraction} \left(\frac{W_{empty}}{W_{TO}} \right) = 0.554$$

$$W_{empty} = (W_{TO}) \left(\frac{W_{empty}}{W_{TO}} \right) = (29574)(0.554) = 16378 \text{ kg}$$

j. Maximum Zero Fuel Weight (MZFW)

To obtain MZFW, the maximum payload weight will be estimated. The payload weight will be primarily dictated by the capacity of 88 passengers and 4 crew members with minimal cargo. Each person and their cargo is estimated to have a mass of 100 kg. The MZFW is calculated to be:

$$\begin{aligned} \text{MZFW} &= \text{Empty Weight} (W_{empty}) + \text{Max Payload} (W_{pay}) \\ \text{Max Payload} &= (88 + 4)(100) = 9200 \text{ kg} \\ \text{MZFW} &= 16378 + 9200 = 27418 \text{ kg} \end{aligned}$$

k. Maximum Take-off Weight (W_{TO})

Finally, to obtain the MTOW, the W_{fuel} will be added. The trip and reserve fuel will be estimated by addition of all the fuel burn weight from all the mission segments. The reserve fuel is estimated to be 15% of the total mission trip fuel. The W_{fuel} is calculated using the following:

$$W_{fuel,total} = W_{fuel,takeoff} + W_{fuel,Climb} + W_{fuel,descent} + W_{fuel,land} + \text{Total Fuel}$$

$$\text{Total Fuel} = \text{Trip Fuel} + \text{Reserve Fuel}$$

$$\text{MTOW} = \text{MZFW} + W_{fuel}$$

$$\dot{m}_f = 0.18$$

$$\text{Cruise Speed} = 667 \frac{\text{km}}{\text{hr}} = 0.185277 \frac{\text{km}}{\text{s}}$$

$$\text{Fuel Burn Rate} = \frac{0.18}{0.185277} = 0.9715 \frac{\text{kg}}{\text{km}}$$

$$\text{Trip Fuel} = (0.9715)(1500) = 1457.25 \text{ kg}$$

$$W_{fuel,takeoff} = (1 - 0.99867) * \text{Assumed } W_{TO} = 39.19$$

$$W_{fuel,climb} = (1 - 0.99196) * \text{Assumed } W_{TO} = 235.13$$

$$W_{fuel,descent} = (1 - 0.9989) * \text{Assumed } W_{TO} = 29.39$$

$$W_{fuel,land} = (1 - 0.99967) * \text{Assumed } W_{TO} = 9.8$$

$$W_{fuel,total} = 1457.25 + 39.19 + 235.13 + 29.39 + 9.8 = 1770.76 \text{ kg}$$

$$\text{Reserve Fuel (5\% of Trip Fuel)} = (0.05)(1770.76) = 88.54 \text{ kg}$$

$$\text{Total Fuel} = 1770.76 + 88.54 = 1859.3 \text{ kg}$$

$$\text{MTOW} = 27418 + 1859.3 = 29277.31 \text{ kg}$$

The table below shows the iterative process of obtaining an accurate MTOW (W_{TO}) by repeating the steps with different assumed W_{TO} values. The final converged iterative solution of $W_{TO} = 30693.07 \text{ kg}$ was obtained with slightly less than 5% variation.

Table 15 :Iterative Process of Obtaining W_{TO}

	Assumed W_{TO} (kg)	$\frac{W_{empty}}{W_{TO}}$	W_{empty} (kg)	Calculated W_{TO} (kg)
1 st Iteration	29574	0.554	16200.94	29277.31
2 nd Iteration	32517.2	0.554	19615.08	30693.07 (Final)

d. FR25 Aerodynamics (Florian)

The aerodynamics characteristics of the FR25 are calculated below to determine the total drag to evaluate the overall flight performance of the FR25.

i. Total Drag Coefficient

The total drag is composed of parasite drag, induced drag, and wave drag. As the FR25 will operate at subsonic speeds, wave drag is neglected. The total drag component was calculated to be:

$$C_D = C_{D0} + C_{Di}$$

1. Parasite Drag

Parasite drag accounts for skin friction and form drag. To calculate the total parasite, drag, the individual drag components contributed by the fuselage, wing, empennage and nacelle are first calculated. The equivalent flat plate area for each component is given by:

$$F_p \stackrel{\text{def}}{=} \Phi_f C_f S_{wet}$$

Where:

Φ_f =Form Factor

C_f = Skin Friction Coefficient

S_{wet} = Wetted Surface

a. Fuselage

The skin friction coefficient, C_f is calculated for turbulent flow using the Reynolds number Re based on the fuselage characteristic length (Torenbeek, 2013)

$$C_f = \frac{0.044}{Re^{\frac{1}{6}}}$$

Given a service ceiling Re of 3.19×10^8 derived from the FR25's cruise velocity of 360 knots, the skin friction coefficient evaluates to:

$$C_f = \frac{0.044}{Re^{\frac{1}{6}}} = 1.82 \times 10^{-3}$$

The form factor $\Phi_{f,fus}$ is obtained from experimental data based on the FR25 surface area ratios. According to the data, the form factor for the FR25 fuselage is:

$$\Phi_{f,fus} = 1.15$$

The total wetted surface area of the fuselage is:

$$\begin{aligned} (S_{fus})_{surf} &= \pi \bar{d}_{fus} (l_{fus} - 1.3d_{fus}) \\ &= 266.6m^2 \end{aligned}$$

where the parameters $\bar{d}_{fus}$ and l_{fus} relate to fuselage geometry.

Finally, the fuselage's equivalent flat-plate area $F_{p_{fus}}$ is:

$$F_{p_{fus}} = 1.15 \times 1.79 \times 10^{-3} \times 266.6 = 0.516m^2$$

b. Wing

The Reynold's number of the wing is calculated as a function of FR25's cruise velocity and service ceiling. However, the number will account for the MAC of the wings. The MAC is calculated to be 2.61 m. The Reynolds number of the wing is:

$$Re = \frac{3.19 \times 10^8}{2.61} = 2.51 \times 10^7$$

Therefore the C_f is

$$C_f = \frac{0.044}{Re^{\frac{1}{6}}} = 2.6 \times 10^{-3}$$

The form factor $\Phi_{f,wing}$ of the wings obtained from experimental data is:

$$\Phi_{f,wing} = 1.30$$

The total wetted surface area of the wing is:

$$\begin{aligned} (S_{wet})_{surf} &= \left\{ 2 + 0.5 \left(\frac{t}{c} \right)_w \right\} \left\{ \frac{Q_w \sqrt{A_w(1 + \lambda_w)}}{k_Q t/c} \right\}^{2/3} \\ &= 133.1m^2 \end{aligned}$$

where the parameters $\frac{t}{c}$, Q_w , $A_w \lambda_w$ and k_Q relate to wing geometry and corrections.

The wing's equivalent flat-plate area $F_{p_{wing}}$ is:

$$F_{p_{wing}} = 1.30 \times 2.6 \times 10^{-3} \times 133.1 = 0.445m^2$$

c. Empennage

The total wetted surface area of the empennage is:

$$\begin{aligned} (S_{wet})_{Emp} &= 2 \times (S_h + S_v) \\ &= 66.78 m^2 \end{aligned}$$

Where S_h and S_v represent the surface area of the horizontal and vertical stabilizers on one side

The empennages' equivalent flat-plate area $F_{p_{emp}}$ is:

$$F_{p_{emp}} = 1.30 \times 2.6 \times 10^{-3} \times 66.78 = 0.223m^2$$

d. Nacelle

The Nacelle, which holds the engines consists of the nacelle and pylon contribute to the overall drag as well. The total wet area of the Nacelle is:

$$(S_{wet})_{Nac} = (S_{nacelle} + S_{pylon}) \\ = 49 m^2$$

The Nacelles' equivalent flat-plate area $F_{p_{Nac}}$ is:

$$F_{p_{Nac}} = 1.30 \times 2.6 \times 10^{-3} \times 49 = 0.164m^2$$

e. Propellers

Finally, as the engine is a turbo propeller, the blades will have to be accounted for. The total wet area of the 12 propellers is:

$$(S_{wet})_{Prop} = (12 \times S_{Prop}) \\ = 7.2 m^2$$

The Nacelles' equivalent flat-plate area $F_{p_{Prop}}$ is:

$$F_{p_{Prop}} = 1.30 \times 2.6 \times 10^{-3} \times 7.2 = 0.024m^2$$

Having found all the flat-plate areas, the total parasite drag for the FR25 is:

$$C_{D0} = \frac{F_{p_{fus}} + F_{p_{wing}} + F_{p_{emp}} + F_{p_{Nac}} + F_{p_{Prop}}}{S_{Ref}} \\ = 0.0191$$

2. Induced Drag

Induced drag occurs due to the pressure difference between the bottom and top surface of the wing during flight. The induced drag is calculated using:

$$C_{Di} = \frac{C_L^2}{\pi AR_w e}$$

Where:

e =Oswald Efficiency

AR_w = Aspect Ratio of Wing

The Oswald efficiency can be found using:

$$e = \frac{e_v}{1 + k_L C_{D0} \pi A_w e_v} \\ e = 0.716$$

where e_v is fuselage mounted engine interference.

vii. Lift coefficient for minimum drag

The best aerodynamic efficiency is achieved at:

$$C_{L_{MD}} = \sqrt{C_{D0}\pi A_w e}$$

$$= 0.731$$

and speed:

$$V_{MD} = \sqrt{\frac{2W}{\rho S C_{L_{MD}}}}$$

$$= 44.7 \text{ m/s}$$

viii. Drag Analysis

Figure 35 illustrates the relationship between total drag and airspeed for the FR25. The total drag is broken into parasite and induced components. The analysis reveals that minimum drag point occurs at approximately 145-150m/s indicated by the red marker on the curve. At this speed, the induced drag and parasite drag components creates an inflection point where the L/D ratio reaches its theoretical maximum of approximately 16.7. While this point is aerodynamically optimal for travelling at the maximum range, the FR25 will not be profitable due to long travelling times and low turnover.

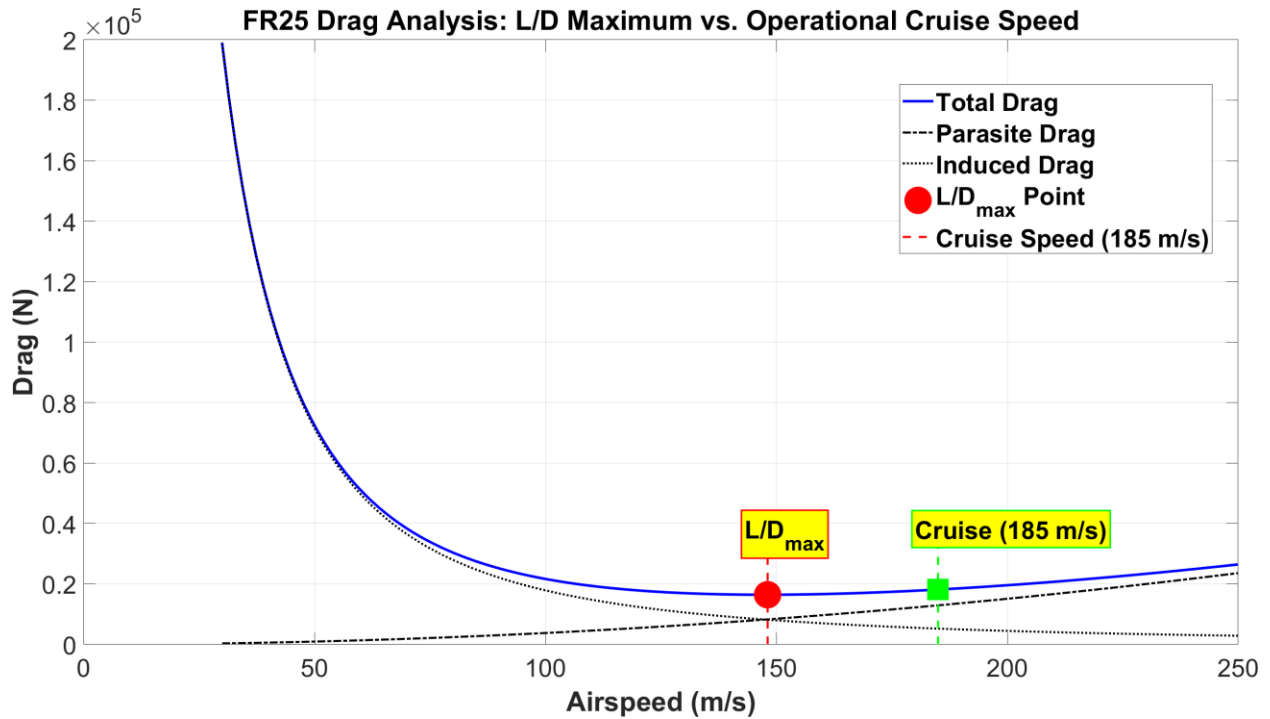


Figure 35: Drag vs Speed Plot for the FR25

Based on this analysis, the FR25 is designed to cruise at:

$$V_{cruise} = 185m/s = 667km/hr$$

This operational choice represents a compromise between aerodynamic efficiency and commercial requirements. The slight efficiency penalty of the operating above the L/D_{max} will be offset by the improved turnover rate.

9. Wing Design (Thaddeus)

a. Introduction

The design of the wing has critical role in determining factors such as aerodynamic efficiency, structural integrity, operational versatility and overall performance of the FR25 hybrid-electric regional turboprop. Considering the aircraft's capacity of 88 passengers and 1500km range, the wing must be capable of providing sufficient lift at low-speed cruise, ensure good short-field performance and adequate space to house the necessary subsystems like cabling and battery cooling. The wing's design should also ensure optimal stall behaviours along with propeller slipstream interactions and maintain its structural integrity under varying load conditions as well as display tolerance to adverse weather conditions like icing and turbulence.

b. Guiding Principles

For the current stage, several guiding considerations are as follows:

i. Cruising Efficiency and Performance

Turboprop regional planes usually fly at a lower speed compared to turbofan jets, at approximately Mach 0.4-0.5, which equates to speeds of around 400 – 600 km/h. Turboprop regional jets also tend to fly at a lower altitude, with a C_L design of about 0.45-0.55 (Roskam, 1985) which leads to flying through higher density of air leading to a higher Reynolds number.

ii. Low Speed Operations

Since the aircraft operates at lower speeds, the wing should be able to provide sufficient lift to ensure safe take-off, cruising and landing. This is essential especially when considering shorter runways for smaller aircraft like regional jets (Nicolosi, 2021). The wing should have airfoils that are responsive to flap deployment.

iii. Propeller Slipstream

Due to the aircraft having turboprop engines, the propeller will cause slipstream interactions on the wing. As such, the wing root is required to handle higher airflow, and the airfoil of choice needs to encourage root-first stall to preserve control. This leads to the requirement of a thicker, more cambered airfoil at the root to handle these additional dynamic pressures.

iv. Operations

Regional aircraft will still meet with varying weather conditions. The airfoil should be robust and tolerant of these changes and minimize effects like icing or roughness therefore creating a reliable wing that does not compromise performance in adverse conditions. As a hybrid-electric aircraft,

the integration of hybrid equipment should also be considered, ensuring internal space for spars, cabling, and other systems necessary whilst minimizing drag.

c. Airfoil Selection

Several NACA 4-digit profiles were considered and narrowed down to the 2xxx and 4xxx series such as NACA 2412 and NACA 4412, as they tend to offer moderate to high cambers which aid in providing the airfoil and wing with increased lift coefficients and lift-to-drag ratio. The NACA profiles are seen in figure 36 NACA 2412 is a widely used airfoil profile in commercial aviation and has proven to be reliable while NACA 4412 with its larger camber and possible larger thickness when varied within the profile family, can be considered for the inner wing, close to the wing root.

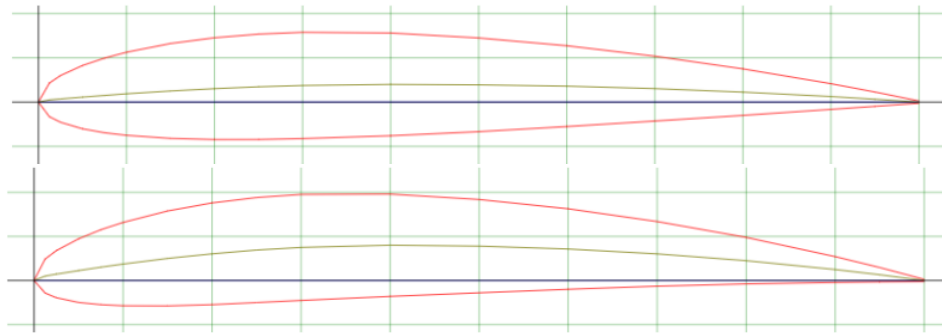


Figure 36: NACA 2412 (Top) and NACA 4412 (Bottom) Airfoils

i. Root (NACA 4415)

The NACA 4415 profile seen in figure 37 offers a high camber and thick section which leads to the provision of stronger lift as well as structural depth for interior design to fit hybrid-electric components for the aircraft. It also possesses favourable mild-stall behaviour in the slipstream of the propeller.

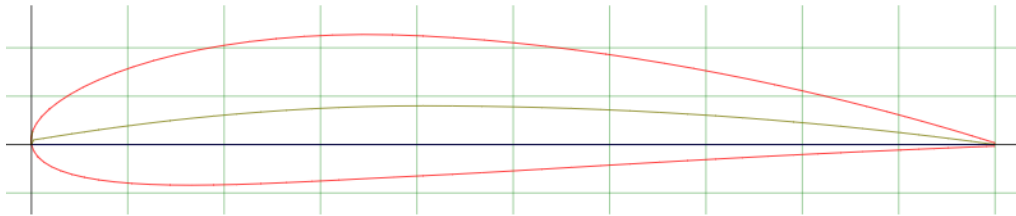


Figure 37: NACA 4415 Airfoil

ii. Mid Span (NACA 2412)

The NACA 2412 is a reliable, moderate camber found abundantly in general aviation as seen in figure 37. This profile is chosen as a mid-wing profile to ease the transition from root profile to tip profile, smoothening the aerodynamic and structural progression from root to tip. This can aid in preventing abrupt changes in lift distribution across the wing.

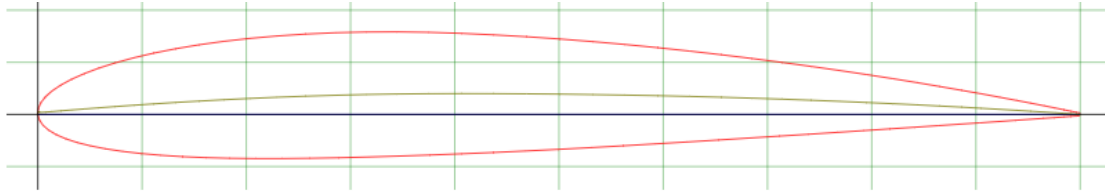


Figure 38: NACA 2412 Profile

iii. Tip (NACA 0012)

The NACA 0012 uses a thin, symmetrical foil at the tip seen in figure 38 can reduce drag and promote docile stalling, thus preserving control over the ailerons during high angle-of-attack operations.

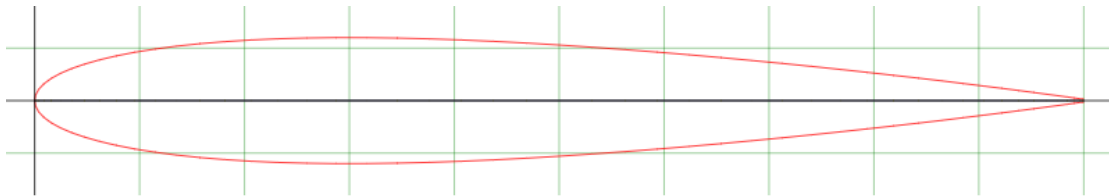


Figure 39: NACA 0012 Profile

This 3-profile design provides good lift from root to tip and offers a smooth transition at the mid-wing section for smoother CAD-modelling, structural integrity and more calculable stall behaviour. The summary of these airfoils is shown in table 16.

Table 16 : Design input and parameters

Position	Airfoil	Function
Root	NACA 4415	- High lift - Spar depth - Thicker / cambered
Mid	NACA 2412	- Transition section - Balanced camber - Reliable
Tip	NACA 0012	- Low drag - Symmetrical - Docile stall

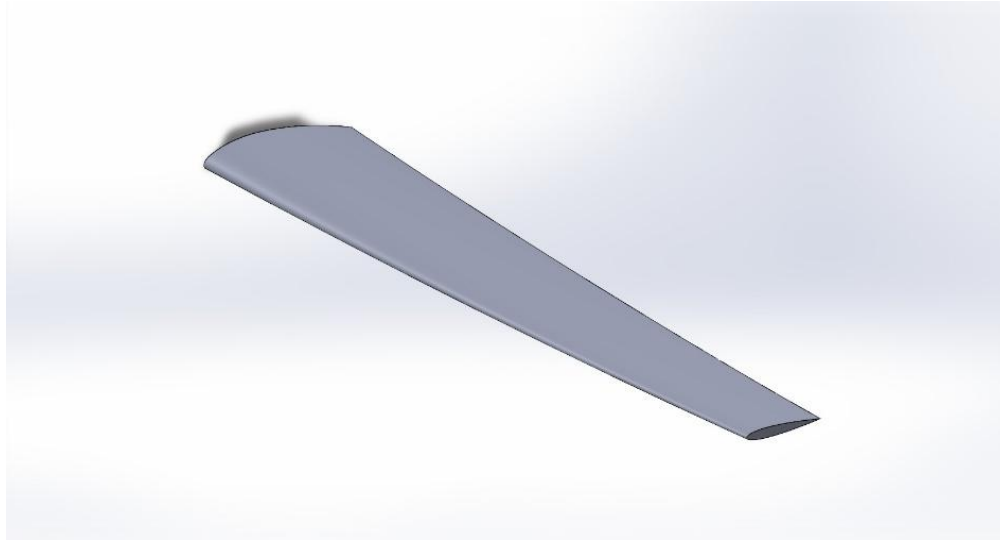


Figure 40: 3-Profile Wing CAD Model

d. Wing Parameters

With the MTOW of 30693.072 *kg* which equates to 3.01×10^5 N, various calculations can be made and thus several parameters can be derived and determined. The purpose of sizing the wing is to ensure that it balances lift with aerodynamic efficiency and its structural feasibility. This also includes having the aircraft to remain compatible with airport constraints.

i. Design Input

A summary of the design input can be found in the table 17 below. Detailed calculations that show the derivation of these values will be done in the following sections.

Table 17 : Design input and parameters

Parameter	Symbol	Value	Unit
Max. Take-Off Weight	W_{TO}	30693.072 $= 3.01 \times 10^5$	N
Landing Weight	W_L	$0.85W_{TO}$ $= 2.56 \times 10^5$	N
Air Density (Sea Level)	ρ_0	1.225	kg/m^3
Approach Speed (Target)	V_{REF}	115	Kts
Stall Speed	V_{Stall}	45.5	<i>m/s</i>
Max. Lift Coefficient (Landing)	$C_{L_{max,L}}$	2.7	-

ii. Preliminary Calculations

Wing loading defined as $\frac{W}{S}$ is a driving factor in determining the wing size and establishes the balance between structural and aerodynamic criteria. It is an appropriate way to start the wing sizing process since an appropriate wing loading directly affects take-off distance, stall speed and climb performance. Wing loading determines how much weight the wing can carry per square metre):

$$\frac{W}{S} = \text{Wing Loading (N/m}^2\text{)}$$

According to published data (Sholz, 2021 & Roskam, 1985), regional turboprops tend to have wing loading values that range between 3.3 – 4.6 kPa (390 – 470 kg/m²), albeit depending on take-off and field length performance. Some examples include:

- ATR 42-500: 3.4 kPa
- ATR 72-600: 3.8 kPa
- Dash 8-Q400: 4.6 kPa

With the given specifications of the FR25 aiming to achieve short-field capabilities along with hybrid propulsion, a wing loading at about 4.22 kPa (3600 N/m²) is selected as the design point. This will allow sufficient lift at lower cruising speeds while maintaining a compact and aerodynamic structure.

iii. Wing Area

Along with the MTOW and wing loading, the wing area can be determined using the following:

$$S = \frac{W}{W/S} = \frac{3.01 \times 10^5}{4200} \approx 72 \text{ m}^2$$

As a design point, the wing area of 72 m² is selected. Compared to other regional turboprops, such as the ATR 72-600 and Dash 8-Q400 with wing areas of 61 m² and 63 m² respectively, this appears to be a larger value but is not too extreme. It can provide the adequate lift required by an aircraft of this weight while having enough internal space for other necessary equipment to be mounted in.

iv. Landing Requirements

Additionally, the wing area needs to satisfy the landing requirements, and this is done by verifying against the stall speed constraint:

$$S \geq \frac{W_L}{0.5\rho_0 V_{Stall}^2 C_{L_{max,L}}}$$

Where:

W_L = Landing Weight

$\rho_0 = 1.225 \text{ kg/m}^3$ (Sea Level Density)

V_{stall} = Stall Velocity

$C_{L_{max,L}}$ = Maximum Lift Coefficient for Landing

In preliminary design, we can assume that as the aircraft approaches landing, it has consumed an approximate of 15% of its take-off weight in fuel (Scholz, 2021 and Roskam, 1985). In real world examples, the maximum landing weight (MLW) and MTOW ratio in the ATR 72-600 and Dash 8-Q400 are approximately 0.97 and 0.95 respectively. Commercial aircraft often have a ratio of between $MLW \approx MTOW \times 0.90 - 0.97$. However, for a preliminary conceptual design, a more conservative value of 0.85 is used to ensure a larger margin for factors such as longer flight durations or variations in payload.

The $C_{L_{max,L}}$ value can be justified through common aircraft values in commercial aviation. The usual $C_{L_{max}}$ values are seen in the table 18 below:

Table 18 : Cl Ranges

Configuration	$C_{L_{max}}$ Range	Remarks
No flap – clean	1.2-1.6	Cruise / Climb
Partial Flap	1.7-2.0	Take-Off / Initial Climb
Full Flap + LE Device	2.2-2.7	Landing / Short-Field

(Scholz, 2021), (Roskam, 1985)

The FR25 deploys Fowler flaps as well as inboard Leading Edge (LE) devices, as such the value of $C_{L_{max,L}} = 2.7$ is a realistic estimation and aligns with current regional turboprops.

Lastly, to determine the stall velocity V_{stall} , several parameters are taken before selecting a representative value. According to FAA, the regulatory link between V_{REF} and V_{stall} is minimally (FAA, n.d.):

$$V_{REF} = 1.3 V_{stall}$$

The V_{REF} of the ATR-72 is publicly stated to be at 113 Knots-Indicated-Airspeed (KIAS) (ATR, 2015). That of the Dash 8-Q400 also has a range of 108-117 knots (Bombardier, 2004). This gives a safe range for regional turboprop to have V_{REF} value between 110-120 knots. Using the available measurements:

$$V_{stall} = \sqrt{\frac{2W_L}{\rho S C_{L_{max,L}}}} = \sqrt{\frac{2(2.584 \times 10^5)}{1.225(72)(2.7)}} = 46.585 \text{ m/s}$$

$$V_{REF} = 1.3V_{stall} = 60.56 \text{ m/s} \approx 118 \text{ kts}$$

This justifies the selection of our current design input of $S = 72.0 \text{ m}^2$ since the V_{REF} that arises from the calculations fit within the specifications of a safe speed. This area is a reference area and is a clean configuration that does not involve flaps or LE devices, that was determined with landing performance constraints such as V_{stall} and $C_{L_{max,L}}$. Despite the use of $C_{L_{max,L}} = 2.7$, the resulting

wing area of $S = 72.0 \text{ m}^2$ is that of a fixed reference area. In this preliminary design and sizing, the effect of flaps and LE devices (i.e. augmentation in lift), is reflected by $C_{L_{max,L}}$ and not through a geometric increase in area. The derived wing area of $S = 72.0 \text{ m}^2$ remains as the common reference area for all flight conditions (Roskam, 2018; Raymer, 2018).

Table 19 : Cl with regards to Wing Area

Flight Condition	Wing Devices Deployment	$C_{L_{max}}$	Wing Area, S	Effective Lift Area
Clean	None	1.2 – 1.6	72 m^2	$\cong S$
Take-Off	Partial	1.7 – 2.2	$72 \text{ m}^2 (ref)$	Slightly $> S$
Landing	Full	2.2 – 2.7	$72 \text{ m}^2 (ref)$	$\approx +5\% S$

v. Aspect Ratio and Span

For regional turboprops with common Mach numbers of 0.40-0.5, they usually adopt aspect ratios (AR) ranging from 10-12. This can be seen in published data of the ATR 72 and Dash 8-Q400 that also suggests this optimal AR range (Nicolosi, 2021). With that, the design point AR that was decided upon was ~ 11.4 . Based on the wing area and AR, the exposed span (b) that the FR25 adopts can be calculated:

$$b = \sqrt{AR \times S} = \sqrt{11.4 \times 72} \approx 28.65 \text{ m}$$

This also gives an exposed half-span of 14.325 m . Additionally, a taper ratio (λ) of 0.5 was selected to allow for gradual lift distribution as well as keeping the structure simple.

$$\lambda = \frac{c_t}{c_r} = 0.5$$

A trapezoidal wing employs this equation:

$$S = \frac{b}{2} (c_r + c_t) = \frac{b}{2} c_r (1 + \lambda)$$

$$\therefore c_r = \frac{2S}{b(1 + \lambda)} = \frac{2 \times 72.0}{28.65(1 + 0.5)} \approx 3.351 \text{ m}$$

This gives a tip chord of:

$$c_t = c_r \times \lambda = 3.351 \times 0.5 = 1.675 \text{ m}$$

The midsection is located at 50% of the half span of each side of the wing which gives a linear taper. With a linear taper, this means that the taper is distributed evenly. Hence, the chord length at the midspan is simply the average between the root chord length and the tip chord length.

$$c_{midspan} = \frac{c_r + c_t}{2} = 2.513 \text{ m}$$

Table 20 : Chord Length Summary

Position	Chord Length (m)	Airfoil
Root	3.351	NACA 4415
Mid	2.513	NACA 2412
Tip	1.675	NACA 0012

To ensure validity, a check back can be done to ensure that it matches the original wing area.

$$S_{check} = \frac{b}{2}(c_r + c_t) = \frac{28.65}{2}(3.351 + 1.675) \approx 72.0 \text{ m}^2$$

Additional parameters can be calculated with these values such as the mean aerodynamic chord (MAC) and the spanwise location of the MAC.

$$MAC = \frac{2}{3}c_r \frac{1 + \lambda + \lambda^2}{1 + \lambda} = 2.606 \text{ m}$$

$$y_{MAC} = \frac{b}{6} \frac{1 + 2\lambda}{1 + \lambda} = 6.367 \text{ m}$$

The thickness to chord ratio can also be determined based on the choice of airfoil. For example, at the root, the current NACA 4415 airfoil profile gives a ratio of $t/c = 15\% = 0.15$. Similarly, the t/c can be determined at the mid-span and tip to be 0.12 for both. To calculate the average thickness to chord ratio across the wing, a simple average can be taken across the three airfoil profiles.

$$\overline{t/c}_{avg} = \frac{(t/c)_{root} + (t/c)_{mid} + (t/c)_{tip}}{3} = \frac{0.15 + 0.12 + 0.12}{3} = 0.130$$

As such, the average thickness to chord ratio across the wing is approximately 13%.

vi. Wing Geometry

The FR25 is expected to operate in the ranges of Mach 0.40-0.50, as such, a sweep in the wings provides very little to no benefit and instead, cause reduction in low-speed lift and negatively affect stall and aileron behaviour (Raymer, 2018; Nicolosi, 2021). A straight wing on the other hand can maximise firstly, low-speed capabilities by providing a higher $C_{L_{max}}$ and therefore shorter distances for take-off and landing (Roskam, 2018). Secondly, it can also improve aileron effectiveness resulting in benign stall progressions. Lastly, it also provides a simple structure where simpler integration of other devices such as flaps and nacelles can be achieved. As such, many regional turboprops operate on a straight-wing configuration such as the ATR-72 and the Dash 8-Q40 (ATR, n.d. and Bombardier, 2020).

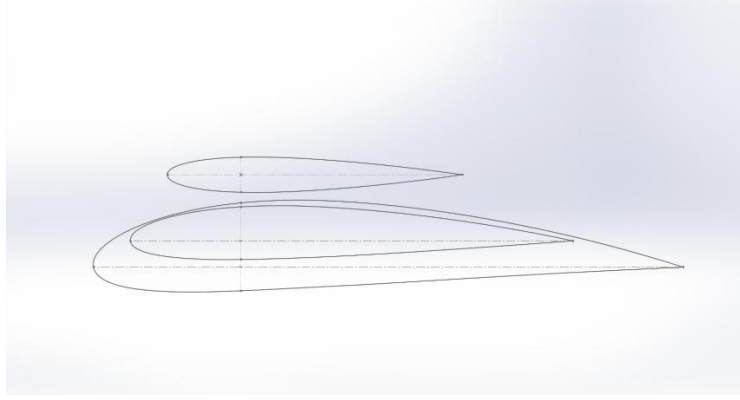


Figure 41: Airfoil Cross Section

As such, the wing's 3 airfoil profiles are aligned. They are aligned at the quarter-chord ($c/4$) line which is the approximate aerodynamic centre of each profile. This provides a consistent reference axis when it comes to stability and control analysis as well as possible future sizing alterations (Anderson, 2011). This also leads to a clearly defined sweep angle ($\Lambda_{0.25}$) of 0° .

vii. **Dihedral**

The wing deploys a polyhedral which refers to a non-standardised dihedral along the wing. The wing has a 1° dihedral at the mid-span and a 3° dihedral at the tip and the effective angle can be computed as (Raymer, 2018; Roskam, 2018):

$$\Gamma_{eff} = \tan^{-1}\left(\frac{z_{tip} - z_{root}}{s}\right)$$

Where:

Semispan, $s = 14.325 \text{ m}$

Root height, $z_{root} = 0 \text{ m}$

Tip height, $z_{tip} = 0.525 \text{ m}$

$$\therefore \Gamma_{eff} = \tan^{-1}\left(\frac{0.525}{14.325}\right) \approx 2.1^\circ$$

High wings often do not have additional dihedral because the wing position already offers a positive dihedral effect which is the tendency to restore roll (Nelson, 1998; Etkin & Reid, 1996). Having the wing above the centre of gravity (CG) naturally leads to a moment that is roll-stabilising, also known as the high wing effect. Additionally, as with the FR25, the fuselage alongside the vertical tail also adds to lateral stability. Therefore, utilising too large of an additional dihedral can lead to an over-stabilising aircraft which may cause negative effects such as sluggish roll response, unwanted Dutch-roll tendencies (coupling with yawing moment). These ultimately prove to have no benefits to low-speed performances and short take-off / landing.

The selected polyhedral angles suit the FR25's mission for many reasons:

1. Sufficient Roll Stability

With a $0 - 1^\circ$ dihedral at the centre panel, it helps to avoid over-restoring moments while the 3° angle at the outboard allows for gentle roll damping which leads to an overall effect of a laterally stable aircraft without sluggish responses (Nicolosi, 2021).

2. Gentle Stall

This polyhedral combined with the taper ratio, $\lambda = 0.5$, and an unswept wing at the quarter-chord, the tip remains at a lower effective angle of attack when the wing approaches stall, which protects the roll control, particularly in the approach phase of the aircraft.

3. Simple Structure and Integration

A relatively flat centre panel ($0 - 1^\circ$) simplifies the installation of many additional parts such as the nacelle, flap tracks and possibly landing gear geometry without compromising on the lateral stability.

4. Gust Response

This polyhedral of effective angle $\sim 2.1^\circ$ leads to a milder but well-damped roll response which is necessary for comfort of passengers and workload of the pilot in the event of turbulence.

e. Washout

Aerodynamic twist, also known as washout, is a common addition to tapered wings to ensure that the root of the wing achieves the critical angle of attack before the wing tip, therefore preventing premature stall at the tip and consequently, preserves the ailerons' effectiveness at high lift. In the case of FR25, an approximate 2.5° washout is incorporated into the wing which gives a gradual reduction in the angle of incidence from wing root to wing tip. This ensures that the wing root (NACA 4415) reaches the critical angle of attack before the wing tip (NACA 0012) does, henceforth encouraging a root-first stall and maintaining aileron effectiveness during low-speed, high-lift operations. This is a common practice among regional turboprops due to their tendency to operate at lower speed whilst requiring maintaining high lift. Regional turboprops also often operate at lower altitudes and use shorter runways where ability to handle the aircraft at near-stall conditions is very important. (Roskam, 2018; Anderson, 2011; Nicolosi, 2021)

f. Overview of Structural Design

i. Structural Design concept

The wing of the FR25 is designed as a 2-spar torsion box structure. This is a standard design practice for transport aircraft due to the high stiffness-to-weight ratio and damage tolerance (Raymer, 2018). Since the FR25 operates as a modular aircraft and will be required to transport cargo, this design is suitable for the aircraft. This design encompasses the following:

- Front spar that is located at about 25% chord; serves as primary load-bearing member
- Rear spar that is located at about 65% chord; forms the aft boundary of the torsion box
- Stringers that act as longitudinal stiffeners along the upper and lower skin panels to increase stiffness and the wing resistance to buckling
- Semi-monocoque stressed wing skin which forms part of the torsion box and carries shear flow loads

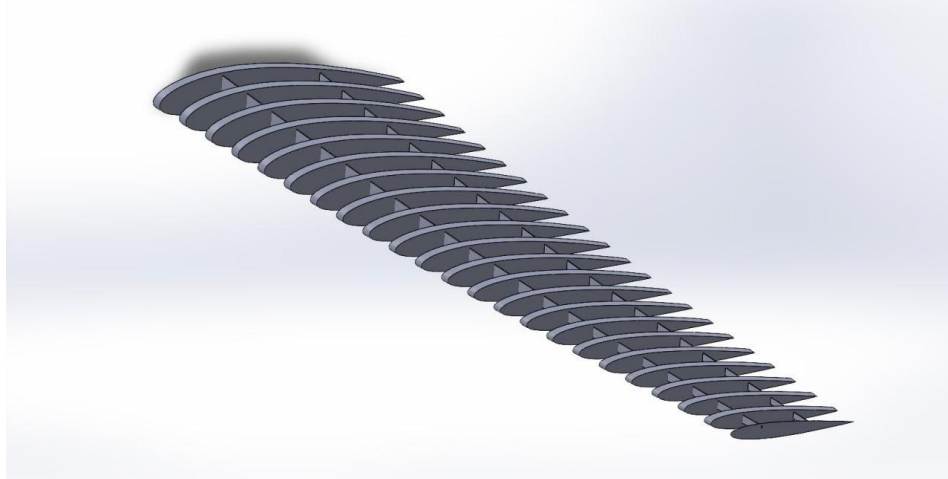


Figure 42: Structural Cross Section

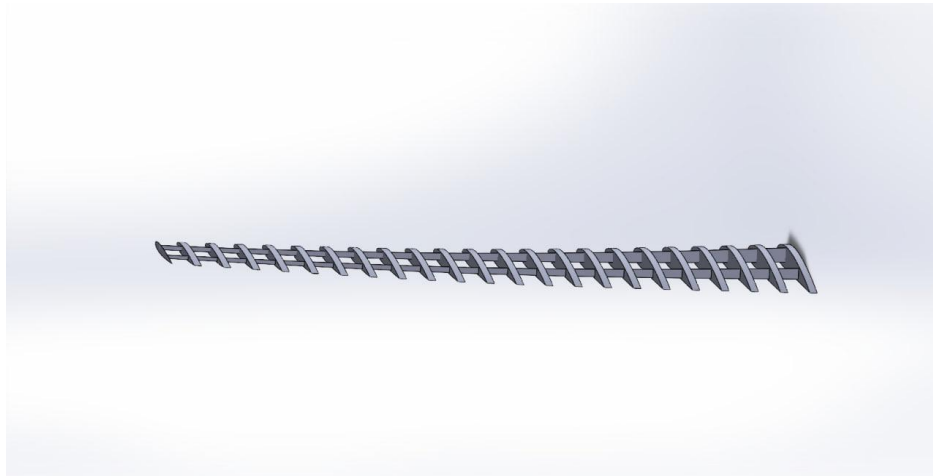


Figure 43: Structural Cross Section view 2

ii. **Load Path**

Aerodynamic forces that act on the wing will cause bending and torsion to occur. This is where the spars play a part to resist the bending, while the skin and stringers form a closed torsion box to carry the torsional shear flow, therefore maintaining the wing's structural integrity and strength. The ribs also transmit distributed loads – aerodynamic and fuel loads – to the spars. This allows lift and inertial forces to be distributed well and evenly across the wing, ensuring efficient load distribution and minimising localised stress.

g. **Fuel System**

The FR25 adopts a design that allows it to store and supply fuel efficiently within the wing while maintaining aerodynamic efficiency, stability and weight balance. Since the FR25 is a hybrid-electric turboprop, it requires a smaller fuel capacity as compared to conventional aircraft of smaller sizes. However, it still demands a safe, well-integrated fuel layout.

The integral wing fuel tank design is where the internal wing volume which is enclosed by the front and rear spars as well as the wing skin and some of the ribs. This allows reduction in weight and drag whilst maximising usable volume and minimising complexity in the structure.

i. Requirements and Estimation

The total fuel mass for the FR25 1859kg and using Sustainable Aviation Fuel (SAF) which has an approximate density of 0.8 kg/L, this corresponds to about 2.09 m³ of fuel in volume.

Since the wings need to be balanced, the fuel must be stored symmetrically across both wings:

$$V_{wing_{1,2}} = \frac{2.32}{2} = 1.16m^3$$

The portion of the half span used for the fuel can be used to estimate the length of the fuel tank. Since the half span equates to 14.325 m, and the outboard which occupies about 30% is usually kept dry for the installation of control cables, aileron linkages, etc.

$$S_{half_{outboard}} = 30\% \times 14.325 \approx 4m$$

This leaves about 70% of the half-span for the tank to occupy. As such, we can estimate the length of the tank to be:

$$L_{tank} \approx 10m$$

Using the thickness of the airfoil and ribs, we can estimate the average height of the tank. The tank height at the root will be larger than that at $L_{tank} = 10m$. At the root, the total thickness is approximately 0.5m and at the tip of the fuel tank, it is approximately 0.25m. (insert CAD dimensions). Leaving space for other fittings, the average usable portion of the thickness is about 60%, therefore the average height of the fuel tank is:

$$H_{avg} = 0.6 \times \frac{0.5 + 0.25}{2} \approx 0.3m$$

Next, the chordwise depth, which refers to the distance between the front and rear spars. With the spars at 25% and 65% of the chord

$$D = (0.65 - 0.25)c = 0.40c$$

The average chord length can be calculated since the wing is tapered from 3.351m → 2m → 1.675m (double check this). Since the chord distribution along a tapered, un-swept wing is linear, the following assumptions can be made:

$$c(y) = c_r - (c_r - c_t) \frac{y}{b/2}$$

At $y = 10m$:

$$c(10) = 3.351 - (3.351 - 1.675) \frac{10}{14.325} = 2.18m$$

Therefore, the chord at the outboard of the fuel tank is around 2.18m.

$$c_{avg} = \frac{3.351 + 2.18}{2} \approx 2.77m$$

Since it occupies the space between the spars – 25% and 65% chord – the effective tank width can be approximated in the chordwise direction to be:

$$D_{avg} = 0.4c_{avg} = 0.4 \times 2.77 = 1.11m$$

Lastly, since not all of the geometric box volume can contain fuel considering skin, spar, rib thickness and other margins, the typical integral tank efficiency is $\approx 0.65 - 0.8$ (Raymer, 2018). Using mid-range value of $\eta = 0.7$:

$$V_{available} = L_{tank} \times H_{avg} \times D_{avg} \times \eta$$

$$V_{available} = 10 \times 0.3 \times 1.11 \times 0.7 = 2.33m^3$$

Using about 70% of the half-span for the integral fuel tanks, the wing box provides $\sim 2.3m^3$ per wing at 70% volumetric efficiency. This comfortably exceeds the $1.045m^3$ volume requirement calculated for 1675 kg of SAF.

ii. Fuel Tank Configuration

Each wing will contain 3 primary fuel cells accompanied by a vent or surge tank:

1. **Centre Tank:** This tank is located at the wing root and supplies the engines with fuel first to reduce wing bending.
2. **Inner Tank:** This is the main operational tank and holds most of the usable fuel and feeds the collector sump. It is located next to the centre tank.
3. **Outer Tank:** The outer tanks provide lateral balance to the wing and aircraft and serves as a reserve and is usually used last among the tanks.
4. **Vent / Surge Tank:** This tank is located nearer to the tip of the wing, and its purpose is to allow pressure equalization and overflow control.

A collector sump is added and located at the lowest point of the centre tank. This is to ensure reliable feeding of fuel to the engine during climb, descent or low-fuel conditions. Anti-slosh baffles and flapper valves can be added to maintain even fuel distribution and prevent abrupt centre of gravity (CG) shifts during manoeuvres and turbulence.

The torsion-box as defined by the space enclosed by the spars serve as the cavity to store fuel. The front and rear spar form the structural boundaries of the fuel tanks while the wing skin that surrounds it act as the tank walls.

Table 21 : Component vs Function Wing Fuel Tank Integration

Component	Function
Spars	Primary load paths; structural boundaries of the fuel tank
Ribs	Divide fuel tank into cells; provide shape retention; carry pressure loads
Stringers	Reinforce skins against buckling
Sealant	Applied along joints and fasteners to prevent leakages

At the tank region, the ribs are perforated to permit fuel transfer and pressure equalising while simultaneously reducing weight. Each rib will contain circular holes. The holes will be arranged such that the fuel flow between the tanks to the engine is smooth and efficient. The holes are there to facilitate fuel transfer in different conditions and aid in drainage as well as venting to prevent over-pressurisation.

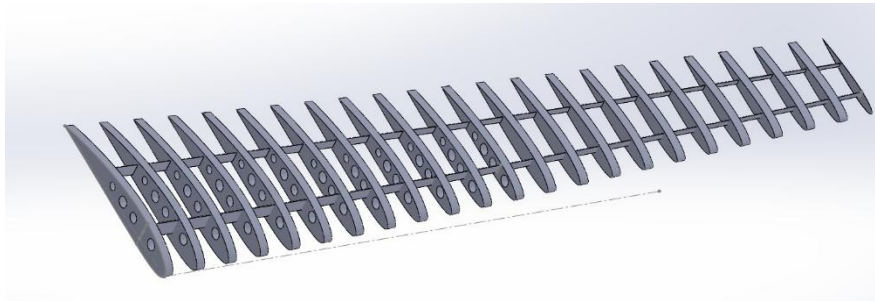


Figure 44 : Holes for Fuel Feeding

Circular holes are adopted in the fuel tank ribs to better facilitate feeding of fuel, drainage and venting while preserving structural efficiency. As opposed to rectangular cut-outs, circular ones pose less risk of stress concentration due to the lack of corners.

Stringers are also added to stiffen the wing, particularly to stiffen the fuel tank skins against local buckling. They also aid in defining the fuel tank's internal geometry. They are designed to run parallel to the span and are aligned longitudinally between the spars. The upper stringer will primarily manage the compressive skin loads, while the lower stringer will handle more tension.

iii. **Fuel Management**

The fuel is stored and used from centre to tip sequentially to maintain desirable and favourable bending moments and CG locations. This will lead to the centre tank being drained first during

take-off, climb and the early stages of cruise. Then, the inner tank feeds next, thus maintaining lateral balance. Lastly, the outer tank is used, minimising tip loading.

Vent lines are also present to ensure pressure equalisation across the tanks, and should overpressure arise, it is released via surge tanks that are located near the tips. The collector sump and boost pumps deliver fuel to the hybrid-electric engine system at regulated pressure.

Since the FR25 is a hybrid-electric configured aircraft, it allows partial substitution of fuel energy with electric power during low-demand flight phases. This in turn extends the aircraft's range while keeping fuel mass modest.

h. Structural Analysis

i. Load Cases

With the MTOW at 30693.072 kg , the weight of the aircraft equates to $W = mg = 30693.072 \times 9.81 = 3.01 \times 10^5 \text{ N}$. Along with other geometrical and structural factors determined prior, load cases can be assumed for the aircraft.

Design Load Factors:

- Cruise Check: $1.0g$
- Positive limit manoeuvre: $2.5g$
- Positive Ultimate: $= 1.5 \times \text{limit} = 3.75g$

Calculating per-half-wing root reactions:

- Root Shear at $2.5g$:

$$V_{root} \approx \frac{nW}{2} = \frac{2.5 \times 3.01 \times 10^5}{2} \approx 376250 \text{ N}$$

- Bending moment at root at $2.5g$:

$$M_{root} \approx \frac{nWb}{8} = \frac{2.5 \times 3.01 \times 10^5 \times 28.65}{8} = 2.695 \times 10^6 \text{ Nm}$$

- Ultimate $3.75g$ moment:

$$M_{root,ult} \approx 1.5 \times M_{root} \approx 4.042 \times 10^6 \text{ Nm}$$

ii. Bending Calculations

The primary bending resistance is handled and provided by the closed wingbox that is formed by the front spar and rear spar, accompanied by the wing skin as well as stringers. To ensure that the wing box can sustain $2.5g$ limit load without experiencing excessive stress, a required section modulus at the root can be estimated.

Taking a conservative allowable bending stress limit:

$$\sigma_{allow} = 150 \text{ MPa}$$

Then:

$$S_{req} = \frac{M_{root}}{\sigma_{allow}} = \frac{2.695 \times 10^6}{150 \times 10^6} = 1.8 \times 10^{-2} m^3$$

This is the target root section modulus for the wing box about its bending axis. The adopted configuration with a relatively deep front spar, rear spar, provides a realistic path to achieve the required section modulus without the wing components being overly thick or require rare materials.

iii. Stress Estimation

Without a full simulation, the stress estimation is not whole but can be verified using simplified closed-form checks to ensure that the proposed spar and wing layout is reasonable under 2.5g.

Using the following values:

- Front spar height: $\sim 0.33m$
- Root Shear (limit, per half wing): $V_{root} \approx 3.76 \times 10^5 N$
- Assuming front and rear webs share vertical shear: $t = 12mm = 0.012m$

The average shear stress per web is:

$$\tau \approx \frac{V_{root}}{2ht} = \frac{3.76 \times 10^5}{2 \times 0.33 \times 0.012} \approx 48MPa$$

When taking conservative assumption that only one web carries the shear stress:

$$\tau \approx \frac{3.76 \times 10^5}{0.33 \times 0.012} \approx 96MPa$$

Both values are within acceptable limits for the webs in shear stress.

i. Further Airfoil Analysis (Jason)

i. NACA Airfoils (Jason)

Further analysis was done on NACA 4415, NACA 2412 and NACA 0012 airfoils. We compared the characteristics of each airfoil by looking at their lift curve ($C_L - \alpha$) and the ratio of lift to drag at varying angles of attack ($C_L/C_D - \alpha$).

To observe the graphs, we will have to first determine the Reynold's number (Re). The parameters of the aircraft at cruise conditions are given in table 22 below:

Table 22 : Aircraft parameters at cruise conditions

Cruise speed	185.2 m/s
Cruise altitude	25,000 ft (7630 m)
Mean Aerodynamic Chord (MAC)	2.61 m
Density (at FL250)	0.549 kg/m ³
Kinematic viscosity (at FL250)	1.921 x 10 ⁻⁵ m ² /s

The Reynold's number at 25,000 ft is calculated to be:

$$Re = \frac{\text{cruise velocity} \times MAC}{\text{kinematic viscosity}}$$

$$Re = \frac{185.2 \times 2.61}{1.921 \times 10^{-5}} = 2.51 \times 10^7$$

The lift curve of the various airfoils is shown below:

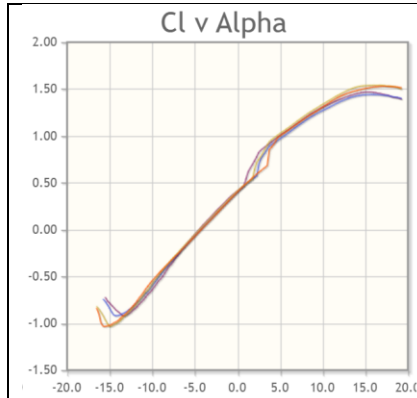


Figure 45: C_L - α graph for NACA 4415

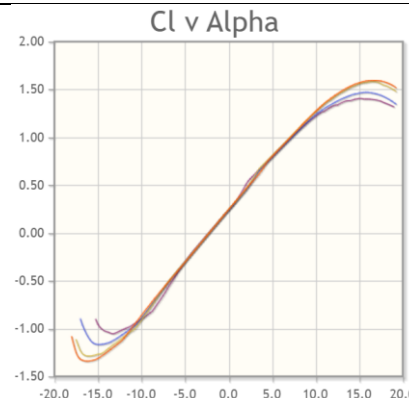


Figure 46: C_L - α graph for NACA 2412

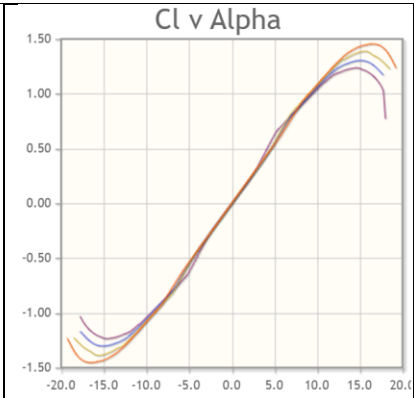
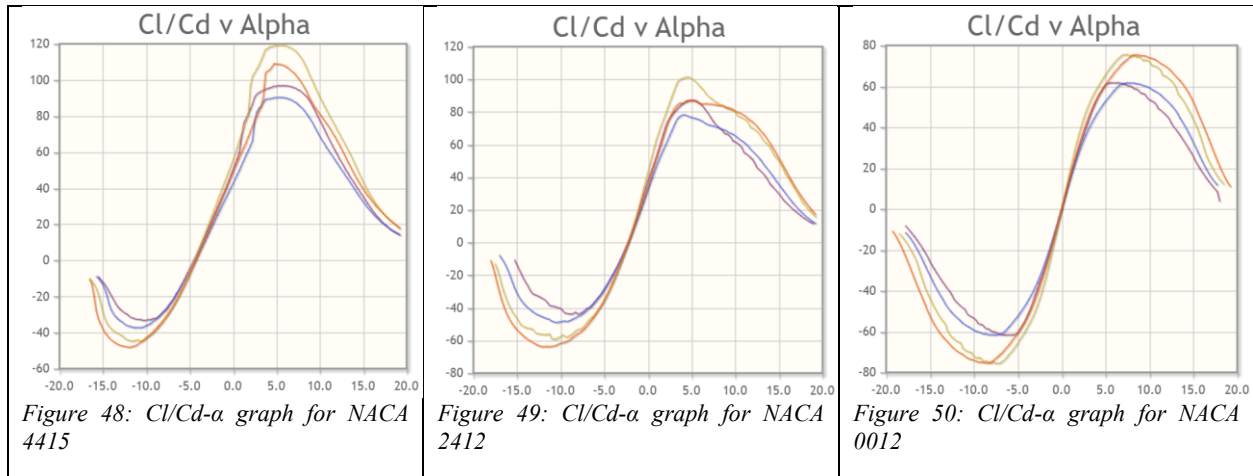


Figure 47: C_L - α graph for NACA 0012

The graphs above are generated via Xfoil which may be inaccurate at the transonic regime which our aircraft will be operating in. However, it can serve as a good reference to how well the various airfoil shapes are in generating lift at varying angles of attack.

From the graphs, we can observe that NACA 4415 airfoil has the highest lift curve slope out of the three airfoils. This means that NACA 4415 can generate the most lift and this is due to the airfoil's positive camber. For typical cruising angle of attack of 0° to 5° , C_L is positive for all three airfoils but most positive for NACA 4415. The critical angle of attack for all three airfoils is approximately at 15° . If the angle of attack goes beyond 15° , flow separation occurs and the aircraft will experience sudden loss of lift and begin to stall.

Other than the lift curve slope, we also looked into the lift-to-drag ratio (C_L/C_D) at the various angles of attack. The $C_L/C_D - \alpha$ graphs are shown below:

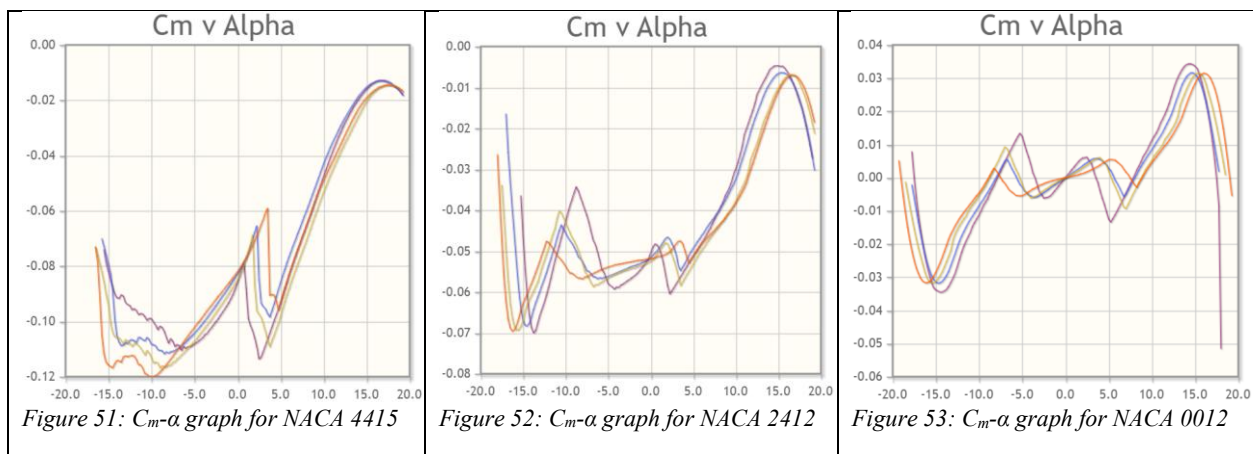


The ratio of lift-to-drag indicates the overall aerodynamic efficiency of the airfoil. By plotting the $C_L/C_D - \alpha$ graph, we will be able to determine the angle of attack where the aircraft is the most efficient for a given lift and speed. By increasing its overall aerodynamic efficiency, especially during cruise, we will be able to increase performance and reduce drag, which results in lower fuel consumption.

From the graphs above, we can observe that NACA 4415 and NACA 2412 airfoils perform the best at angle of attack of 5° , while NACA 0012 airfoil has the best performance when the angle of attack is around 7.5° .

This information can be useful when deciding the most optimal angle of attack during cruise conditions to maximise the aircraft's overall efficiency.

To design a good aircraft, we need to analyse more than the lift generated and the aircraft's aerodynamic efficiency. We will also have to consider the static stability of the aircraft, which can be determined from the C_m - α curves. The C_m - α graphs of the different airfoils are shown below:



From the graphs above, we can observe the angles of attack where the airfoil is longitudinal statically stable. The gradient of the graph determines the static stability of the airfoil. A negative

gradient indicates that the airfoil is longitudinal statically stable. For NACA 4415 and NACA 2412 airfoils, the region with negative gradient is roughly between 0° and 5° angle of attack. The graph for NACA 0012 airfoil shows that the region with negative gradient is approximately between 2° to 8° , indicating that the airfoil is longitudinal statically stable at higher angles of attack.

However, it is also important to note that the airfoil and wing shape is not the most crucial factor when considering the aircraft's longitudinal static stability.

ii. Blended Wing (Florian)

An analysis was conducted on the blended structure of the 2d airfoil to verify the suitability of the wing design as well as its efficiency.

1. C_l vs Angle of attack

The lift coefficient variation across the angle of attack demonstrated effectiveness of the FR25 wing's multi-aerofoil design strategy. The FR25 cruise point as seen in the red marker on figure 54 occurs at 1.15° with a lift coefficient of 0.445.

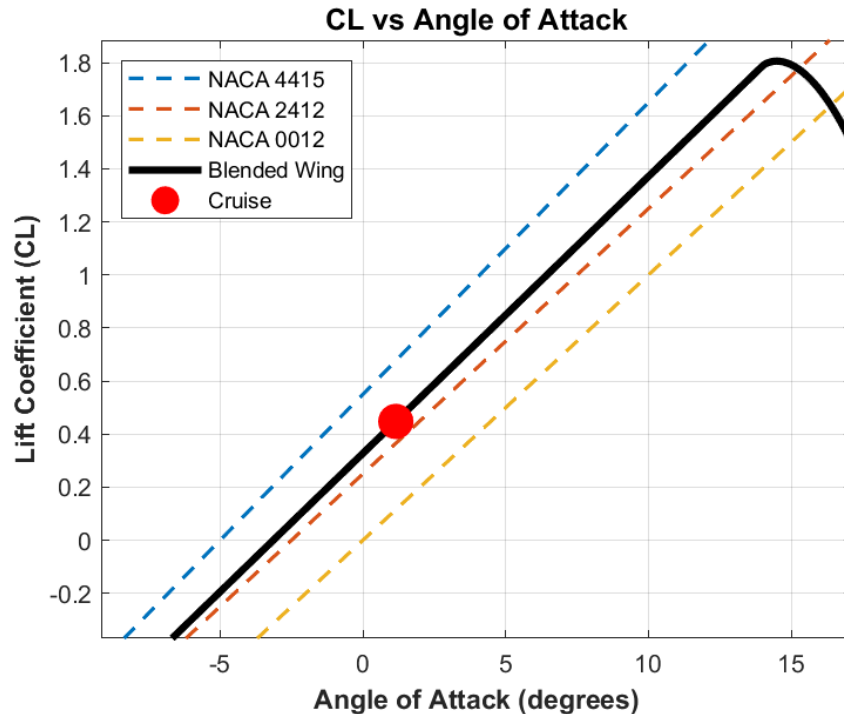


Figure 54: C_l vs Angle of Attack plot for the FR25

2. C_d vs Angle of attack

The drag coefficient variation with angle of attack shows a parabolic drag curve with a minimum drag coefficient of 0.0289 as seen in figure 55. This occurs near the cruise angle of attack. This plot demonstrates good efficiency compared to conventional regional turboprop aircraft.

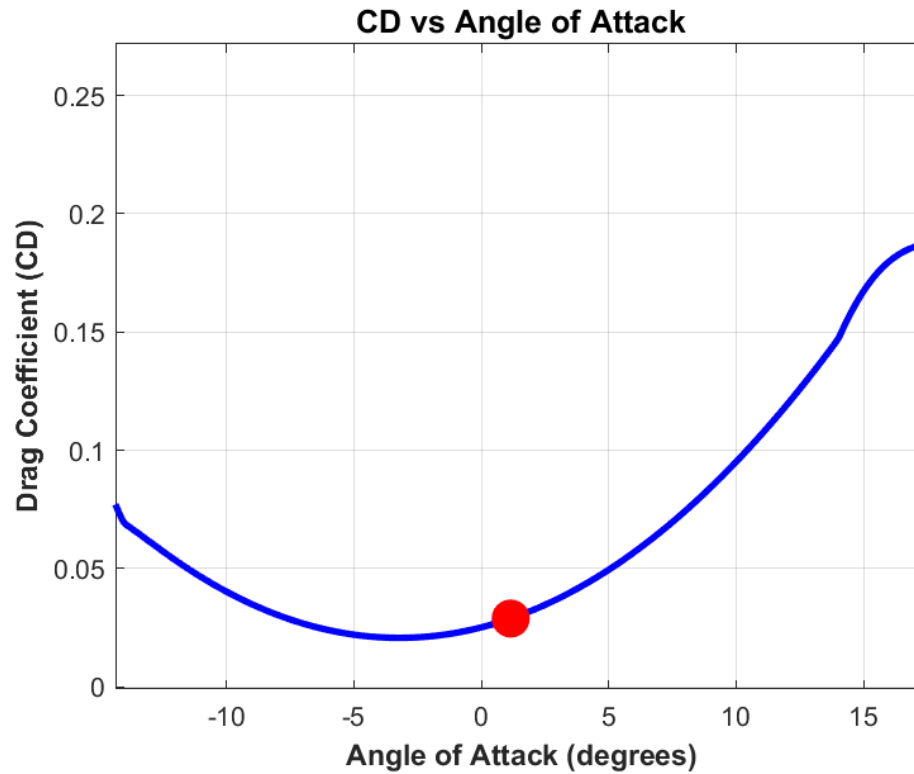


Figure 55: *Cd vs Angle of Attack plot for the FR25*

3. *Aerodynamic Efficiency*

The peak of the L/D curve occurs at about 3.8° with an L/D value of 17-18 as seen in figure 56. The cruise operating point at 1.15° angle of attack yields an L/D ratio of 15.52. This provides good efficiency during normal commercial cruise operations. This L/D value of 15.52 is competitive with modern regional jet aircraft and is comparable to industry standards.

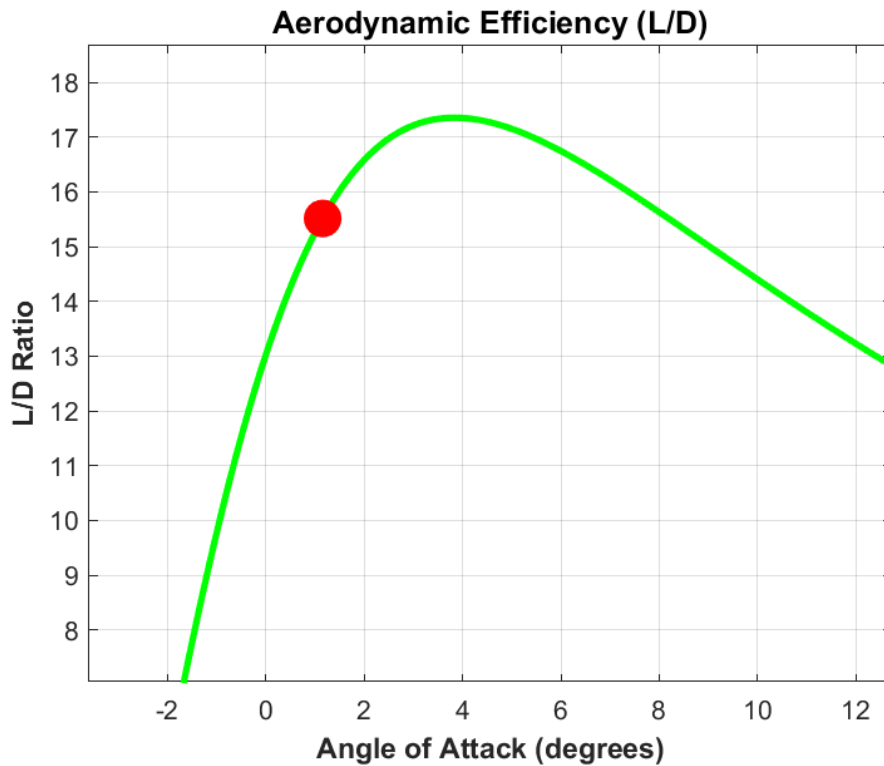


Figure 56: L/D vs Angle of Attack plot for the FR25

4. Drag Polar

The drag polar representation plots drag coefficient against lift coefficient and provide a characterization of the FR25's aerodynamic performance envelope. As seen in figure 57 the cruise point occurs at $C_d = 0.0289$ and $C_l = 0.445$. This point balances between lift production and drag penalty.

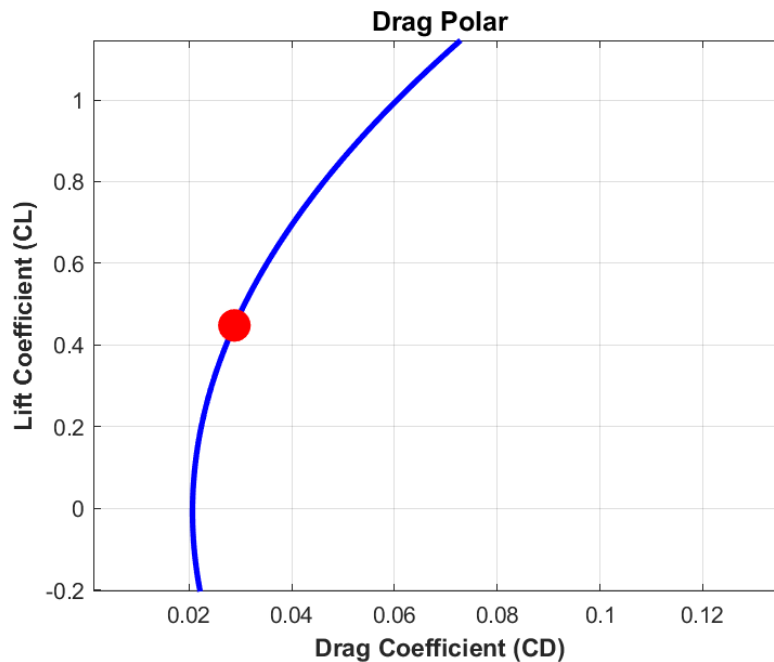


Figure 57: Drag Polar of the FR25

j. Wing Analysis

Since FR25's wing is unique as it incorporates three different airfoil shapes with varying chord lengths across the wingspan, Flowrider conducted a detailed analysis to determine its performance. We utilized ANSYS (Fluent) and conducted numerical simulation for the wing under cruise conditions.

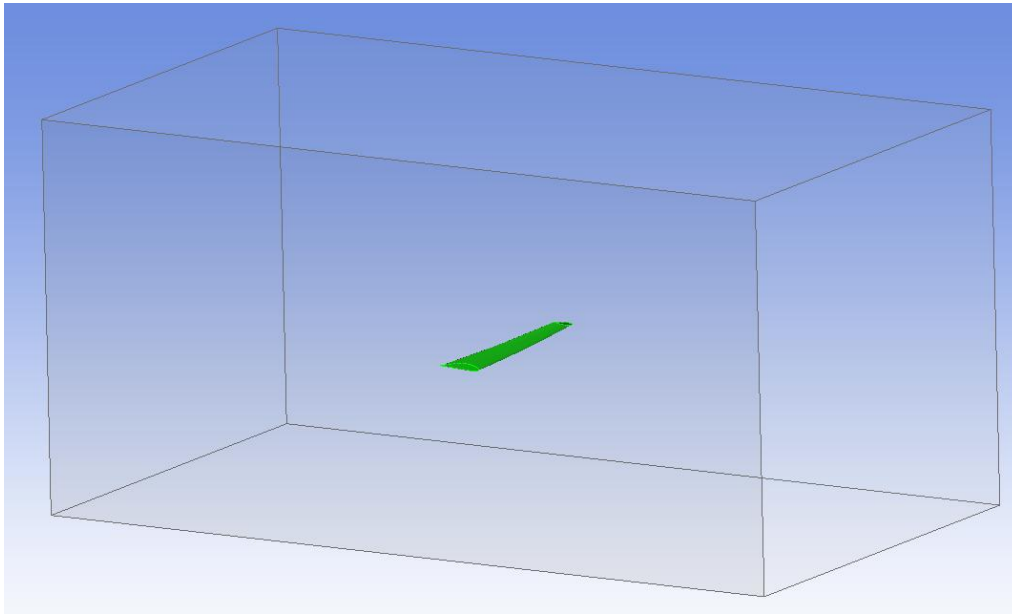


Figure 58: Drawing of fluid domain over wing

First, a fluid domain was drawn over our wing to simulate the flow over the surface of the wing. The domain should be large enough to prevent any disturbances to the flow during the simulation, and the fluid used in the simulation is air. A meshing before simulation was made as seen in figure 58.

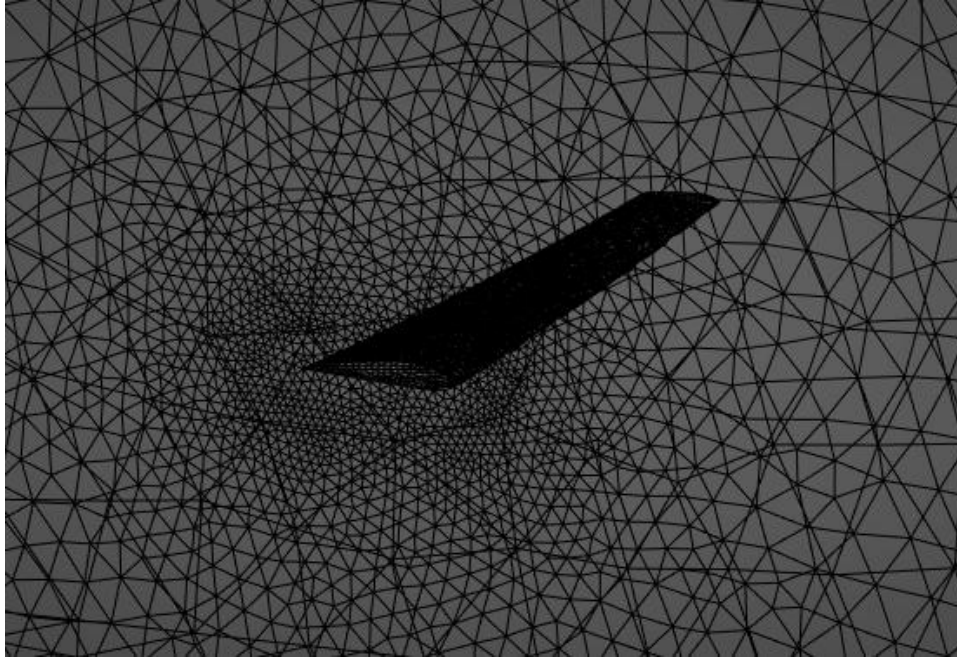


Figure 59: Meshing of wing in fluid domain

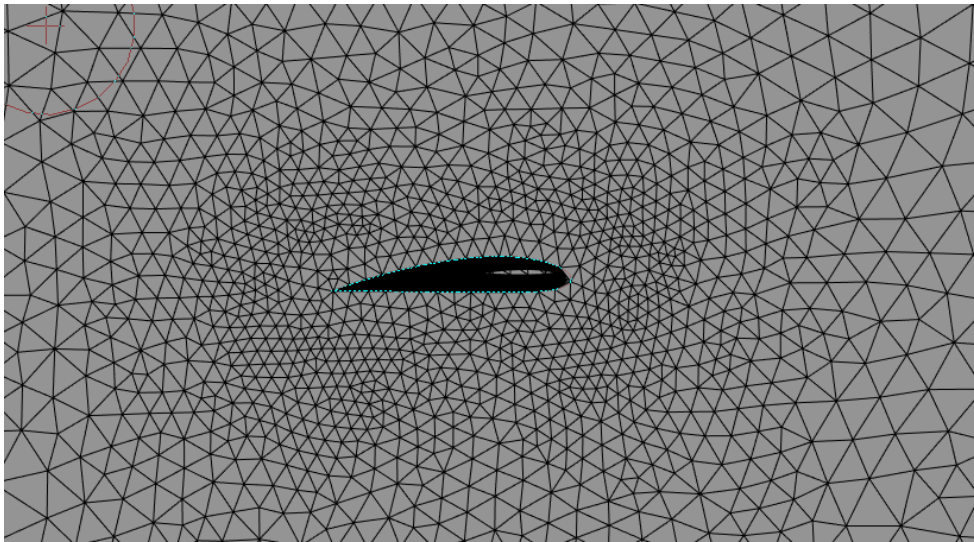


Figure 60: Close-up of the meshed wing

Before conducting the simulation, the airflow speed of the inlet needed to be calculated. To ensure geometric and dynamic similarity, several parameters were modified to ensure that the Reynold's number remained constant. The parameters used for Fluent analysis are shown in [table 23](#) below.

Table 23 : Parameters for Fluent analysis

Reynold's number	2.51×10^7
Wing scale	1:5
Half wing area	7.2 m ²
Mean Aerodynamic Chord (MAC)	2.61 m
Scaled MAC	0.52 m

Density of air (at FL250)	0.549 kg/m ³
Kinematic viscosity of air (at FL250)	1.921 x 10 ⁻⁵ m ² /s
Current cruise velocity	185.2 m/s

With the known parameters above, we can calculate to find the new cruise velocity:

$$v_{new} = \frac{Re \times Kinematic\ viscosity}{MAC\ (scaled)}$$

$$v_{new} = \frac{2.516 \times 10^7 \times 1.921 \times 10^{-5}}{0.52}$$

$$v_{new} = 929.47\ m/s$$

The simulation was conducted on Fluent at cruising altitude conditions and at 1000 iterations with a convergence of 0.001 for each value as seen in figure 61.

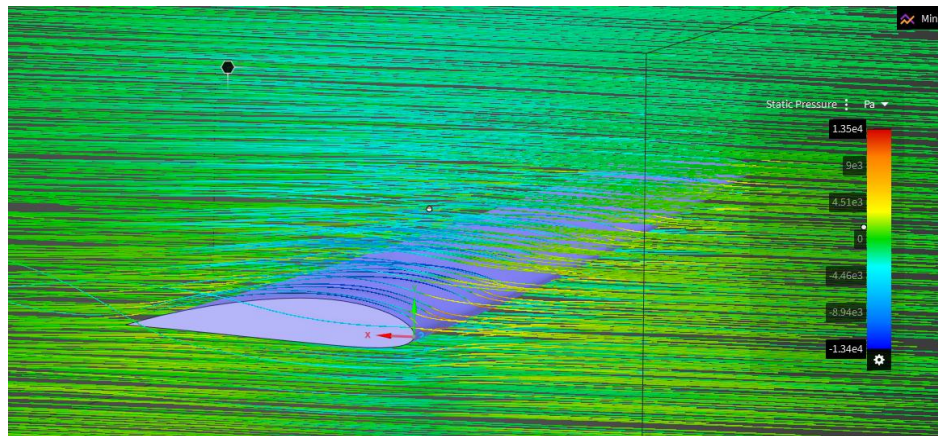


Figure 61: Velocity streamlines over wing design

The data obtained from Fluent is tabulated in table 24 below.

Table 24 : Data obtained from Fluent

AoA in degrees	AoA in rad	Vx (m/s)	Vy (m/s)	Lift (N)	Drag (N)	Cl	Cd	Cl / Cd
-4	-0.0698	924.1334	-64.6217	139282	14987.5	0.0821	0.0088	9.2932
-2	-0.0349	925.8257	-32.3305	150612	14642.0	0.0888	0.0086	10.2863
0	0.0000	926.3900	0.0000	161604	14266.9	0.0953	0.0084	11.3272
2	0.0349	925.8257	32.3305	172445	13840.0	0.1017	0.0082	12.4599
4	0.0698	924.1334	64.6217	183614	13344.5	0.1083	0.0079	13.7595
6	0.1047	921.3151	96.8341	194499	12740.9	0.1147	0.0075	15.2657
8	0.1396	917.3744	128.9286	205787	12122.9	0.1213	0.0071	16.9751

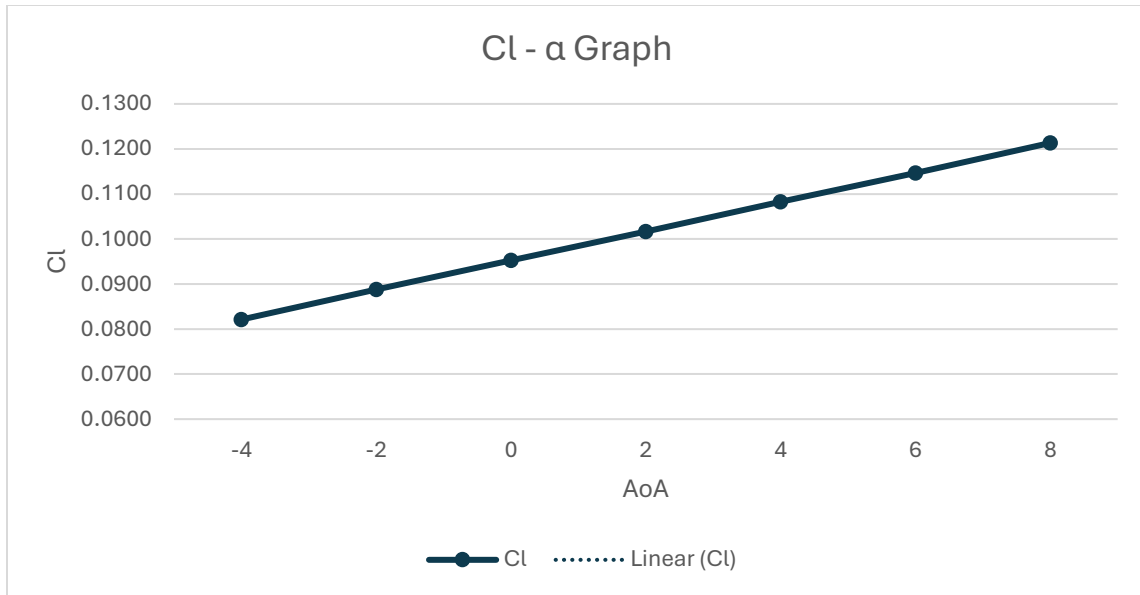


Figure 62: Cl vs Angle of Attack plot

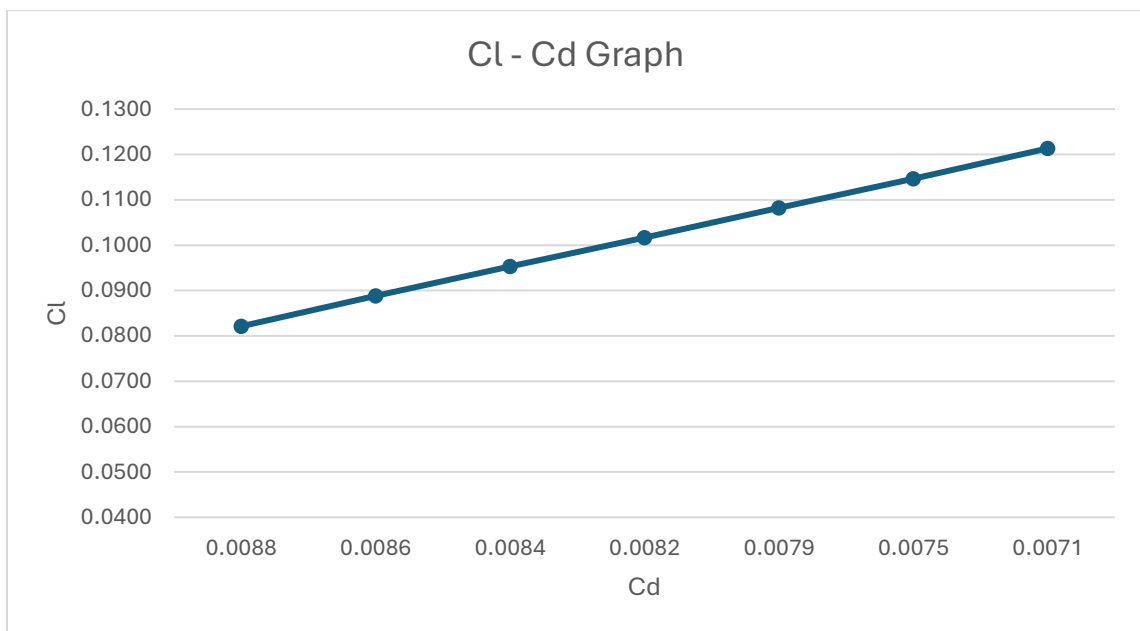


Figure 63: Cl vs Cd plot

From Fluent analysis, the pressure distribution over the wing can also be observed. This represents the lift and drag forces acting on the wing during cruise.

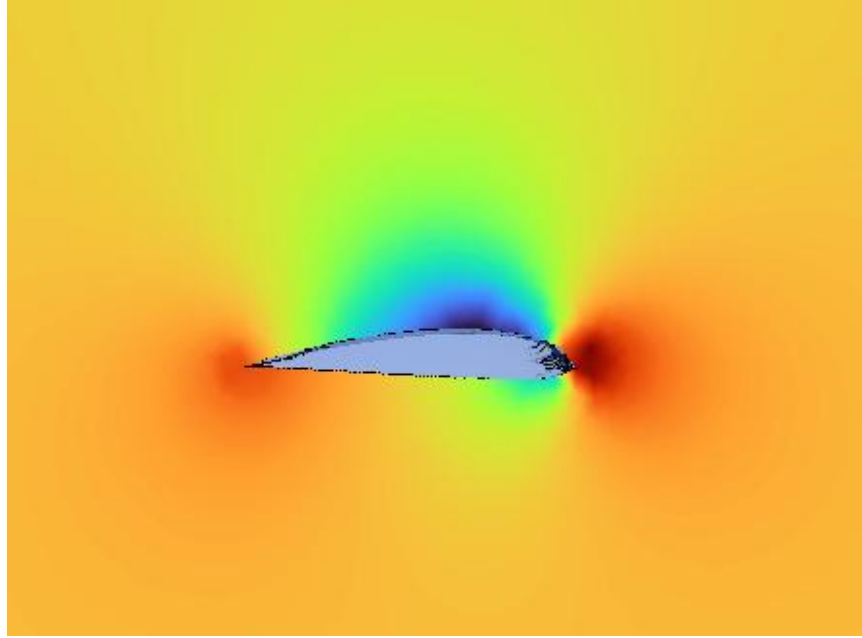


Figure 64: Pressure distribution over wing

The blue areas represent the regions with low pressure while the red areas represent the regions with higher pressure. Observing the data from Fluent in table 25, the lift and drag forces at cruise conditions can be compared with the MATLAB - calculated data above in Blended Wing Analysis (Section 9.j.ii).

At cruise, with an angle of attack of 2° :

Table 25 : Comparing Fluent data and blended wing analysis data

	Fluent analysis	Blended wing analysis
Lift coefficient (C_L)	0.1017	0.42
Drag coefficient (C_d)	0.0082	0.03
L/D	12.46	15.5

Comparing the values in both analyses above, we can observe some differences. However, these differences can be attributed to several factors. These factors include:

1. High fluid velocity in Fluent analysis

To ensure that the Reynold's number remains constant in the analysis of the scaled-down wing, the fluid flow velocity is increased to 929.47 m/s, which is close to the Mach number. As the velocity of air increases on the upper surface of the wing, shock waves may potentially form as the velocity exceeds the Mach number. This may result inaccurate readings obtained during Fluent analysis.

2. Blended wing analysis in 2D

The MATLAB-calculated values obtained in blended wing analysis is done on a 2D multi-aerofoil shape. Hence the data may not be the most accurate as well.

10. Engine Selection (Heong Luck)

a. Overview of Selected Engine

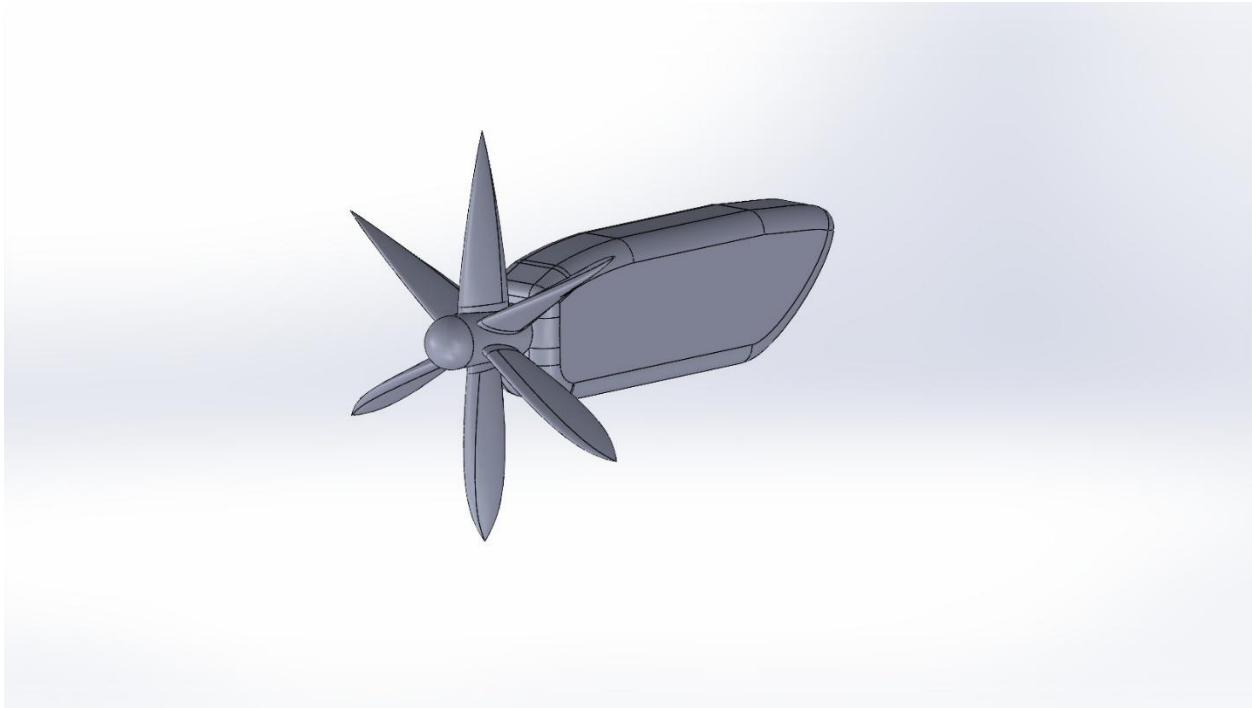


Figure 65: CAD Model of proposed engine

The engine design is highly critical in meeting the goals of sustainable regional air travel. The propulsion system accounts for the largest share of the turboprop's emission and fuel consumption. Therefore, the choice of engine design will directly shape the aircraft's performance, efficiency and environmental impact. Careful consideration of engine design and selection is therefore essential in ensuring technical feasibility as well as meeting the future sustainability target of regional travel.

To address these requirements, a hybrid-electric turboprop engine is therefore selected as seen in figure 65. The selected engine for this aircraft is a parallel hybrid-electric turboprop, which combines the reliability of conventional turboprop gas turbine engine with an electric motor that can work together to deliver required shaft horsepower. With the assistance of the electric motor in this parallel hybrid-electric engine, fuel consumption as well as overall emission can be reduced.

Currently, hybrid-electric engines are under research, testing and development phase by RTX. The engine itself is developed by Pratt & Whitney, while the electric motor, capable of providing up to 1 megawatt (1 MW) of electrical power, is developed by Collins Aerospace (Collins). Collins and Pratt & Whitney are both business units of RTX. This hybrid-electric engine combines both conventional turboprop engines and an electric motor to boost fuel efficiency and reduce emissions. This research and development by RTX provide strong proof of concept that hybrid-electric propeller engines are both feasible and future-ready.

With the 1MW electric power and the turboprop engine delivering reliable thrust, this offers additional power to the aircraft during its take off and climb phase of the flight (Pratt & Whitney, 2021). During those phases of flight, the engine will be operating at or near its maximum power, burning the most fuel per minute. By using a hybrid-electric engine for this aircraft, the electric motor can assist the engine during those phases of flight, reducing the power required to be produced by the engine, thus helping to save fuel and reduce emission.

This hybrid-electric engine is also well-suited to the aircraft mission profile, as a combination of both propeller and electric motor power can help the aircraft to take off from a shorter runway within a regional airport. Furthermore, the hybrid-electric engine has reduced noise emission, enabling lower noise pollution near airports during take off, an essential consideration for airports with curfew due to its proximity to residential areas (Niccolò Martello, 2023).

By integrating a hybrid-electric turboprop system with a modular cabin design for our aircraft, the design combines both sustainable propulsion and flexible operations within the regional market.

b. Specification of Engine

Table 26 :Engine Specifications

Parameter	Engine Specification (per engine)
Height	2.4m
Width	0.76m
Length	1.1m
Dry weight	718kg
Mechanical power	2492kW (3342shp)
Electric motor power	890kW (1194shp)
Combined power (P_s)	3382kW (4536shp)
Propeller diameter	3.93m
Number of blades	6 blades
Type of blade	Scimitar propeller blade
Nacelle frontal area	1.9m ²
Pylon and fairing weight	150kg

The data in table 26 shows the engine specifications, including mechanical and electrical power available at the propeller shaft. The mechanical power of the shaft is 2,800kW (3,755shp), the electrical power is 1000kW (1,341shp) and combining the mechanical power and the electrical power available to the shaft, we get 3,800kW (5,096shp) for one engine. However, the effective propulsive power is to be determined by factoring in the propeller's efficiency to obtain the actual combined power for each engine.

The propeller efficiency (η_{pr}) of the Scimitar propeller is as follows:

$$\eta_{pr} = 0.89$$

$$\eta_{pr} = \frac{P_p}{P_s}$$

where P_p is the propulsive power of the engine

$$\begin{aligned} P_p &= \eta_{pr} \times P_s \\ P_p &= 0.89 \times 3800 = 3382 \text{ kW} \\ P_t &= 3382 \times 2 = 6764 \text{ kW} \end{aligned}$$

Converting from kW to shp, we get:

$$6764 \div 0.7457 = 9071 \text{ shp}$$

From the calculations above, after taking propeller efficiency into consideration, the effective propulsive power of each engine is 3,382kW. And for the FR25, it is powered by two engines. Therefore, having a combined propulsive power of 6,764kW.

Furthermore, the Thrust Specific Fuel Consumption (TSFC) shows the relation of the fuel flow rate to the thrust output. However, in the case of the FR25, which is a hybrid modular turboprop aircraft, the power output of each engine is shaft power that will drive the propeller, rather than direct jet thrust. Therefore, the appropriate measure of TSFC in the case of a turboprop engine is the Power Specific Fuel Consumption (PSFC). The expression can be given as follows:

$$PSFC = \frac{\dot{m}_f}{P_s}$$

where $\dot{m}_f$ is the mass flow rate of the fuel and P_s is the shaft power of the engine (kW). The PSFC will have a unit in terms of kg of fuel burned per kW of power per hour (kg/kW/h). P_s is used instead of P_p because the PSFC measures the efficiency of the engine, where P_s is the actual output of the engine. If P_p is used instead, then the fuel efficiency will be calculated with propeller efficiency, flight speed and altitude. This will make PSFC inconsistent for aircraft that are using the same engine. Therefore, shaft power, (P_s) is used when calculating PSFC.

Next, the fuel flow rate, $\dot{m}_f$ can be obtained by making a comparison to aircraft within its range. For the ATR 72-600, it has a propulsive power of 3,691kW (4,950shp). And that has a fuel consumption of 762kg/h during cruise. By converting the units to kg/s, ATR 72-600 has a fuel flow of 0.212 kg/s.

Now, the Dash 8-400 can be taken into comparison in terms of its fuel consumption during cruise. The Dash 8-400 has a propulsive power of 7,563kW (10,142shp). It also has a fuel consumption of about 816 kg/h during cruise. By converting that to the fuel flow rate in kg/s, it is about 0.227 kg/s.

The FR25 has a propulsive power of 6,764kW (9,071shp) and is powered by both mechanical power and electrical power. With the help of the electrical motor to achieve the propulsive power requirement, the fuel consumption during cruise will be able to be reduced by up to 20%. Since the FR25 has a MTOW of about 30693.072 kg, the fuel consumption baseline will be justified to be 20% lower than the fuel consumption of its competitors. The following calculation will justify the fuel flow rate of the FR25:

$$\dot{m}_f = 0.227 \times 80\% \approx 0.18 \text{ kg/s}$$

With the fuel flow rate, $\dot{m}_f = 0.18 \text{ kg/s}$, the fuel consumption at cruise can be calculated to be:

$$\dot{m}_f \times 3600 = 0.18 \times 3600 = 648 \text{ kg/h}$$

These values reflect the FR25's expected operational efficiency and align with the intended operational objective of lowering emissions and improving operational efficiency through its improved fuel efficiency.

Given the fuel flow rate, $\dot{m}_f$ is 0.18 kg/s , the PSFC can then be calculated to be:

$$PSFC = \frac{\dot{m}_f}{P_s} = \frac{0.18}{3800 \text{ kW}}$$

Converting the units to kg/kWh :

$$PSFC = \frac{\dot{m}_f}{P_s} = \frac{0.18 \times 3600}{3800} = 0.171 \text{ kg/kWh}$$

Furthermore, in contrast to the traditional turboprop engines, which are mainly powered by purely fuel, the FR25 will incorporate a hybrid-electric propulsion system. This system will integrate an electric motor and a high-power lithium-ion battery pack on the aircraft. This electric motor will supplement the turbine's mechanical power output during high power phases of flight where power is critical in helping the aircraft to take off, climb and during go-around. The electric motor of the hybrid system will provide additional thrust while reducing overall fuel consumption and emissions.

On top of that, since the FR25's mission is mainly regional flights, where regional airports are near to residential areas and authorities may impose noise emission standard or nighttime curfews to minimise community disturbance. By using electric motor to assist the hybrid system, noise can be reduced significantly, and exhaust emissions can also be lowered as well during departure and arrival, where these are critical phases of flights which use higher engine settings. The FR25 can thus have a quieter take off, and with the help of the electric motor, can allow the FR25 to have a steeper climb rate, reducing noise emissions near regional airport and allow traffic movements in the night at airports with curfews to boost travel and operational efficiency.

During the cruise phase of the flight, the engines will be operating at their optimal efficiency point. In that phase, the electric motor will remain in a standby mode, contributing little to no power output to the propellers. This is because during cruise, the mechanical power from the turbine will be enough to keep the FR25 within its operational airspeed. So, during cruise, the electric motor will function in regenerative mode, using the mechanical energy to recharge the onboard lithium-ion battery so that the battery is ready for deployment again if more power is needed during the descend.

The specification of the battery can be shown as follows:

Table 27 : Electric Power Specification

Parameters	Units	Value
Electrical power	$P_{electric}$	2,000 kW
Specific power	$P_{specific}$	5 kW/kg
Specific energy	$E_{specific}$	250 Wh/kg
Battery voltage	V	800 V

With reference to the table above, the battery mass, m , can be calculated as follows:

$$m = \frac{P_{electric}}{P_{specific}}$$

$$m = \frac{2000}{5} = 400 \text{ kg}$$

With the calculations above, the mass of the battery is 400kg, and this is the value for the battery to supply to both the electric motors in each engine.

With the tables above, the battery capacity can also be determined. The calculations are as follows:

$$\text{Total Stored Energy} = m \times E_{specific}$$

$$\text{Total Stored Energy} = 400 \times 250 = 100000 \text{ Wh} = 100 \text{ kWh}$$

With the value of the total stored energy, the battery capacity can be calculated as follows:

$$\text{Battery Capacity} = \frac{\text{Total Stored Energy}}{\text{Voltage}}$$

$$\text{Battery Capacity} = \frac{100000 \text{ Wh}}{800 \text{ V}} = 125 \text{ Ah}$$

With the battery capacity defined, the C rating will then be defined. C rating of a battery is a measure of the battery's charge and discharge rate relative to the battery capacity. A higher C rating means that the battery can be charged and discharged faster while the lower C rating means that the charge and discharge rate will be slower. High C rating is preferred when powerful energy burst is required for a short period of time. This is consistent with what the engines of the FR25 need, with powerful burst of energy to boost the take off and climb segment of the flight. The calculations shown below will show how the C rating can be obtained for the take-off phase of the flight. First, obtain the current drawn from the battery for the take-off phase.

$$I = \frac{P}{V} = \frac{2000 \text{ kW}}{800 \text{ V}} = 2500 \text{ A}$$

Computing for the C rating, the following calculation can be done.

$$C = \frac{I}{Q} = \frac{2500A}{125Ah} = 20C$$

The battery will be designed with a peak 20C rating. 20C rating will be used only for take-off phase only to give the FR25 the boost it needs to accelerate and perform its initial climb. This high discharge rate is only sustainable for short duration boost, providing the FR25 with some additional boost and early climb of about additional 2000kW of electric power to supplement the mechanical power. This high-power boost setting is managed by the Battery Management System (BMS), where it will discharge at its peak C rating of 20C rating.

Once the aircraft reaches a steady climb rate, the power demand will decrease and the BMS will revert the battery back to its nominal rate of 3C. This is to prevent the battery from dropping to below 20% as battery should always be kept above 20% state of charge. At this stage, the engine not only will maintain the climb thrust, but also drive the generator to recharge the battery. This means that the battery will simultaneously provide the additional electric power, assisting the climb without compromising the battery health.

This C rating of 3C for the battery provide a balance between power delivery and thermal management, allowing the FR25 to continue its climb, while remaining within the safe operating limit of the battery, preventing overheating of the battery due to high charge and discharge rate. This approach allows the FR25 to exploit the battery's peak power capability for high power demand phases without exceed its safe design limit. Since the high discharge rate only last for the initial take off and climb phase, reverting to 3C once the positive climb rate is established can help enhance the lifespan of the battery as well.

The C rating of 3C on the battery also ensure that the electric motor assist and deliver the required supplementary power for the climb, while maintaining the battery state of charge at above 20%, as the lithium-ion batteries should not be drained below 20%. This will satisfy the function of the hybrid-electric system, where the battery can provide effective short burst of power for short duration and ensuring that the battery can be sustainable for use in long-term.

The placement of the battery will be installed withing the lower fuselage, nearer to the front of the aircraft to help balance the centre of gravity of the aircraft. Furthermore, by placing it nearer to the front of the aircraft, the battery is away from the flammable fuel, which is mostly concentrated near the aircraft centre tank and the wings.

The weight of the battery calculated is 400kg. That will be the pure weight of the battery itself. However, batteries need to be properly protected due to its fire risk. All the protective equipment for the hybrid battery will be discussed below, where the final weight of the battery will also be determined.

c. Fuel Type and Emission

To align with the IATA's commitment to fly net zero by 2050, the FR25's engines need to be designed to run using Sustainable Aviation Fuel (SAF), offering flexibility during global transition to greener fuels (IATA, n.d.).

Compared to conventional jet fuel, the SAF can reduce carbon emission by up to 80%. The SAF is a blend of existing jet fuel with renewable or synthetic hydrocarbons produced from sustainable sources such as used cooking oil, animal fats or agricultural residues. This blending process ensures that the SAF are compatible with the current aircraft engines and fuel systems to help reduce the carbon emissions. Currently, the regulations state that the maximum blend of 50% is allowed due to potential issues with existing jet engines that are ageing. Newer engines, such as the propeller engines designed for the FR25 will incorporate the use of 100% SAF to align the engines to produce net zero carbon emissions by 2050.

The ability to operate on 100% SAF ensures that the FR25 remains future proof as the aviation industry standard evolves. Since the aviation authorities are targeting a net zero carbon emissions by 2050, airlines operating the FR25 will be able to meet the sustainability requirements of the aviation without the need to perform engine modifications or retrofitting since it is compatible with 100% SAF. This will enhance the FR25 commercial appearance, particularly for the regional turboprops.

As a proof of concept, giant aerospace engine manufacturer, Pratt & Whitney, had successfully tested the GTF Advantage engine with 100% SAF. As the maximum blend of 50% is in place in the aviation industries today, engine companies are working hard to validate their engines to proof that they can operate with 100% SAF, as SAF is the key to achieving a net zero carbon emissions by 2050 (Pratt & Whitney, 2022)

d. Hybrid-Electric Configuration and Operation

The following few flowcharts will show how the FR25's parallel hybrid-electric propulsion system will operate under different phases of flight, including taxiing, take off and climb, cruise, descent, and in the event of a go-around. In each of the flowchart, the direction of power flow between the turboprop engine, generator, hybrid battery, electric motor and the propeller itself, highlighting how both the mechanical power and electric power will work together to achieve the performance required during its specific phases of flight.

When the aircraft is on ground and taxiing, the power required is relatively low. The mechanical engine can provide power to the propeller and charge the hybrid battery through the generator at the same time, allowing the battery to provide electrical power to power the propeller as well. This is demonstrated in figure 66.

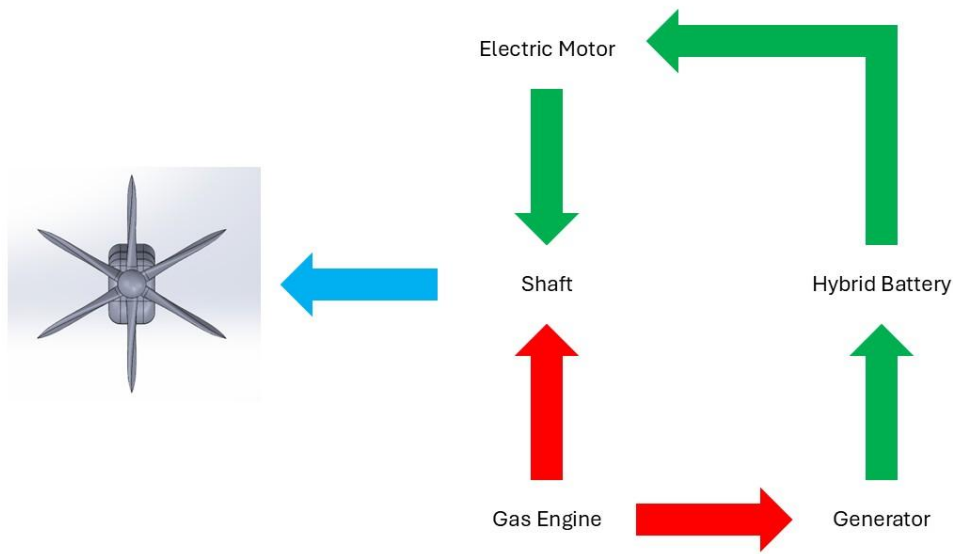


Figure 66: Engine Power Configuration on Ground

During take-off and early climb phase, the engine and electric motor will work together to drive the propeller and deliver the take-off thrust required by the aircraft to maintain a steady climb rate. Since this phase is a high-power demand phase of flight, the engine will purely generate mechanical power to power the propeller, it will not charge the hybrid battery in this phase of flight as seen in figure 67.

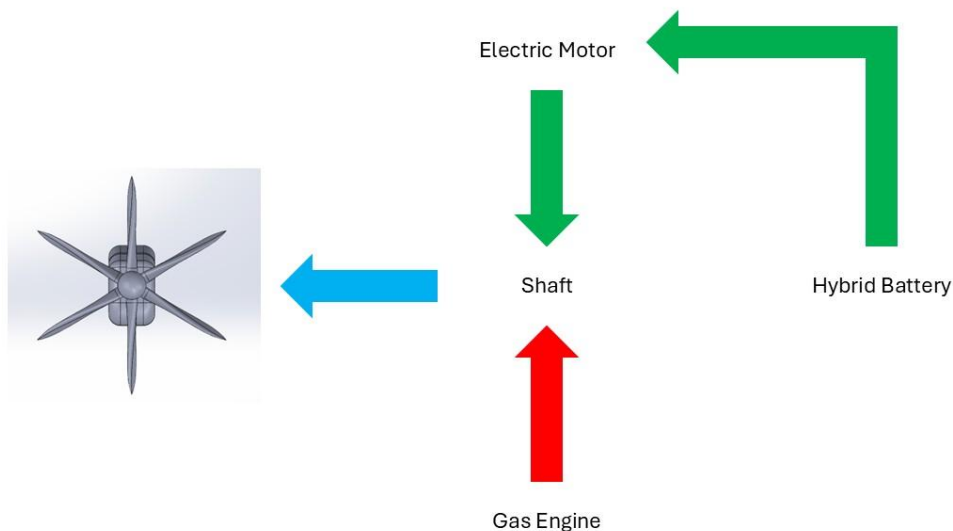


Figure 67: Engine Power Configuration During Take off Phase

Once the aircraft establishes a positive rate of climb, the electric power will gradually be reduced from its peak discharge level, allowing the hybrid system to transition into a more balanced mode, where the C rating of the battery is returned to 3C rating. This will prevent excessive thermal stress on the battery. During this period, the turboprop engine will utilise power from both the mechanical and electric power, with the mechanical power charging the battery through a generator and at the same time delivering electrical power, ensuring that the hybrid system is working in tandem, delivering steady thrust while restoring the state of charge of the battery after the initial take off and climb phase.

During cruise, the FR25 will maintain the flight using its hybrid electric propulsion system in a steady and efficient manner. Both the turboprop engines and the electric motor will contribute to the propeller thrust. At this stage, the generator is also actively using the engine power to recharge the battery, allowing for the balanced configuration to sustain the FR25 during the cruise phase of flight.

During descent, as the power required during this phase of flight is low, the propeller can also utilise both mechanical power and electrical power to assist the aircraft to line up and come into land. Since the power required during this phase of flight is low, the mechanical engine can also charge the hybrid battery while providing power to the propeller. The electrical power utilised here is mainly to reduce the noise emission from the engine, as this regional aircraft will likely be utilised on regional routes.

Should there be a missed approach, the pilot will increase the power to take off power, and the system will go into the configuration that is the same as during take off phase, where the C rating will temporarily increase to its peak power of 20C and deliver the maximum power for the aircraft to successfully go around. .

The hybrid-electric system of the FR25 will operate through an automated power management system, and this will help the pilots by eliminating the need for manual switching of power delivery. That means based on the throttle input and flight phase, the system will automatically determine the optimal power distribution between the mechanical power and the electric power.

This automation ensures a smooth and efficient transition among all the different phases of flight, where the power required are different based on the pilots' throttle input. The system, which is managed by the Battery Management System, will therefore not add additional workload to the pilot and thus will not require the pilot to go through additional training to operate the hybrid aircraft, allowing seamless transition from traditional engine driven turboprop to the FR25's hybrid-electric driven turboprop.

e. Battery Safety

The safety of the FR25 will need to be critically considered due to the incorporation of hybrid-electric propulsion system, where the lithium-ion battery that powers the electric motor can pose serious fire risk when the battery is not properly managed.

Fire in the air, which is also called in-flight fire, can be one of the most hazardous situations where the crew can face. Thermal runaway is a chain reaction within a battery cell that can be very difficult, or near impossible to stop once it has started. It can occur when the temperature of the battery reaches a certain temperature that will trigger a chemical reaction to occur inside the battery. This chemical reaction will produce even more heat, driving the battery temperature higher, causing more chemical reaction and overheat, causing a fire that is near impossible to put out.

There are several ways that can cause a thermal runaway in the battery. One of the main causes of thermal runaway is the battery is not properly maintained, which can lead to internal short circuit. Another cause of thermal runaway is improper cooling of the battery, causing the battery to overheat and triggering the chain reaction for thermal runaway (Dragonfly, 2022).

To mitigate the potential fire risk from the battery, the FR25 battery will be housed in a reinforced, fire resistance enclosure for insulation. This housing will be equipped with active coolant to regulate the temperature during charge and discharge cycle, where battery will likely be warmer. Venting mechanisms will also be added to this housing enclosure to air circulation within the housing enclosure, preventing any buildup of pressure.

The housing enclosure will also be reinforced to withstand a high gravity load. This is to provide mechanical integrity and crash protection during high impact landing if in the event of a crash. The material uses high-strength and lightweight materials such as aluminium alloy to withstand high impact loads during abnormal landing or crash events, without deforming excessively to protect the battery from damage. This reinforced enclosure will act as a rigid containment, ensuring that the FR25's battery system is well protected from displacement or short circuit in the event of high-G loading, meeting the aviation safety certification.

Furthermore, there will be thermal zoning and module separation within the battery cells. This is done by separating battery modules with thermal barriers that can prevent heat transfer between battery cells. This is essential approach to ensure that any localised failure will not spread to adjacent battery cells.

For safety reason, class D fire extinguishers will also be mandated to be carried on the aircraft. This is because class D fire extinguishers use dry powder to smother the flames and absorb the heat from the fire, effectively preventing the fire from spreading and reigniting. Other classes of fire extinguishers are not suitable for battery fire because firstly, water type fire extinguishers may exacerbate the fire by reacting with the lithium within the battery. Foam type and carbon dioxide type of fire extinguishers are not capable of reducing the heat from battery fire to extinguish the fire. Therefore, the class D fire extinguishers will be the most suitable to fight battery fire should there be a need. This extinguisher will be in the form of a sprinkler system, where it will be controlled by the smoke detector in the battery compartment. If smoke is detected, the sprinkler will activate the class D sprinkler to help extinguish the battery fire.

Fire suppression blanket will also be placed on the aircraft to help prevent the spread of the fire during the event where the fire is not able to extinguish. The blanket will be made from high-temperature-resistant material and will act as a secondary containment layer that will isolate the

affected battery modules, and restrict oxygen flow to the fire, starving the fire from oxygen. This will stop the spread of the fire and limit the smoke and toxic fumes from being released into the fuselage. Although with many fail-safe system built in place for the battery, the event of an inflight fire caused by battery thermal runaway be extremely rare. This additional layer of fire protection will ensure that in the event of a fire, the fire suppression blanket will help to contain the fire if the fire is unable to be extinguished.

Finally, the Battery Management System (BMS) will gather crucial data from the battery, such as temperature, voltage and current through the sensors. This BMS is a fail-safe system, and it will serve as a control centre for the battery modules. It has a four-level emergency response procedure if a fault is detected by the BMS.

Level 1 – Abnormal reading detected

- In the event of abnormal readings, the BMS will initiate a reduced electrical power, isolate the affected battery module and try to maintain the overall power output. Take off and climb can still be initiated, but the BMS will closely monitor the battery parameters to ensure that the abnormal readings do not diverge.

Level 2 – Persistent abnormal reading

- If the abnormal reading persists and does not return to normal operation standard, the BMS will deploy enhanced cooling and will advise the pilot to cut off the power from the electrical system. Should this happen during the take off and climb phase, pilots must be prepared to level off and hold altitude if the mechanical power is not sufficient to ascend the aircraft. They should monitor the parameters. Should the reading fail to return to normal within a safe period, the flight crew are instructed to request an immediate diversion to nearest suitable airport because during the climb phase, there is a high demand for power and by shutting down the electrical power, the aircraft will be climbing at a much slower rate. During the cruise phase, pilots will only need to monitor the parameters on the BMS since mechanical power from the engine is sufficient to keep the aircraft in cruise conditions.

Level 3 – Critical overheating or smoke detection

- If the thermal sensors detect signs of overheating, or smoke detector in the battery housing detects smoke, the BMS will trigger a higher urgency of alert to the cockpit, issuing an audible and visual cockpit warning to inform the pilot of the impending fire. The BMS will isolate the entire battery system while activating emergency cooling and venting mechanisms to dissipate the heat and pressure. Once level 3 protocol is triggered, the pilots are obliged to land the aircraft as soon as possible at the nearest airport due to the danger of in-flight fire.

The FR25's battery design and safety protocol will align with aviation framework for the use of lithium-ion batteries. These standards emphasize fault tolerance, containment, and a fail-safe operation, ensuring that the hybrid-electric propulsion system can meet the same reliability standard as compared to conventional propulsion system, reducing emissions but meeting the standards of aviation.

On top of that, the FR25's hybrid battery system is designed for performance and its uncompromised safety in flight. While the lithium-ion battery pack itself weight only 400kg based on the calculations done above, the total mass increases to 800kg once all the protective systems are incorporated. The breakdown of the weight of the battery system is shown in table 28.

Table 28 : Breakdown of Mass of Battery System

Component	Description	Mass (kg)	% of Total
Battery	The battery pack where it will supply power to the electric motor	400kg	50%
Battery Management System (BMS)	Sensors, control units, cable and wires, electronics and device housing	40kg	5%
Containment housing with active coolant system	Fire-resistant casing with venting system and pressure relief panels, incorporated with cooling system that circulate the coolant, allowing for a active liquid cooling loop around the housing of the battery	200kg	25%
Thermal barrier	Thermal barriers are placed between battery cells to prevent heat transfer and propagation of thermal runaway	60kg	7.5%
Mounting and structural reinforcement	Damp out vibration and ensure the battery is well protected during a crash	60kg	7.5%
Wiring	Wires running from the battery to the electric motor in the engine	40kg	5%
Total	-	800kg	-

To conclude the battery safety, the FR25's battery system is designed to ensure absolute reliability and safety operation during flight. While the battery itself only weight 400kg, the inclusion of the Battery Management System, coolant circulation, thermal barriers, reinforced housing, mounting structures and wiring will increase the total weight of the battery system to 800kg. Each of these

features will serve critical purposes, from temperature control to energy absorption during high impact loading events. Although these safety features increase the mass of the battery system, they are very important in ensuring the battery's safe operation.

f. Propeller Blade

The propeller blades play an important role in the FR25's propulsive principle, serving as the primary interface for converting engine power into useable thrust. This means that the propeller performance directly affects the power efficiency of the aircraft. Therefore, the design of the propeller blades must balance aerodynamic dynamics and structural integrity.

The propeller blades will consist of six scimitar blades propeller configuration, chosen to maximise the thrust generation by pushing the propeller efficiency to the highest. Scimitar blades are chosen for its ability to provide enhanced performance. These enhancements are in the form of minimizing the formation of shock wave near the tip, which will result in reduced drag on the aircraft. The reason behind this is due to the curved shape of the propeller blades, which will help to minimise the strength of the sonic shock wave formation, allowing the propellers to provide more optimal performance at higher power and RPMs. (Ray, 2025)

Furthermore, operating the blades near the speed of sound is also the source of noise emitting from the turboprop aircraft. By reducing the formation of shock waves, the curved blade with a swept tip scimitar blade can significantly reduce noise pollution without affecting the engine power and propeller RPMs.

The propeller blade design also plays a significant role in the fuel efficiency of the turboprop. With the enhanced aerodynamic design of the scimitar blade chosen for the FR25, the turboprop can achieve higher propeller efficiency, potentially helping airline companies to save on flying cost. (Hartzell, n.d.)

The propeller blades will also feature a mechanism, which will allow the pitch of the propeller blades to be varied. This blade angle can be adjusted, which changes the angle of attack of the blade. This mechanism will allow the blade to be set at the correct blade angle according to the flight condition.

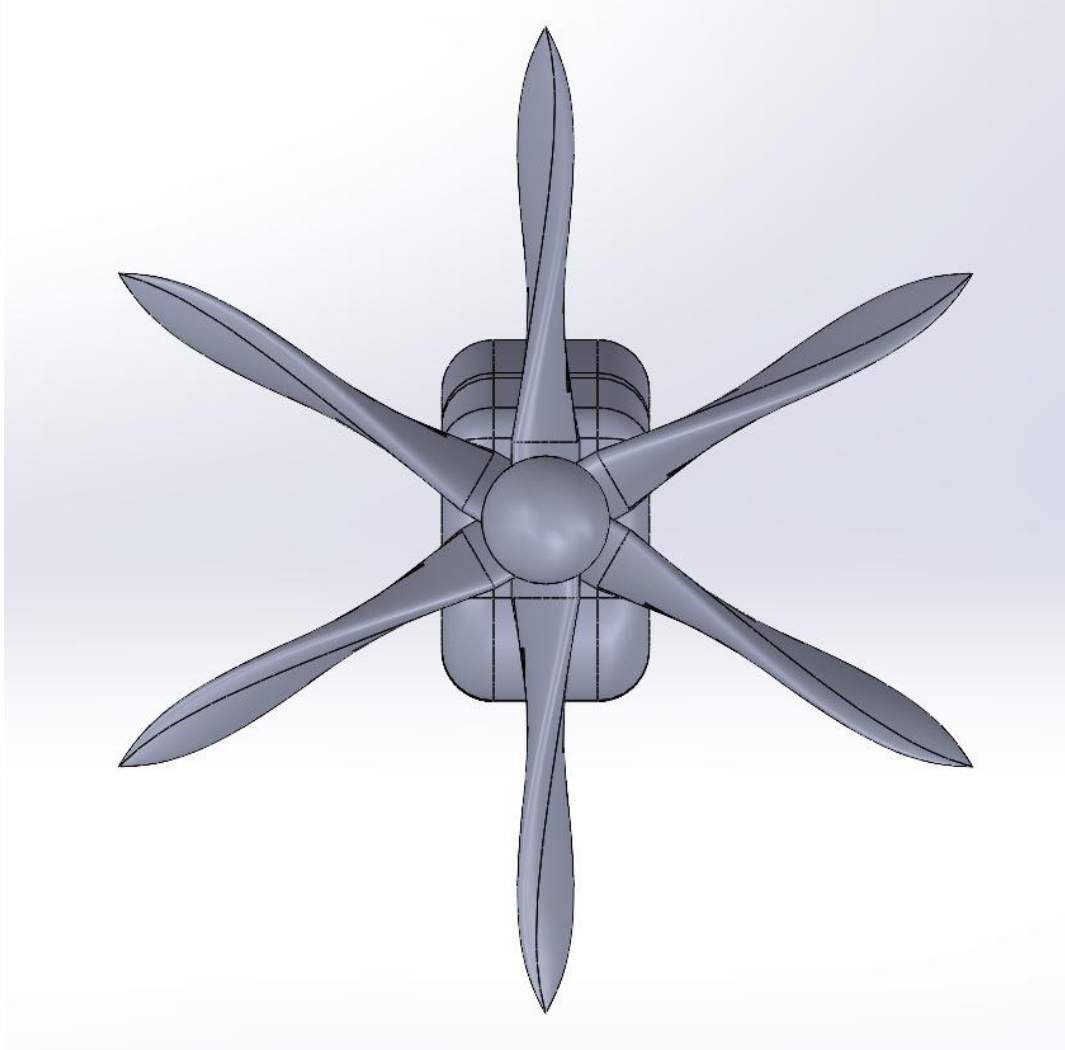


Figure 68: Propeller Angle at Low Speed

As seen in the figure 68 above, the propeller blades angle is at a shallow angle. This is consistent with the aircraft being at low-speed configurations, such as taxiing and take off and climb phase of the flight. This configuration allows the propeller to take a larger aerodynamic portion of the air, generating higher thrust and rapid acceleration. As the aircraft approaches the cruise region of the flight, the propeller blades will gradually transition to a coarser pitch, as seen in the figure 69 below.

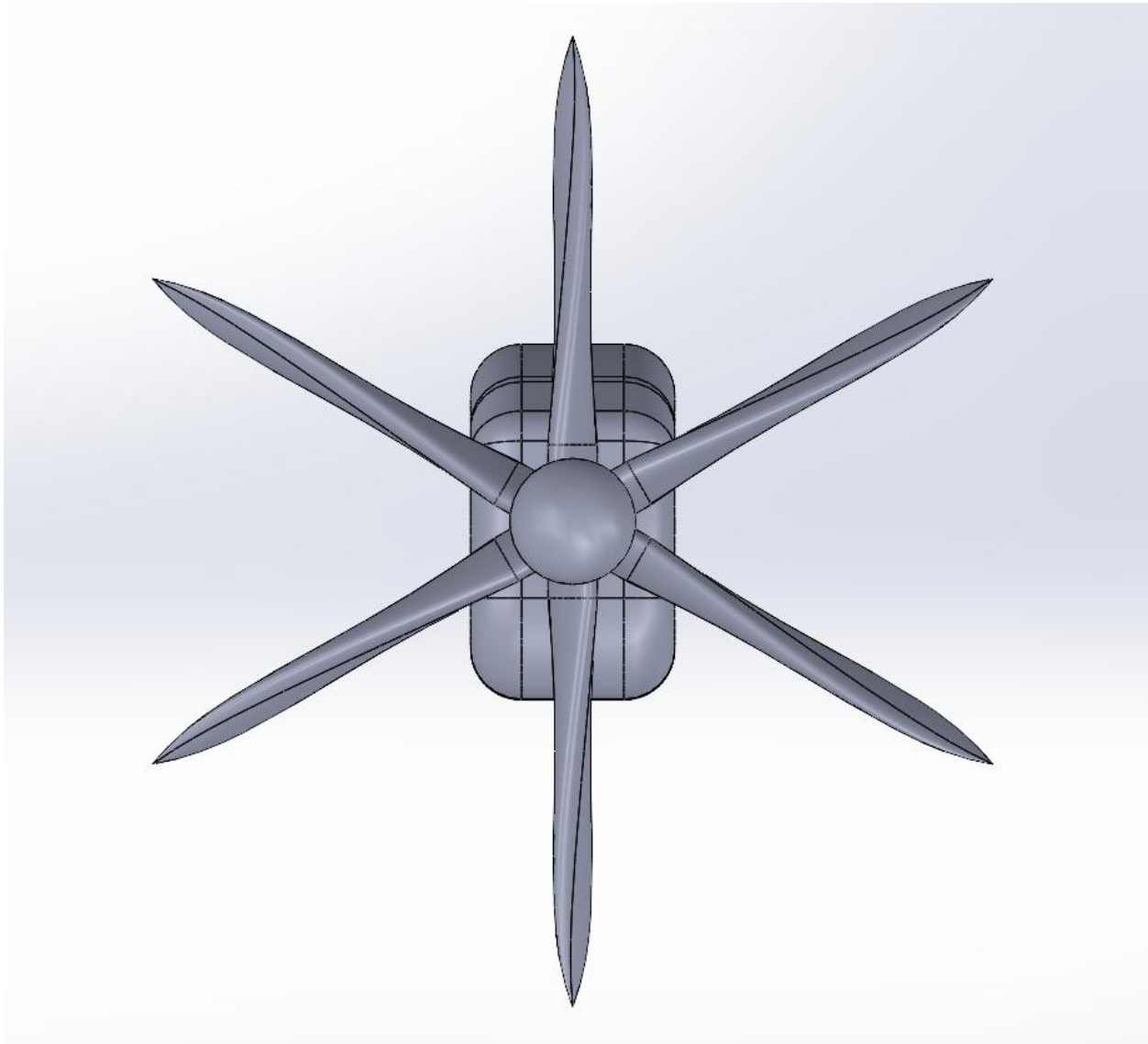


Figure 69: Propeller Angle at High Speed

The propeller pitch blades are at a coarser pitch, increasing the angle of attack of the blade and reducing rotational drag when the airspeed is high, such as during cruise. By increasing to a coarser pitch, each blade will take a larger aerodynamic bite per revolution, matching the higher inflow velocity. This also reduces rotational drag and allow the propellers to generate the same amount of thrust power as at lower RPM, improving the propulsion efficiency and lower the fuel and noise levels.

g. Auxiliary Power Unit (APU)

Although the FR25 features a hybrid propulsion system, the APU will still be required as part of the FAA regulations. This is because the APU will help provide electricity to the aircraft when the

engine is off, such as providing electricity to the air-conditioning system, cabin lighting and cockpit display.

The APU also serves as a backup electrical power in the rare event of a dual engine failure. When the engines fail inflight, many crucial flight instruments will not function as engines are the one supplying power to the instruments. This is where the APU will kick in to supply power to key instruments and flight controls, preventing a total loss of aircraft controllability and can help support a safe recovery and diversion procedures. For the FR25 the inclusion of the APU will ensure compliance with the FAA regulations and enhance the aircraft's overall reliability during emergencies.

h. Comparative Analysis

To determine the propulsion system that is suitable for the FR25's mission profile, which is short to medium haul regional flight, a few turboprops were being evaluated based on their power output, MTOW.

Table 29 : Comparative Analysis between Turboprops

Engine Model	MTOW (kg)	Power (kW)	PSFC (kg/kWh)	Propeller Efficiency
Pratt & Whitney PW100 used by ATR 72-600	23,000	4,045	0.23	0.87
Pratt & Whitney PW150A used by Dash 8-400	29,574	7,562	0.24	0.85 to 0.88
Pratt & Whitney PW38HE used by FR25	30693.072	6,764	0.171	0.89

Based on the comparison in table 29, the FR25's hybrid-electric turboprop offers an optimal balance between power-to-weight ratio, and a better fuel efficiency for regional operations. While the ATR 72-600 and the Dash 8-400 engines provide reliable mechanical performance, their fuel consumption and noise emission made their aircraft slightly less favourable for regional operations. The FR25 hybrid configuration, delivering a combined output of 3,382kW per engine, equate to 6,764kW for both engines, can achieve comparable power output from the propeller with a projected 20% to 30% lower in fuel consumption due to the hybrid assistance during take-off and climb. This hybrid engine also has lower noise emissions, which is highly suitable for regional airports that have noise emissions standard to adhere to. The balance of performance, efficiency and technological advancement incorporated into the engine enhance the feasibility of the engine to be used on the FR25.

i. Cost Breakdown of Engine

Table 30 : Cost Breakdown

Components	% of total	Value (USD)
Core Engine (Compressor, combustor, turbine, battery, hybrid system)	45%	\$675,000
Propeller Gearbox	18%	\$270,000
Propeller Control System	12%	\$180,000
Fuel Pump Control System	7%	\$105,000
Propeller (Blades and Hubs)	13%	\$195,000
Testing and Certification	5%	\$75,000
Total	100%	\$1,500,000

Table 30 above shows the cost breakdown of one engine. It is priced reasonably but will lower in value upon mass ordering of multiple engines following the FR25 development program timeline.

11. Aircraft Material Selection and Analysis (Nikunj)

The following are the materials we use in our aircraft for the respective components:

a. Fuselage Material

The fuselage is constructed primarily using carbon fibre reinforced polymer (CFRP). This material choice is supported by its successful use in commercial programs such as the Boeing 787 (about 50% CFRP by weight) and the Airbus A350 (around 53% CFRP), both of which have accumulated millions of flight hours without any reported fatigue failures in composite fuselage structures (Boeing, 2024; Airbus, 2024). Regional aircraft such as the D328eco, which incorporates roughly 25% composites, further demonstrate CFRP's suitability for smaller-scale and high-cycle operations (Deutsche Aircraft, 2025). The proven performance of the ATR 72-600 in demanding environments, including high temperatures, coastal humidity, and unpaved runways, reinforces the material's durability under challenging operating conditions (ATR Aircraft, 2024).

The fuselage structure consists of CFRP skins supported by Aluminium–Lithium 2050-T84 frames and stringers. CFRP provides excellent fatigue resistance, eliminates corrosion, and delivers a 20–25% weight reduction compared to traditional metallic designs, resulting in improved fuel efficiency (Toray, 2025; MMPDS, 2025). The Al–Li 2050-T84 alloy offers high strength-to-weight ratios (tensile strength 455 MPa, yield strength 395 MPa, elastic modulus 76 GPa) and superior fatigue performance (140 MPa at 10^7 cycles), ensuring a robust load-bearing capability (NASA, 2020; MMPDS, 2025). Together, these materials provide a well-balanced fuselage design that combines efficiency, durability, and maintainability, which is an optimal choice for our aircraft's operations across regions like the African continent.

b. Wing Material

The wing structure employs CFRP for the main load-bearing components, such as the upper and lower covers, spars, and wing box, while aluminium-lithium alloy 2050-T84 is used for secondary elements including ribs and floor beams. This configuration follows industry practices proven in aircraft like the Boeing 787 and Airbus A350, both of which feature fully composite wings (Aviation Week, 2020; Airbus, 2021). Regional platforms such as the ATR 72-600 (with roughly 30% carbon fiber content) and the D328eco also validate this hybrid approach for short-haul, high-frequency routes (ATR Aircraft, 2024; Deutsche Aircraft, 2025).

CFRP provides the necessary fatigue resistance and weight efficiency (specific strength 774 kN·m/kg, elastic modulus 130 GPa unidirectional), while the aluminium-lithium alloy offers high strength-to-weight ratios and ease of repair using existing maintenance infrastructure (Toray, 2025; NASA, 2020). This combination achieves a practical balance between advanced material performance and maintainability, an important consideration for operations in regions with limited technical resources.

c. Tail/Empennage Material

The empennage structure follows a similar design philosophy to the wing. CFRP is used for the primary tail surfaces, while aluminium-lithium alloys form the internal substructures, with titanium (Ti-6Al-4V) reinforcements added in high-stress areas. Fully composite tail assemblies, as seen in the Boeing 787 and Airbus A350, have shown excellent in-service performance, while the D328eco demonstrates this configuration's suitability for regional aircraft (Aviation Week, 2020; GKN Aerospace, 2021).

In certain applications, such as leading edges, fiber-metal laminates like GLARE, successfully applied on the Airbus A380, can enhance impact resistance and damage tolerance (Wikipedia, 2004; CompositesWorld, 2025). The CFRP structure offers high stiffness-to-weight ratios (elastic modulus 60-70 GPa quasi-isotropic), which are essential for effective center-of-gravity control, while the metallic subcomponents ensure compatibility with conventional fittings and attachments. Titanium reinforcements (tensile strength 895-1000 MPa, service temperature up to 400°C) provide superior corrosion resistance, an important factor for aircraft operating in humid and coastal African environments (Carpenter Technology, 2023; MMPDS, 2025).

d. Lifting Device Material

The high-lift devices (such as flaps and slats) primarily use aluminium-lithium alloy 2050-T84, with titanium Ti-6Al-4V used at high-wear and load-bearing interfaces. These materials have been proven in service on platforms such as the Airbus A380 and Boeing 787, which both incorporate similar Al-Li alloys (2099, 2196, and 2198) for critical structures (Airbus, 2025; Boeing, 2024). Regional aircraft like the ATR 72-600 have also demonstrated the alloy's robustness in frequent take-off and landing operations, even on unpaved airfields (ATR Aircraft, 2024).

Compared to conventional aluminium, Al-Li 2050 offers around 10% weight savings (density 2.70 g/cm³ vs 2.78 g/cm³), improved fracture toughness (35-45 MPa√m), and better resistance to fatigue and corrosion (NASA, 2020; MMPDS, 2025). Titanium parts provide additional protection against

abrasion and environmental wear, particularly in dusty or coastal operating conditions. This combination ensures reliability and maintainability—two key requirements for regional service.

e. Interior Systems and Cabin Materials

The cabin interior uses lightweight sandwich panels made from phenolic resin-bonded aramid honeycomb cores with FST (fire, smoke, and toxicity)-compliant face sheets. This configuration follows practices used in the Boeing 787, Airbus A350, and other commercial platforms optimized for modular cabin configurations (Gill Corporation, 2023; FAA, 2025). Depending on the application, the honeycomb core density ranges from 29 kg/m³ for ceiling and wall panels to 96 kg/m³ for floor panels, all meeting FAR 25.853 flammability standards (FAA, 2025; Gill Corporation, 2023).

Seat structures use aluminium-lithium frames with composite components certified under EASA ETSO-C127c specifications (EASA, 2022). All visible surfaces use thermoplastics such as polyphenylsulfone (PPSU), which provide high heat resistance and durability (BASF, 2023). Together, these materials enable a light, robust, and easily reconfigurable interior suitable for mixed passenger and cargo operations typical of low-density African routes.

The table 31 summarises the material properties of the selected materials for the FR25. Table 32 summarises the cost of the materials which becomes important when calculating the cost of the entire aircraft.

Table 31 : Component Material Properties

Material	Density (g/cm ³)	Elastic Modulus (GPa)	Tensile Strength (MPa)	Yield Strength (MPa)	Fatigue Strength (MPa)	Fracture Toughness (MPa√m)	Corrosion Resistance
CFRP T800/Epoxy	1.55	130 (UD), 60 (quasi)	1 200	800	No fatigue limit (comp.)	45–60	Complete immunity
Al–Li 2050-T84	2.70	76	455	395	140 (10 ⁷ cycles)	35–45	Good
Ti-6Al-4V (AMS 4928)	4.43	114	895–1 000	830–950	515 (10 ⁷ cycles)	75–115	Excellent
GLARE-3	1.96	60–70	400–500	—	Excellent	—	Good

Table 32 : Component Material Costs

Material	Cost per kg (USD)	Market Price Range	Volume Considerations	Cost Trend
CFRP T800/Epoxy	50–150	\$50–150 (aerospace grade)	Volume-dependent	Decreasing
Al–Li 2050-T84	25–50	\$25–50	Growing adoption	Stable
Ti-6Al-4V (AMS 4928)	22–66	\$10–30/lb (\$22–66/kg)	Market sensitive	Increasing

GLARE-3	45–65	3–10× aluminium cost	Specialty, low volume	Stable/↓
Phenolic/Aramid Honeycomb Core	8–15	Depends on density	High volume (interiors)	Stable

12. Structural Weight Calculations (Nikunj)

The following calculation for the structural components of the aircraft are based on chapters in the Advanced Aircraft Design book by Egbert Torenbeek (Torenbeek, 2013). However, the Torenbeek formulas are derived for a predominately metal wing. Therefore, to account for the FR25's use of advanced materials and composite structures, material correction factors were derived from published and peer-reviewed performance data of in-service composite aircraft.

a. Material Correction Factors:

- **CFRP Correction Factor ($f_{CFRP} = 0.56$):**

The correction factor for carbon fibre reinforced polymer (CFRP) structures is taken as the ratio of CFRP laminate density to that of aluminium 2024. CFRP T800/epoxy has a typical density of 1.55 g/cm³ (Toray, 2025; Torenbeek, 2013), while Al 2024 is 2.78 g/cm³ (MMPDS-25, 2025):

$$f_{CFRP} = \frac{1.55}{2.78} = 0.5575 \approx 0.56$$

- **Aluminium-Lithium Correction Factor ($f_{Al-Li} = 0.98$):**

For aluminum-lithium alloys, the correction factor is the ratio of Al-Li 2050 density (2.71 g/cm³) to Al 2024 (2.78 g/cm³) (NASA, 2020, Table 2):

$$f_{Al-Li} = \frac{2.71}{2.78} = 0.9748 \approx 0.98$$

- **Titanium Correction Factor ($f_{Ti} = 1.59$):**

For titanium alloys (e.g., Ti-6Al-4V), the corresponding correction factor uses a density of 4.43 g/cm³ (Carpenter Technology, 2023):

$$f_{Ti} = \frac{4.43}{2.78} = 1.594 \approx 1.59$$

b. Fuselage

The following is the equation to calculate the weight of the fuselage:

$$W_{fus} = C_{shell} d_{fus}^2 \{l_{fus} + l_{ref}\} + \Omega_{fl} n_{ult}^{0.5} d_{fus} l_{fus}$$

Where the calibration factors for bodies with a single-deck cabin are $C_{shell} = 60 \text{ Nm}^{-3}$ and $\Omega_{fl} = 160 \text{ Nm}^{-2}$, diameter of fuselage $d_{fus} = 3.1\text{m}$, length of fuselage $l_{fus} = 33.03\text{m}$, $l_{ref} = 1.50\text{m}$, and the ultimate load factor is, $n_{ult} = 3.75$.

Therefore,

$$W_{fus} = 60(3.1^2)\{33.03 + 1.5\} + 160(3.75^{0.5})(3.1)(33.03)$$

$$W_{fus} = 51635.31 \text{ N}$$

$$M_{fus} = \frac{W_{fus}}{g}$$

$$M_{fus} = \frac{51635.31}{9.81} = 5263.54 \text{ kg}$$

$$\therefore M_{fus} \approx 5263.54 \text{ kg}$$

Note,

$$n_{ult} = 1.5 * n_{lim}$$

Where the limit load factor is determined by the FAR regulations (Torenbeek, 2018) and its value is $n_{lim} = 2.5$.

$$\therefore n_{ult} = 1.5 * 2.5 = 3.75$$

However, $M_{fus} \approx 4801.5 \text{ Kg}$ is for a metal fuselage and our aircraft fuselage comprises of a composite structure. The following is the calculation, using the correction factors derived earlier, for a composite fuselage:

Material Breakdown:

- CFRP skins: ~60% of structural weight
- Al-Li 2050-T84 frames and stringers: ~35%
- Titanium fittings/cutout reinforcements: ~5%

Therefore, adjusted fuselage weight,

$$M_{fus,composite} = M_{fus} \times (\%CFRP * f_{CFRP} + \%Al - Li * f_{Al-Li} + \%Ti * f_{Ti})$$

$$M_{fus,composite} = 5263.54 \times (0.6 * 0.56 + 0.35 * 0.98 + 0.05 * 1.59)$$

$$M_{fus,composite} \approx 3992.39 \text{ kg}$$

c. Wing

The weight of the wing is calculated using the following equation:

$$W_w = 0.0013 n_{ult} (W_{MZF} W_{MTO})^{0.5} \left(\frac{n_{cp} b_w}{b_{ref}} \right) \frac{A_w}{\left(\frac{t}{c} \right)_w (\cos^2 \Lambda_w)} + \Omega_s S_w$$

Where, weight of aircraft at zero fuel condition $W_{mzf} = 225747.38 \text{ N}$, maximum take-off weight $W_{mto} = 287210.41 \text{ N}$, the dimensionless lateral coordinate of the centre of pressure $n_{cp} = 0.439$, wingspan $b_w = 28.65 \text{ m}$, aspect ratio $A_w = 11.4$, average thickness ratio weighted by the chord $\left(\frac{t}{c}\right)_w = 0.13$, sweep angle $\Lambda_w = 0$, and wing planform $S_w = 72 \text{ m}^2$. The factors $b_{ref} = 100 \text{ m}$ and $\Omega_s = 210 \text{ Nm}^{-2}$, and their values are dependent on structural topography, materials, geometry, and actuation of high-lifting devices, scale effects, and many other factors (Torenbeek, 2013).

$$W_w = 0.0013 (3.75)(225747.38 * 287210.41)^{0.5} \left(\frac{0.439 * 28.65}{100} \right) \left(\frac{11.4}{(0.13)(\cos^2 0)} \right) + (210)(72)$$

$$W_w = 28811.05 \text{ N}$$

$$M_w = \frac{W_w}{g}$$

$$M_w = \frac{28811.05}{9.81} = 2936.90 \text{ kg}$$

$$\therefore M_w \approx 2936.9 \text{ kg}$$

$M_w \approx 2936.90 \text{ kg}$ is for a metal wing. For a composite wing:

Material Breakdown:

- CFRP skins, spars, wings box: ~70% of structural weight
- Al-Li 2050-T84 ribs and floor beams: ~25%
- Titanium fittings: ~5%

Therefore, adjusted fuselage weight,

$$M_{w,composite} = M_w \times (\%CFRP * f_{CFRP} + \%Al - Li * f_{Al-Li} + \%Ti * f_{Ti})$$

$$M_{w,composite} = 2936.90 \times (0.7 * 0.56 + 0.25 * 0.98 + 0.05 * 1.59)$$

$$M_{w,composite} \approx 2104.29 \text{ kg}$$

d. Tail/Empennage

The weight of the empennage considers the horizontal tail weight as a fraction of the wing weight and the vertical tail as a fraction of the fuselage weight (Torenbeek, 2013). It is calculated using the following equation:

$$W_{emp} = r_h W_w + r_v W_{fus}$$

Where $r_h = 0.10$ and $r_v = 0.07$ are factors that are subject to calibration with similar aircrafts.

$$W_{emp} = 0.1 (28814.05) + 0.07(47102.35)$$

$$W_{emp} = 6178.59 \text{ N}$$

$$M_{emp} = \frac{W_{emp}}{g}$$

$$M_{emp} = \frac{6178.59}{9.81} = 629.83 \text{ kg}$$

$$\therefore M_{emp} \approx 629.82 \text{ Kg}$$

$M_{emp} \approx 629.82 \text{ kg}$ is for a metal wing. For a composite empennage:

Material Breakdown:

- CFRP primary surfaces: ~65% of structural weight
- Al-Li 2050-T84 substructures: ~30%
- Titanium reinforcements: ~5%

Therefore, adjusted fuselage weight,

$$M_{emp,composite} = M_{emp} \times (\%CFRP * f_{CFRP} + \%Al - Li * f_{Al-Li} + \%Ti * f_{Ti})$$

$$M_{emp,composite} = 629.82 \times (0.65 * 0.56 + 0.30 * 0.98 + 0.05 * 1.59)$$

$$M_{emp,composite} \approx 464.49 \text{ kg}$$

e. Landing Gear

The weight of the landing gear is calculate using the following formula:

$$W_{lg} = W_{MTO} \left(0.039 \left(1 + \frac{l_{fus}}{l_{ref}} \right) \right)$$

Where $l_{ref} = 1100m$.

$$W_{lg} = 287210.41 \left(0.039 \left(1 + \frac{33.03}{1100} \right) \right)$$

$$W_{lg} = 11537.55N$$

$$M_{lg} = \frac{W_{lg}}{g}$$

$$M_{lg} = \frac{11537.55}{9.81} = 1176.10 \text{ kg}$$

$$\therefore M_{lg} \approx 1176.10 \text{ Kg}$$

f. Power Plant and Pylons

The weight of the powerplant is calculated as follows:

$$W_{pp} = 0.25T_{TO} + N_{eng}(W_{pp})_{fix}$$

Where thrust of the one engine $T_{TO} = 52,706.5N$, number of engine $N_{eng} = 2$, and $(W_{pp}) = 8000 \text{ N}$.

$$W_{pp} = 0.25(52706.5) + 2(8000)$$

$$W_{pp} = 29176.625 \text{ N}$$

$$M_{pp} = \frac{W_{pp}}{g}$$

$$M_{pp} = \frac{29176.625}{9.81} = 2974.17 \text{ kg}$$

$$\therefore M_{pp} \approx 2974.2 \text{ Kg}$$

g. Systems, Furnishing, and operational items

Weight of systems:

$$W_{sys} = 270 l_{fus} d_{fus} + 150 l_{fus}$$

$$W_{sys} = 270(33.03)(3.1) + 150 (33.03)$$

$$W_{sys} = 32600.61 \text{ N}$$

Weight of furnishings:

$$W_{fur} = 12 l_{fus} d_{fus} (3d_{fus} + 0.5N_{deck} + 1) + 3500$$

Where N_{deck} is number of cabin decks, which is 1.

$$W_{fur} = 12(33.03)(3.1)(3(3.1) + 0.5 + 1) + 3500$$

$$W_{fur} = 16770.13N$$

Weight of operational items:

$$W_{oper} = 400N * N_{seats}$$

$$W_{oper} = 400 * 88 = 35200 N$$

Therefore,

$$W_{sys+fur+oper} = 32600.61 + 16770.13 + 35200$$

$$W_{sys+fur+oper} = 84,570.74N$$

$$M_{sys+fur+oper} = \frac{W_{sys+fur+oper}}{g}$$

$$M_{sys+fur+oper} = \frac{84,570.74}{9.81} = 8620.87 kg$$

$$\therefore M_{sys+fur+oper} \approx 8620.87 kg$$

Since our aircraft incorporates modern, lightweight interior materials, we apply a 5% weight reduction to the systems, furnishing, and operations baseline. This adjustment is supported by studies showing that phenolic and aramid honeycomb panels—widely used for floors, walls, and monuments—are on average 15% lighter than older panel technologies (FAA, 2000; Gill Corporation, 2002). Similarly, the use of aluminium-lithium alloys in seat frames and tracks typically results in an 8% weight saving compared to conventional aluminium. Considering that approximately half of the systems/furnishing/operations weight benefits from these advanced materials with individual savings of 15–30%, while the other half remains unchanged, the weighted average group-level savings realistically falls in the 4–8% range. Therefore, a 6% overall reduction provides a conservative and defensible estimate for the impact of modern aircraft interior materials.

$$M_{sys+fur+oper,composite} \approx 8281.5 * 0.94$$

$$\therefore M_{sys+fur+oper,composite} \approx 8103.61 kg$$

Table 33 : Structural Component weights (adjusted for material composition)

Structural Component	Mass of Component (kg)
Fuselage	3992.39
Wing	2104.29
Tail/Empennage	464.49
Landing Gear	1176.10
Power Plant and Pylons	2974.2
System, Furnishings, and Operations	8103.61
Batteries	800
Total	19615.08

13. Air Flight Envelope (Nikunj):

a. Introduction and Purpose:

The flight envelope or the V-N diagram represents the operational limits where the aircraft can safely operate. It defines the combinations of airspeed and load factor that establish the structural design requirements of an aircraft. For the FR25, the V-N diagram was constructed at sea level based on guidelines provided by the Federal Aviation Regulations (FAR) part 25.

b. Manoeuvre Load Factors

Note n_p and n_u are Proof and Ultimate Load Factors respectively.

Table 34 : Manoeuvre Load Factors

Load factor n	Equation	Calculated Value	Description
n_{max}	$n_{max} = 2.1 + \frac{24000}{W_{MTO} + 10000};$ (W_{MTO} in lbs) (note $2.5 \leq n_{max} \leq 3.8$)	$n_{max} = 2.42$ $n_{max} \approx 2.5$ (rounded for certification)	Maximum Positive Load Factor
n_{min}	$n_{min} = -1$	$n_{min} = -1$	Maximum Negative Load Factor
n_p	$n_p = 1.25n_{max}$	$n_p = 3.125$	Proof Load Factor
n_u	$n_u = 1.5n_{max}$	$n_u = 3.75$	Ultimate Load Factor

c. Manoeuvre Speeds

Table 35 : Manoeuvre Speeds

Manoeuvre Speeds	Equation	Calculated Value	Description
V_a	$V_a = \sqrt{\frac{2W_{MTO}n_{max}}{\rho SC_{L_{max}}}}$	$V_a = 110.418s^{-1}$ $\approx 110.42 ms^{-1}$	Design Positive Corner Speed
V_c	(Cruise speed)	$V_c = 667 kmph$ $= 185.28 ms^{-1}$	Design Cruise Speed
V_d	$V_d = 1.4V_c$	$V_d = 259.39 ms^{-1}$	Design Dive Speed
V_e	$V_e = V_c$	$V_e = 667 kmph$ $= 185.28 ms^{-1}$	Design Cruise Speed
V_f	$V_f = \sqrt{\frac{2W_{MTO}n_{min}}{\rho SC_{L_{min}}}}$	$V_f = 98.761 ms^{-1}$ $\approx 98.76 ms^{-1}$	Design Negative Corner Speed

d. Gust Load Factors and Speeds

For gust load factors the aircraft is first assumed to be in the steady level flight. The gust speeds (u) are 18.47m/s, 13.65m/s, and 6.83m/s at 25,000ft as given by the FAR (Part 25.341). The initial load factor is 1. Upon entering gust, the new load factor has the following formula:

$$n_{new} = 1 \pm n_{gust}$$

Where the gust load factor is,

$$n_{gust} = \frac{1}{2} \frac{\rho V S C_{l_\alpha} u}{W}$$

e. Flight Envelope/V-N Diagram

The dashed lines represent the gust lines. The red line represents the flight envelope of our aircraft.

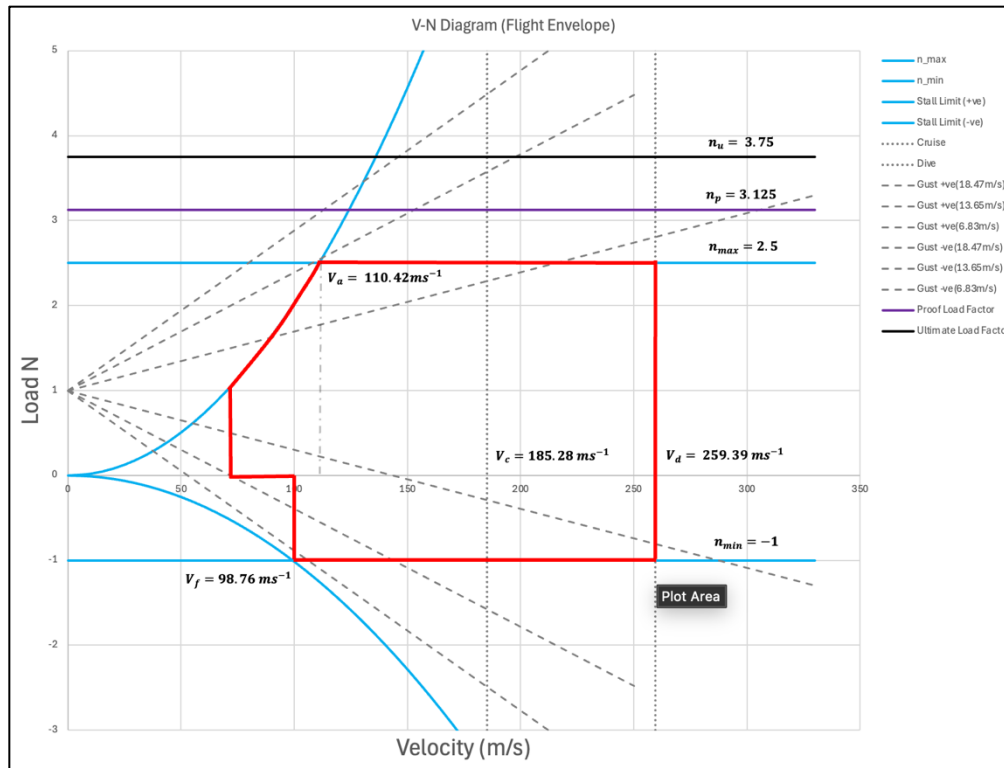


Figure 70: Flight Envelope

14. Avionics Systems and Cockpit Design (Florian)

a. Introduction

Modern aircraft are equipped with onboard avionics system that play a pivotal role in the ensuring the comfort and safety of passengers. The avionics system integrates communication, navigation, display management and control systems. The evolution of avionics technology has transformed aircraft operations from analog instrument-based systems to fully integrated digital architectures.

Avionics system serves multiple critical functions. They enhance the safety of flight through real-time monitoring and alerting, reducing pilot workload through automation and enable advanced capabilities such as all-weather operations. The certification of these systems must comply with the FAR/CS-25 for transport aircraft, DO-178C for software development and DO-254 for hardware design assurance.

Human factors considerations are one of the highest priorities in avionics design. It ensures that pilots have high situational awareness to improve decision-making processes. The cockpit design for the FR25 will consolidate information and present only important or critical information to the pilots.

This section presents the avionics architecture for a modern transport aircraft that incorporates state-of-the-art systems including an advanced turbulence detection and mitigation system designed to enhance passenger safety and comfort. It will also ensure reduced structural loading on airframes.

b. Avionics System Architecture

The FR25 will employ the Integrated Modular Avionics (IMA) architecture based on ARINC 653. This architecture provides significant advantages over traditional federated systems while maintaining the safety levels required for critical flight operations. A key advantage of the IMA is its reduced weight compared to traditional system architectures.

i. IMA Architecture

The IMA Architecture consists of several key elements which includes the core processing modules (CPM), Partition Management and Network architecture.

The IMA consists of four CPMs for redundancy as seen in figure 71. The CPMs provide the computational backbone for the avionics system. Each CPM shares from the computer recourse which can house multiple avionics applications simultaneously (Prisaznuk, P. J.,2008). The ARINC 653 operation system ensure that each partition receives its allocated processing time and memory with fault prevention.

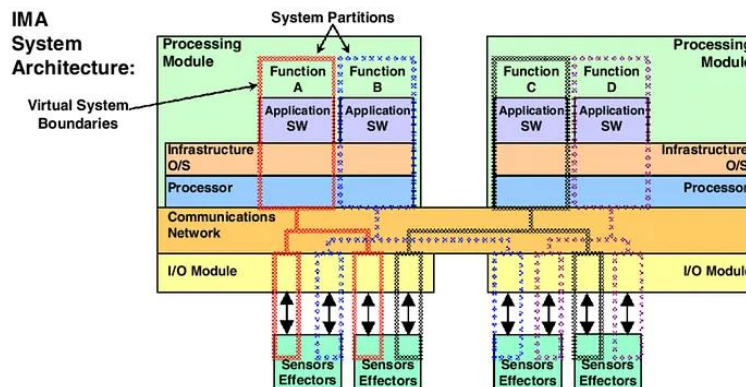


Figure 71: IMA Architecture Block Diagram (Kose, 2019)

Applications are segregated into partitions based on criticality levels aligned with the DO-178C design assurance levels (DAL). Each partition operates as a virtual flight computer that has guaranteed access to processing time and protected memory space. The DAL includes:

- DAL A: Flight control functions and primary navigation
- DAL B: Communication systems and secondary navigation aids
- DAL C: Flight management optimization functions

Lastly, the Network Architecture employs ARINC 664 Part 7 (AFDX) as the primary data network. AFDX provides deterministic, high bandwidth communication between avionics modules with guaranteed latency and bandwidth allocation (J Li, 2009).

ii. Safety and Redundancy

To meet the standards of FAR 25.1309 on equipment systems and installations, the system will implement a quadruple redundancy for critical functions using a quad-redundant configuration with cross-channel monitoring as seen in figure 72. The sensor cable network is designed using a one-to-four distribution scheme allowing each FCC to receive identical sensor signals. The voter is deployed upstream of each actuator. it receives control commands for the respective actuator from all four flight control nodes via the CAN bus and selects the final command using redundancy management algorithms (Wang, A, 2025). The system architecture achieves a failure probability of less than 10^{-9} per flight hour for failure conditions.

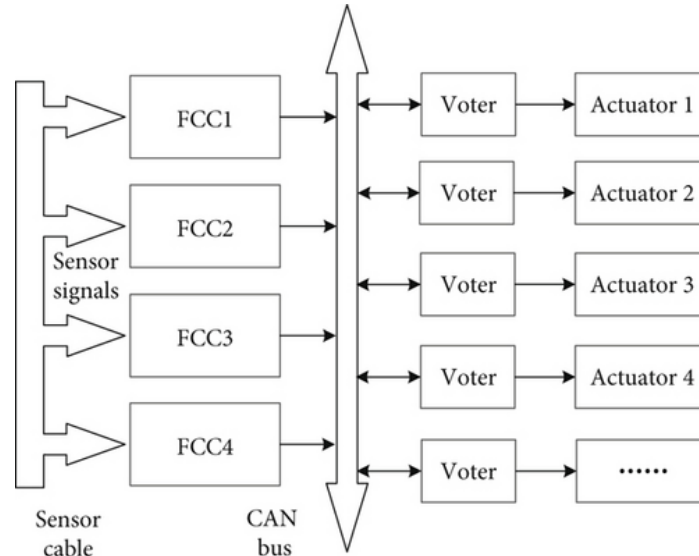


Figure 72: System diagram of a quad-redundant flight control system (Wang, A, 2025)

iii. Resource Allocation

The processor will be maintained below 70% during normal operations with 30% for reserve capacity for future growth allocation.

c. Data Communication and Processing

Data communication and processing within the avionics system of the FR25 will employ multiple standards. These standards are optimized for different applications as well as legacy compatibility requirements.

i. ARINC 429 Standard

ARINC 429 will be utilized in most systems onboard the FR25. ARINC 429 continues to remain the most reliable point-to-point data bus for sensors, actuators and standalone equipment (O Saotome, 2021). The unidirectional protocol features one transmitter and up to 20 receivers per bus using 32-bit words transmitted at either 12.5 kbits/s or 100 kbits/s. Low speed transmission will be used for navigational data while high speed transmission will be utilized for flight controls and dynamic data.

The FR25 will utilize ARINC 429 for the:

- Inertial Reference System position and Attitude Data
- Radio Altimeter Altitude

- Weather Radar Information
- Air Data Computer Outputs
- Engine indication parameters
- Landing gear and flight control surface position feedback

ii. ARINC 664 (AFDX) Network

The AFDX network provides high-speed, deterministic communication for the IMA modules (Pasquier,2025). AFDX operates at 100 Mbps full duplex extending ethernet with aviation specific enhancements. These include bandwidth allocation for traffic adjustments, redundancy to ensure continued operation after a single network failure, guaranteed message transmission timing.

iii. Data Processing Architecture

The Flight Management Computer (FMC) serves as the central processing unit for navigation, flight planning and performance optimization. The FR25 will employ a dual-FMC operation in active- standby mode with automatic switchover. The data will be optimized with a Kalman filter before going to the CDU.

iv. Data Format Standards

Data formatting needs to be standardized to ensure interoperability:

- Binary Coded Decimal (BCD) for display data
- Two's complement binary for computational data
- Binary Angular Measurement (BAM) for angles
- ARINC 424 format for navigation database encoding

d. Communication System

The FR25 communication system provides comprehensive connectivity within and outside the aircraft. Communication outside the aircraft include communication with air traffic control (ATC), Airline operation centres and other aircrafts during all phases of flight. Figure 73 shows an overview of all avionics systems onboard the FR25.

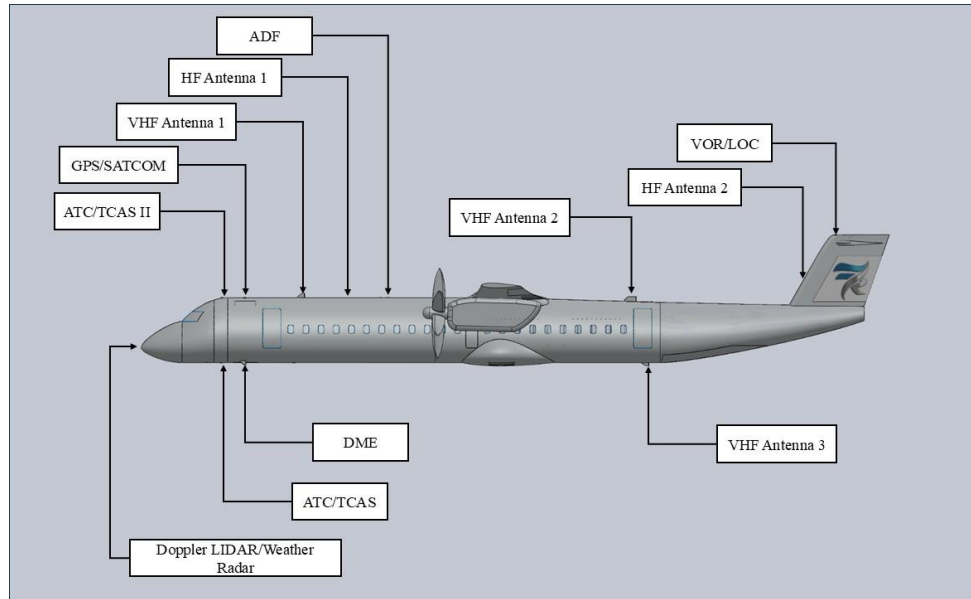


Figure 73: FR25 onboard avionics

i. VHF Communication System

The FR25 will be equipped with three independent VHF transceivers that cover the frequency range of 118.000 – 136.975 MHz with 8.33 KH channel spacing. This follows ICAO requirements for European Airspace. The system provides line-of-sight (LOS) communication with a range of 200 nautical miles at cruise altitude (M Tooley, 2024). The communication management function allows frequency pre-selection and automatic tuning for flight waypoints.

ii. HF Communication System

There will be two onboard HF transceivers operating in the 2-30 MHz band enabling long-range communication beyond VHF LOS coverage. The system will employ automatic antenna tuning across the HF spectrum. While HF usage has declined with Satellite systems, it remains a required backup for oceanic operations.

iii. Satellite Communication (SATCOM) System

SATCOM system provides global connectivity using the Inmarsat and Iridium satellites networks (M Tooley, 2024). L-band antennas will be mounted on the upper fuselage to maintain connectivity during the flight envelope. The system supports voice communication, high-speed data links for cockpit and cabin internet connectivity and integration with Aircraft Communications Addressing and Reporting System (ACARS) for message delivery.

iv. ACARS

ACARS provides automatic datalink messaging between the aircraft and ground stations. The system operates through multiple media. The message categories include:

- Airline Operational Control (AOC)
- ATC
- Administrative Communications (AAC)

The system automatically transmits OOOI events for flight tracking and schedule management. Engine parameters and aircraft health data are continuously transmitted for real-time condition monitoring.

v. Emergency Communication

In the event of an emergency, the FR25 is equipped with dedicated emergency frequencies, emergency locator transmitter (ELT) with 406 MHz satellite alerting and 121.5 MHz Homing signal and cockpit voice recorder with 25-hour capacity interfaced to all communication systems.

e. Navigation System

i. Flight Management System (FMS)

The dual FMS serves as the main navigation system integrating all navigation sensors and providing guidance to the autopilot. The FMS architecture comprises of four main subsystems as seen in figure 74.

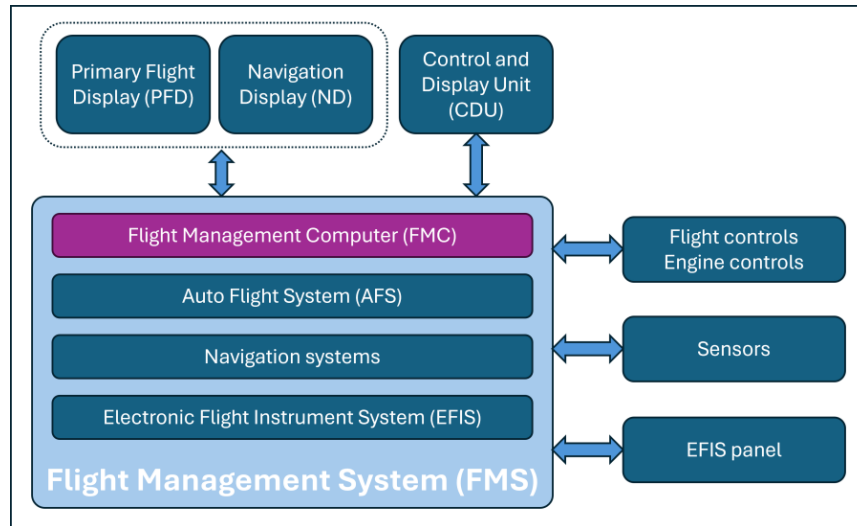


Figure 74: FMS subsystems (Skybrary,2020)

1. Flight Management Computer (FMC)

The FMC performs navigation sensor integration using optimal algorithms and flight plan management procedures to optimize for minimum fuel or cost.

2. Auto Flight System (AFS)

The AFS generates steering commands for lateral navigation (LNAV) and vertical navigation (VNAV). It is integrated with autopilot for automatic flight control.

3. Navigation systems

The system uses ARINC 424- compliant navigation database updated on a 28-day cycle. It contains waypoints, airways, procedures and navigational aid information.

4. *Electronic Flight Instrument System (EFIS)*

The FR25 contains dual EFIS to provide pilots with interface for flight plan entry and modification. Pilots will monitor performance data, navigation and system status.

ii. *Primary Navigation Sensors*

There will be two main primary navigation systems on board the FR25. Primary Navigation serves as an important feature for safety and situational awareness for pilots.

1. *Inertial Reference System (IRS)*

The first primary navigation sensor onboard the FR25 is the IRS. The IRS will consist of three independent inertial reference units (IRU). The IRUs will provide position, velocity, attitude and heading data through the ring laser gyroscope and accelerometers (Skybrary,2025). The IRS operates independently without external reference.

2. *Global Navigation Satellite System (GNSS)*

The FR25 contains two GNSS receivers for worldwide coverage.

iii. *Ground Based Navigational Aids*

1. *VOR/DME Systems*

The FR25 contains three VOR receivers that will provide lateral navigation reference from ground-based VHF Omnidirectional Range Stations. Two DMEs interrogators provide slant range distance to DME ground stations. The system allows the FR25 conventional VOR airway navigation and VOR/DME approach where applicable (Winick, A. B, 2005).

2. *Instrument Landing System (ILS)*

Dual ILS receivers onboard the FR25 provide precision approach guidance through localizer receivers in the 108.1 MHz range for lateral guidance with $\pm 2.5^\circ$ course deviation. Glideslope receivers in the 335 MHz range provide 3° vertical glide path guidance (FAA, 2025). Marker beacon receivers detect outer, middle, and inner markers providing position confirmation during approach.

iv. *Safety Systems*

1. *Traffic Collision Avoidance System (TCAS)*

TCAS II version 7.1 monitors airspace using interrogations of other aircraft transponders. This ensure that the FR25 is protected against mid-air collisions independent of ATC. The System generates Traffic Advisories (TA) when aircraft enters proximity thresholds. Resolution Advisories (RA) command specific vertical manoeuvres when collision exceeds threshold, The TCAS RA and RA bounds can be seen in figure 75.

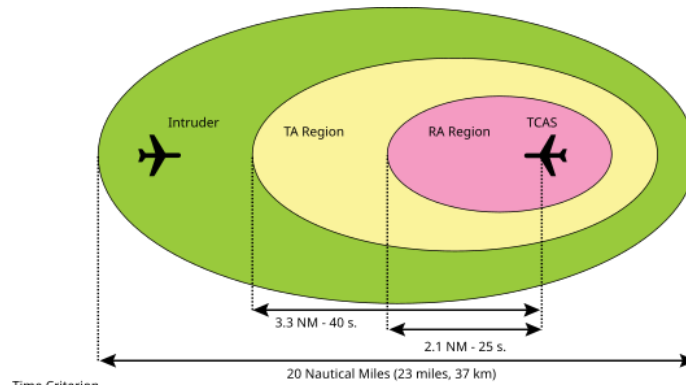


Figure 75: TCAS subsystem

2. Enhanced Ground Proximity Warning System (EGPWS)

The EGPWS integrates GPS position with a worldwide terrain and obstacle database to provide predictive terrain conflict alerting (Ref).

3. Weather Radar

The nose-mounted X-band weather radar provides real-time detection of precipitation and convective activity ahead of the aircraft (Leonardo, 2022). The system features:

- Automatic antenna stabilization referenced to the aircraft pitch and roll
- Detection range exceeding 320 nautical miles
- Turbulence detection mode using Doppler processing to identify velocity variations
- Windshear detection during approach and take-off
- Ground mapping mode for navigation backup

Display modes include weather intensity (green, yellow, red, magenta coding), turbulence areas, and terrain mapping with weather overlay capability as seen in figure 76.

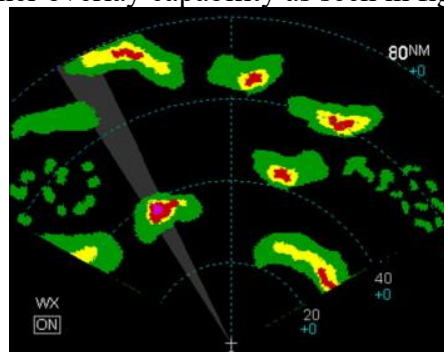


Figure 76: Weather Radar Display onboard the FR25 (Wiggins, 2014)

v. Advanced Turbulence Detection and Mitigation System

A key technological breakthrough onboard the FR25 is its advanced turbulence detection and mitigation system. The high likelihood of adverse weather conditions in the future due to global warming means that aircraft need to be fitted with robust weather and turbulence detection

systems. The system aims to enhance passenger safety and comfort while reducing structural loads. The FR25 incorporates state-of-the-art turbulence detection with automated load alleviation.

1. **Forward-Looking LIDAR System**

The nose of the FR25 will be mounted with a doppler LIDAR system that provides pilot with clear air turbulence capability not normally possible with conventional radar (RTCA, 2020). The system emits laser pulses at 1.55 micrometre wavelengths, detecting backscatter from atmospheric aerosol particles. The doppler shift analysis will indicate air velocity variation indicating if there is turbulence. The table 36 provides specification for the system.

Table 36 : Doppler Lidar Specification (Boeing, 2017)

Feature	Specifications
Detection Range	14 – 17.5 km
Laser Output	3.3W Continuous Operation
Beam Diameter	150mm + 50mm dual-beam configuration
Weight	84 kg
Update Rate	1 Hz

The LIDAR scans ahead of the aircraft along the flight path, measuring vertical and horizontal wind components. Clear air turbulence appears as sharp velocity gradients in the measured airflow. The system provides advance warning through cockpit audio and visual alerts, enabling crew preparation actions: seat belt sign activation, cabin crew securing, and possible flight path adjustment.

The system that will be mounted onto the FR25 comes from a collaboration between Japan Aerospace Exploration Agency (JAXA) and Boeing. Since 2010, the two companies have collaborated and tested LIDAR technology on board. The product is currently under FAR review and will likely be airworthy before the production of the first FR25 prototype.

2. **Active Gust Load Alleviation System (GLAS)**

The FR25 features an active GLAS that reduces turbulence effects by up to 80% through coordinated flight control surface deflection. The system architecture comprises of the following:

Forward Pressure Sensors: The wings leading edge will consist of a five-hole differential pressure probe that measures approaching airflow disturbances. At cruise speeds, this probe provides approximately 100 milliseconds of advance warning before the turbulence reaches the wings (F,Xie, 2025).

Turbulence Cancelling Control System: The FR25 will have a dedicated processor to analyse the sensor data and LIDAR inputs to predict vertical acceleration, pitch and roll disturbances caused by incoming turbulence. The system then calculates optimal control surfaces deflection to generate counteracting aerodynamic forces.

Flight Control Integration: The GLAS integrates with the fly-by-wire flight control system through the primary flight control computers. Control logic ensures that pilot inputs and autopilot commands take precedence over turbulence mitigation

3. System Operating Modes

The FR25's turbulence detection will operate in different modes depending on the frequency and severity of turbulence. There will be 3 main operating modes. These are the Detection Mode, Mitigation Mode and Strong Turbulence Mode.

Detection Mode: In detection mode, the LIDAR continuously scans for turbulence ahead of the aircraft. When significant turbulence is detected beyond the gust alleviation system's response capability, the flight crew receives advanced warning. Upon detection, the cockpit will display the turbulence location, intensity and distance.

Mitigation Mode: Mitigation mode operates when turbulence is unavoidable. In such a scenario, the GLAS will activate automatically.

Strong Turbulence Mode: In severe cases of turbulence exceeding the systems 80% mitigation capability, the following will occur:

- Maximum available control authority applied for load reduction
- Structural load limits monitored to prevent control-induced overstress
- Wing bending moment reduction of 30-40% protects structure
- Pilot retains override capability through normal flight control inputs

4. Performance Benefits

The Advanced Turbulence Detection and Mitigation System operate continuously during flight with no increase in pilot workload. The FR25 has built in test capability that verifies the system health before each flight. The LIDAR and GLAS components are certified to DO-178 DAL B, with system failure reverting to conventional turbulence handling (Ref)/ The integrated turbulence system provides multiple operational advantages such as:

Passenger Comfort improvement: Reduction in vertical acceleration during moderate turbulence.

Injury Prevention: Advanced warning enables cabin crew and passengers to be informed of turbulence well ahead of time. Severity knowledge will possibly reduce turbulence-related injuries.

Fuel Efficiency: 5-10% fuel savings by eliminating altitude changes and routing deviations for turbulence avoidance.

This system represents a significant advancement in aircraft safety and passenger comfort, addressing one of aviation's most persistent operational challenges (Golding, 2002).

f. Cockpit Layout and Design

The cockpit layout employs a modern glass cockpit configuration optimized for crew efficiency, situational awareness and safety during operations. The touchscreen displays reduced the number of mechanical buttons needed on board the FR25. The design of the cockpit is seen in figure 77.



Figure 77: Cockpit layout on FR25

i. **Primary Flight Display (PFD)**

Each pilot station will feature a 10-inch diagonal LCD PFD that displays integrated flight instrumentations. The PFD consolidates the traditional “6-T” instrumentation into a single intuitive display

ii. **Multifunction Display**

Each pilot has access to a 12-inch diagonal MFD providing navigation and systems information. The display supports multiple selectable formats. The MFD includes:

Navigation Display: A moving map presentation with aircraft position, flight plan route with waypoints and airways, terrain elevation shading with peak heights, weather radar overlay with intensity coding, traffic display from TCAS and ADS-B.

Turbulence Detection Display: A dedicated page shows the forward LIDAR turbulence detection with color-coded intensity regions, distance markers to turbulence areas, recommended avoidance headings/altitudes, and GLAS system status.

Weather Radar: Full-screen weather display showing precipitation intensity, turbulence detection regions, windshear alerts during approach, and terrain with weather overlay.

Systems Synoptics: Hydraulic, electrical, fuel, pneumatic, and flight control system pages showing component status and flow diagrams with color-coded status.

Checklist Display: Electronic checklist presentation with automatic item progression and completion confirmation.

iii. *Engine Indicating and Crew Alerting system (EICAS)*

Two 10-inch displays in the centre instrument panel provide engine parameters and crew alerting functions. Alert messages appear in priority order with system synoptic recommendations. The system presents an electronic checklist procedure automatically.

15. Constraint Analysis (Florian)

Constraint analysis is an important step to determine feasible combinations of wing loading (W/S) and thrust-to-weight (T/W) that satisfy key performance requirements across all phases of flight. The constraints of FR25 has been evaluated based on take-off, climb, cruise, landing and manoeuvre conditions. The approach verifies the selection of our wing and engine sizing to safely and efficiently meet all regulatory and mission objectives.

a. Aircraft Specifications

The FR25 specifications are shown in table 37.

Table 37 : FR25 Specifications for constraint analysis

Specification	Data
Take-Off Weight	31000.18 kg
Take-Off Speed	64 m/s
Cruise Speed	185 m/s
Cruise Altitude	25,000 ft
Take-Off Distance	1700 m
Wingspan	28.7 m
Wing Area	72 m ²
Engine Power	3382 Kw

b. Take Off

During take-off, the FR25 will need enough runway distance to safely accelerate, lift off, and reach a height of 50ft within 1,700 meters. Since the aircraft is at its maximum weight during take-off, the engine and wing sizing must be properly sized. The take-off constraint is calculated by the required thrust-to-weight ratio using the following equation:

$$\frac{T_{V_2}}{W_{TO}} = 0.78 \left(\frac{N_{eng}}{0.89 N_{eng} - 1} \frac{\frac{W_{TO}}{b_w^2}}{l_{TO} \rho_{TO} g} \right)^{0.5} + \frac{42(m)}{l_{TO}} \frac{N_{eng}}{0.89 N_{eng} - 1}$$

The equation ensures that the FR25 meets regulatory take-off requirements by providing the minimum thrust-to-weight ratio to achieve safe climb-out. Figure 78 shows the constraint for take-off where values above the gradient indicate safe operation

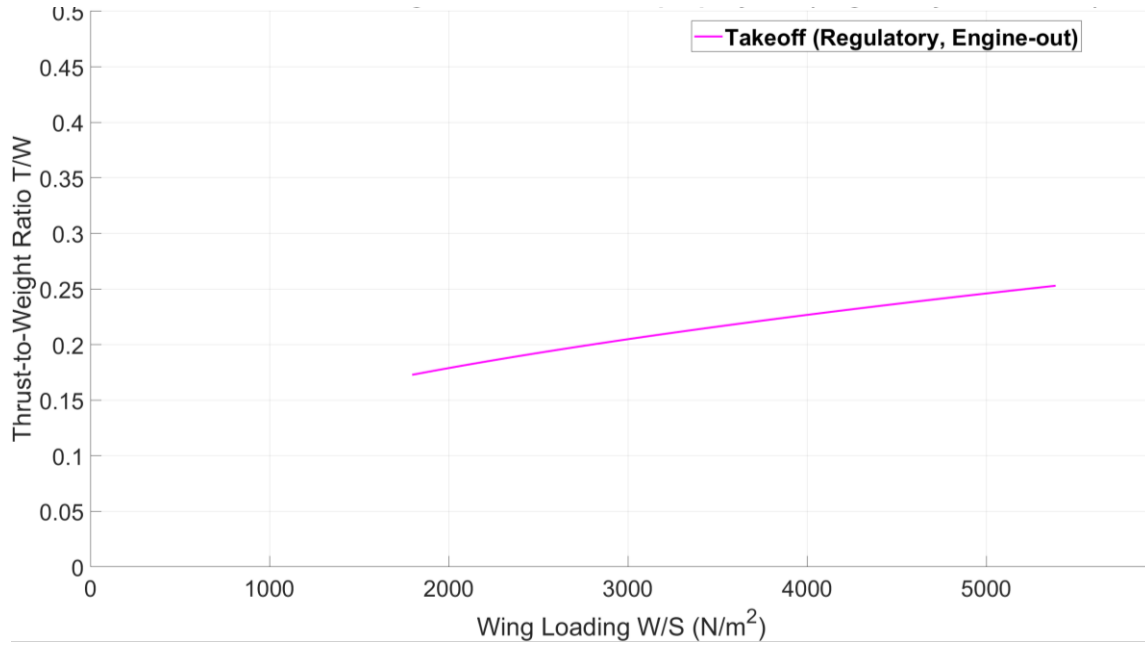


Figure 78: Take off constraint of FR25

c. Climb

During climb, the primary objective is to ensure the aircraft has sufficient thrust to safely clear obstacles such as nearby buildings or populated areas. According to FAA FAR 25.121, for a two-engine aircraft, the minimum required climb gradient during the second segment climb is 2.4%. To satisfy this, the thrust-to-weight ratio must meet or exceed approximately 0.186. This relationship is expressed as:

$$\gamma_2 = \frac{(1 - N_{eng}^{-1})T_{V_2}}{W_{TO}} - \left(\frac{C_D}{C_L}\right)v_2$$

The equation is used to plot the Climb constrain is figure 79 where values above the gradient indicate safe operation:



Figure 79: Climb constraint of FR25

d. Cruise

The cruise phase represents the longest portion of flight for commercial aircraft and therefore has a substantial impact on overall mission efficiency and fuel consumption. The cruise altitude and speed significantly influence powerplant thrust requirements. For the FR25, cruise operations are planned at FL300 and Mach 0.7. To determine the thrust-to-weight ratio required to sustain cruise at these conditions, Raymer Equation 9.3 is employed:

$$T_{TO} = \frac{T/\delta}{\tau} = \frac{\bar{D}/\delta}{\bar{\tau}} = \frac{1}{\bar{\tau}} \left[q_{sl} \left\{ (C_{D\rho} S)_{fix} + \widetilde{C_{D\rho}} S_w \right\} + K_L \frac{(W_{MTO}/\delta)^2}{q_{sl} S_w} \right]$$

Figure 80 depicts the cruise constraint plotted on the wing loading versus thrust-to-weight diagram. The region above the cruise curve satisfies the cruise requirements.

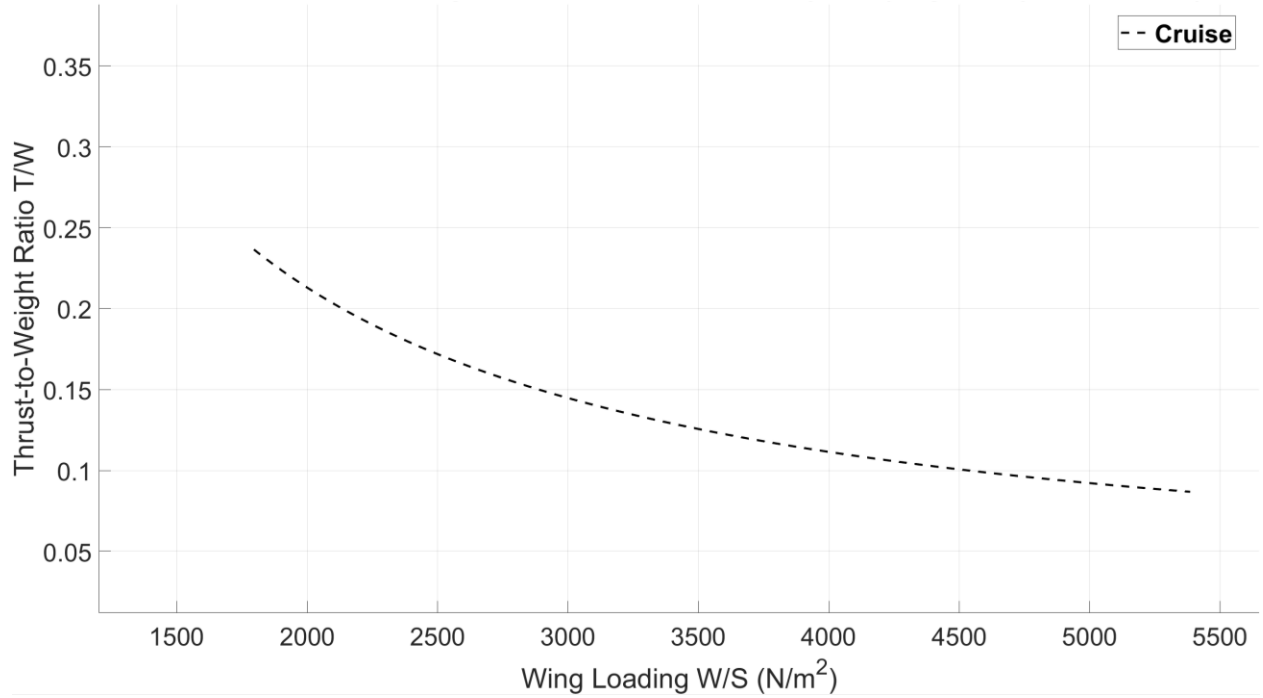


Figure 80: Cruise constraint of FR25

e. Manoeuvre

The manoeuvre constraint ensures the aircraft can perform sustained turns at a specified load factor without loss of altitude or airspeed, which is critical for operational safety and flexibility. For the FR25, a load factor of $n=2.5g$ is used to evaluate manoeuvrability, representing typical operational requirements for transport category aircraft. The sustained turn constraint is given by:

$$\frac{T}{W} = n \left[\frac{C_{D0}}{C_L} + K n^2 C_L \right]$$

Figure 81 displays the 2.5g sustained turn constraint plotted against wing loading. The curve exhibits a monotonically decreasing trend as wing loading increases from approximately 1,500 N/m² to 5,500 N/m².

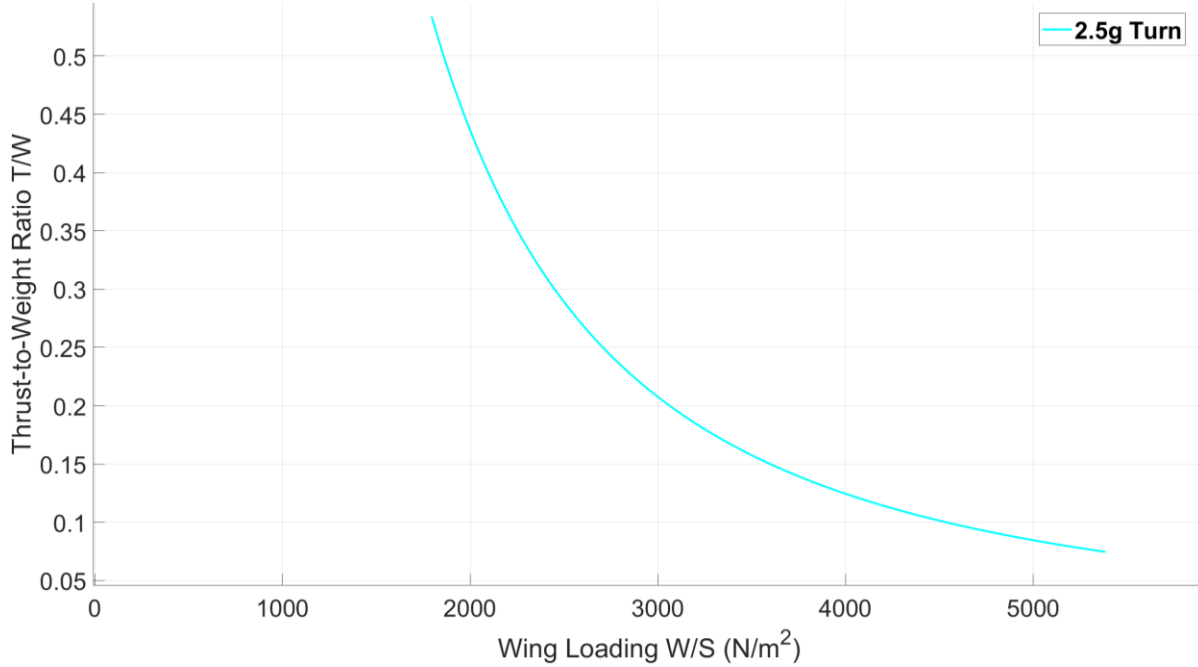


Figure 81: Manoeuvre constraint of FR25

f. Landing

During landing, the FR25 deploys landing gear, full flaps, and slats, which increase maximum lift coefficient and drag. These aerodynamic modifications enable the aircraft to land at lower speeds. The increased drag allows higher engine power settings, enhancing engine responsiveness during the critical landing phase. Since fuel is largely consumed by this point in the mission, the aircraft is significantly lighter than at take-off, reducing landing distance requirements.

The landing constraint is therefore governed not by runway distance, but by the stall speed limitation. The relationship between wing loading and stall speed is critical because an inadvertent stall during landing would be catastrophic. The landing constraint establishes a maximum allowable wing loading for a given stall speed and landing configuration, evaluated at sea-level conditions:

$$\frac{W}{S} \leq 0.335 \rho_{sl} V_{stall}^2 C_{L,max}$$

Figure 82 presents the regulatory landing constraint as a vertical line on the wing loading versus thrust-to-weight diagram at approximately 3,500 N/m². This vertical constraint represents the maximum allowable wing loading determined by the stall speed limitation during landing operations.

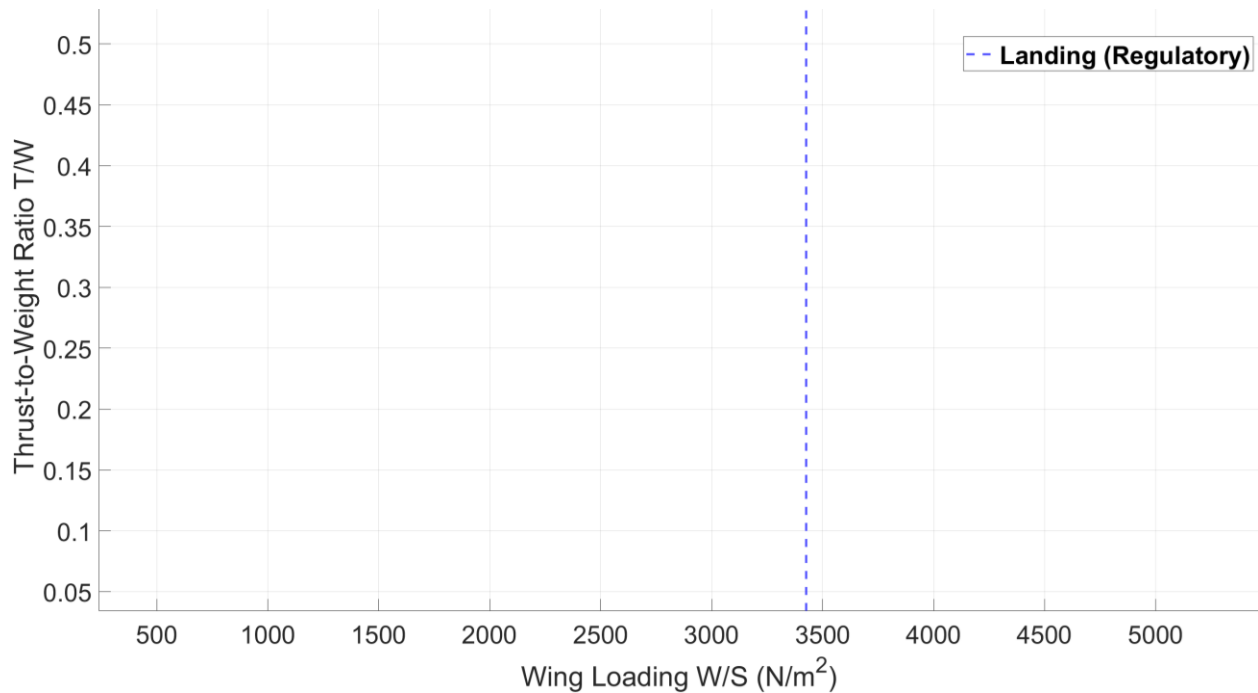


Figure 82: Landing constraint of FR25

g. Analysis

Figure 83 presents the complete aircraft constraint diagram for the FR25 turboprop hybrid, showing all major performance and regulatory constraints plotted on a wing loading (W/S) versus thrust-to-weight ratio (T/W) diagram. The diagram defines the feasible design space and identifies the critical constraints that drive aircraft sizing.

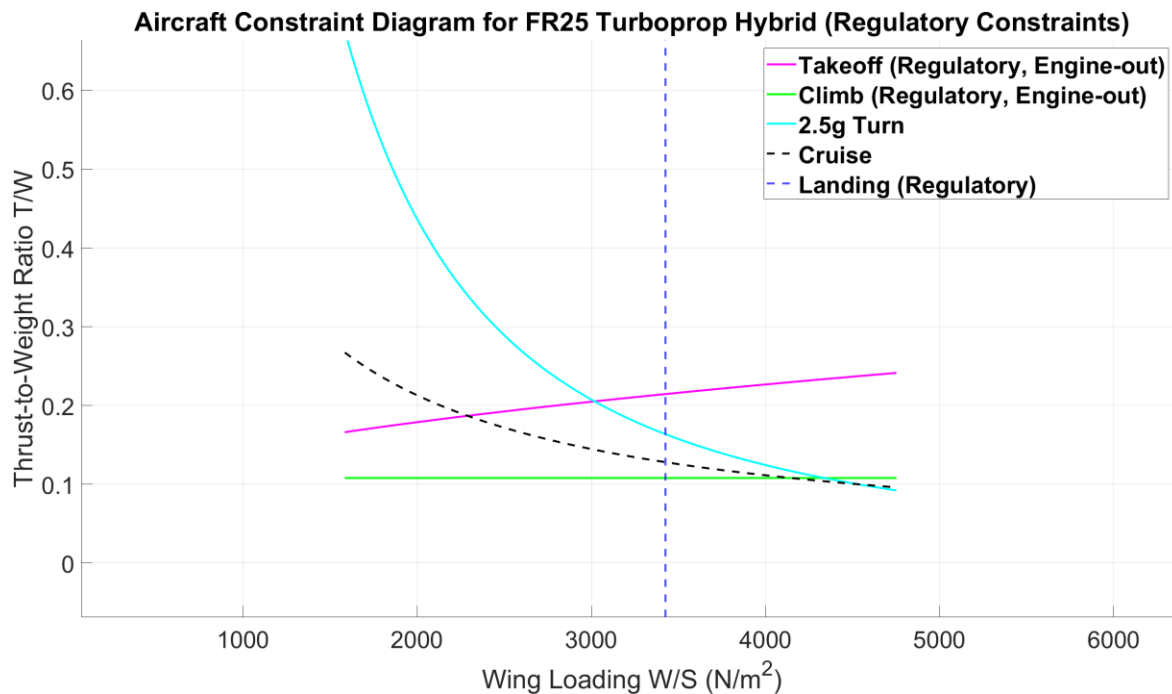


Figure 83: Complete constraint diagram of FR25

Figure 84 presents the constraint diagram for the FR25 with the design point overlaid. This plot shows that the selected aircraft configuration satisfies all regulatory and performance requirements. The FR25 design point is located at a wing loading of 3,167.83 N/m² and a thrust-to-weight ratio of 0.3475, within the feasible design region bounded. At this design point, the aircraft meets all constraint requirements with a combined margin of 0.1388 (66.5%), indicating robust design robustness.

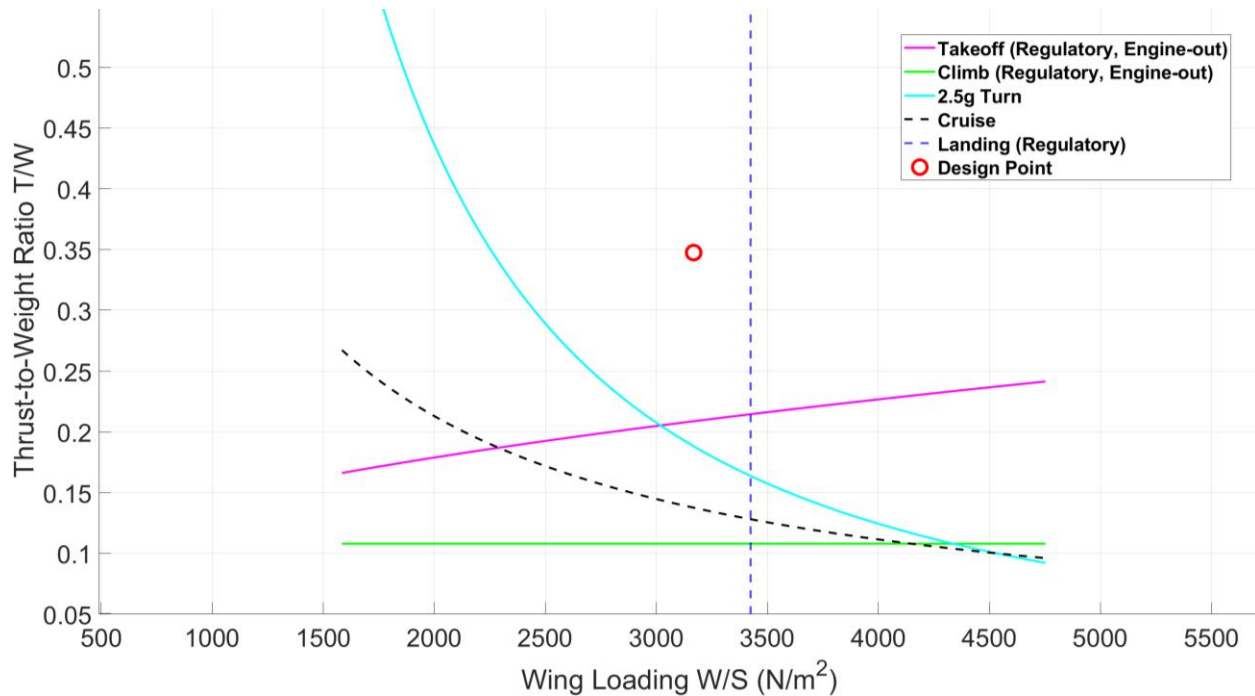


Figure 84: Complete constraint diagram with FR25 Design point

The most critical constraint for the FR25 is the take-off regulatory requirement which requires a minimum thrust-to-weight ratio of 0.2088 at the design wing loading. The design point exceeds this requirement by a margin of 66.5%. The climb constraint has a minimum T/W of 0.1080. The 2.5g turn constraint requires 0.1882 and the cruise constraint requires 0.1376. These values are satisfied by the available T/W of 0.3475. Additionally, the design wing loading of 3,167.83 N/m² remains within the regulatory landing limit of 3,425.27 N/m². A summary of these values is seen in figure 85.

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DESIGN FEASIBILITY ANALYSIS - FR25
=====

DESIGN PARAMETERS:
  Design Wing Loading (W/S):      3167.83 N/m²
  Actual Thrust-to-Weight (T/W):  0.3475

CONSTRAINT ANALYSIS AT DESIGN POINT:
  Takeoff:   Required T/W = 0.2088 | Margin = +0.1388 [✓ PASS]
  Climb:     Required T/W = 0.1080 | Margin = +0.2395 [✓ PASS]
  Turn:      Required T/W = 0.1882 | Margin = +0.1593 [✓ PASS]
  Cruise:    Required T/W = 0.1376 | Margin = +0.2099 [✓ PASS]

LANDING CONSTRAINT:
  Maximum Allowable W/S:      3425.27 N/m²
  Design W/S:                 3167.83 N/m²
  Margin:                     257.43 N/m² [✓ PASS]

OVERALL FEASIBILITY:
  Maximum Required T/W:      0.2088
  Available T/W:             0.3475
  Total Margin:              0.1388 (66.5%)

>>> ✓✓✓ DESIGN IS FEASIBLE ✓✓✓ <<<

Most Critical Constraint: Takeoff
=====

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Figure 85: Constrains analysis summary

16. Conclusion

The FR25 represents a shift in regional aviation, combining cutting-edge technology with market-driven design to address the evolving demands of flying in 2040 and beyond. The aircraft's unique selling points that position it for the future include:

1. Environmental Leadership through Hybrid-Electric Propulsion

The FR25's hybrid-electric turboprop propulsion system transforms how regional aircraft operate. By integrating a 2,000-kW electric motor system with turboprop engines, the FR25 achieves a paradigm-shifting 20% reduction in fuel consumption compared to conventional turboprops of similar capacity.

Furthermore, the FR25 is designed to operate on 100% SAF. This commitment to alternative propulsion and sustainable fuels results in significant environmental benefits. This will cause a reduction of approximately 159 kg of CO₂ emissions per flight compared to conventional turboprops.

2. Operational Flexibility through Modular Design

The FR25 introduces a revolutionary modular cabin architecture that challenges the industry's traditional approach to aircraft utilization. Unlike the fixed configurations of competing platforms such as the ATR 72-600 and Dash 8-400, the FR25 transitions between multiple operational profiles without the need for permanent ground-based conversion.

3. Advanced Safety and Passenger Experience

The FR25 incorporates an ATDMS featuring LIDAR-based turbulence detection capabilities. These state-of-the-art avionics system provides pilots with real-time turbulence awareness. This enables proactive flight path adjustments and significantly enhancing passenger comfort. The system is valuable for regional operations over challenging terrain and tropical regions where convective turbulence is common like in Africa and Asia-pacific.

The FR25 addresses some of most pressing issues for the future. For operators in emerging markets seeking to expand capacity while managing environmental impact and operational costs, the FR25 represents the optimal platform for the next two decades of regional aviation.

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18. Annex

Annex A: ATR 72-600 vs Dash 8-400 Specifications

Aircraft	ATR 72-600	Dash 8-400
Payload (kg)	7,500	8,625
Number of Passengers (pax)	72	78
Range (max payload, km)	1,370	2,700
Maximum Take Off Weight (kg)	23,000	29,574
Maximum Ramp Weight (kg)	-	29,347
Maximum Landing Weight (kg)	22,350	28,010
Maximum Fuel Capacity (kg)	5,000	5,318
Cruise Speed (knots)	260	360
Cruise Altitude (ft)	25,000	25,000
Take Off Distance (m)	1,315	1,300
Landing Distance (m)	915	1,293
Length of Aircraft (m)	27.17	32.83
Height of Aircraft (m)	7.65	8.34
Wing Area (m ²)	61	63.1
Wingspan (m)	27.05	28.42
Aspect Ratio	12	12.8
Engine Type	Pratt & Whitney Canada PW127XT-M	Pratt & Whitney PW150A
Number of Engines	2	2
Maximum Power per Engine (SHP)	2,750	5,071
Propeller	Hamilton Standard 568F	Dowty R408 6-blade
Fan Diameter (m)	3.93	4.1

```

% Load data from Excel
filename = 'Aircraft Data.xlsx';
data = readtable(filename);

% Extract relevant columns
MTOW = data("MTOW_Kg_");
Max_Range = data("MaximumRange_Km_");
AircraftNames = data("AircraftModel");

% Identify FR25 index
fr25_idx = strcmp(AircraftNames, 'FR25');

% Create scatter plot
figure;
scatter(MTOW(~fr25_idx), Max_Range(~fr25_idx), 80, 'filled', 'MarkerFaceColor', 'b');
% other aircraft
hold on;
scatter(MTOW(fr25_idx), Max_Range(fr25_idx), 120, '^', 'filled', 'MarkerFaceColor',
'r'); % FR25

% Improved label placement: staggered vertical offsets + leader lines for close
points
baseOffset = 300; % base vertical offset (tune for dataset
scale)
altOffset = baseOffset * 0.6; % alternate offset for staggering
labelXShift = 200; % horizontal shift to the right for labels

% Compute pairwise distances to detect crowded points
pts = [MTOW(:), Max_Range(:)];
D = squareform(pdist(pts)); % requires Statistics and Machine Learning
Toolbox
crowdThresh = prctile(D(D>0), 10); % 10th percentile of non-zero distances as
crowd threshold

for i = 1:numel(MTOW)
    % Determine offset strategy
    if fr25_idx(i)
        vOffset = 1.5*baseOffset; % make FR25 stand out
    else
        vOffset = (mod(i,2)==0) * altOffset + (mod(i,3)==0) * (-altOffset); % simple
stagger
    end

    % If point is in a crowded neighborhood, increase offset and draw a leader line
    isCrowded = any(D(i, :) > 0 & D(i, :) < crowdThresh);
    if isCrowded
        vOffset = vOffset + baseOffset; % push label further away
        % Draw a subtle leader line from point to label position
        plot([MTOW(i), MTOW(i) + labelXShift], ...
            [Max_Range(i), Max_Range(i) + vOffset], ...
            '-', 'Color', [0.6 0.6 0.6], 'LineWidth', 0.5);
    end

    % Place label

```

```

    text(MTOW(i) + labelXShift, Max_Range(i) + vOffset, AircraftNames{i}, ...
         'FontSize', 15, 'HorizontalAlignment', 'left', 'VerticalAlignment',
         'middle');
end

% Linear fit and confidence band
[p,S,mu] = polyfit(MTOW, Max_Range, 1);
xFit = linspace(min(MTOW), max(MTOW), 100);
[yFit, delta] = polyval(p, xFit, S, mu);

% Slightly wider confidence band
scale_factor = 1.2; % 20% wider band
fill([xFit fliplr(xFit)], [yFit + scale_factor*delta fliplr(yFit -
scale_factor*delta)], ...
     [1 0.8 0.8], 'EdgeColor','none','FaceAlpha',0.5);

% Regression line
plot(xFit, yFit, 'r-', 'LineWidth', 2);

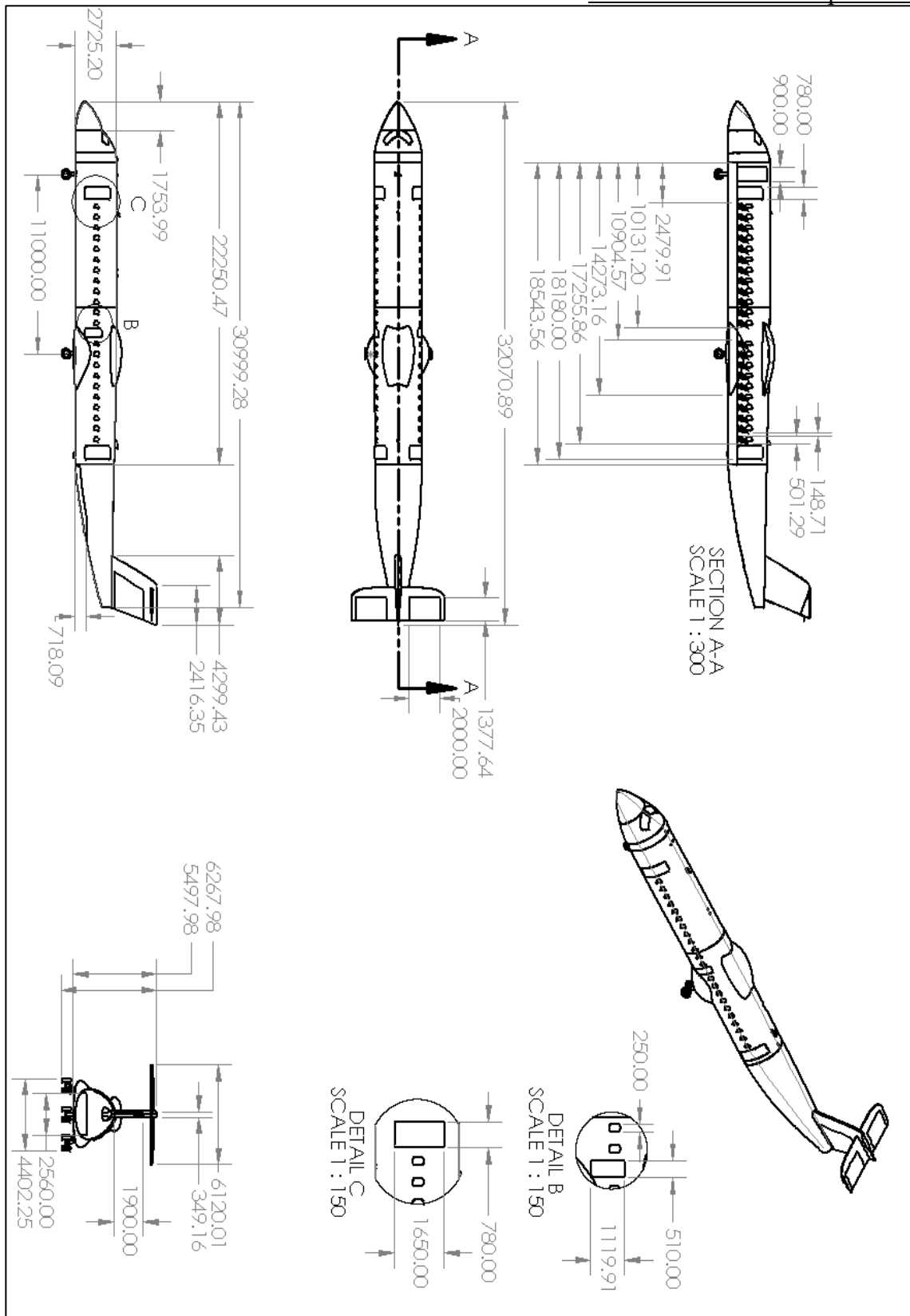
% Axes formatting
ax = gca;
ax.FontSize = 18; % larger tick labels

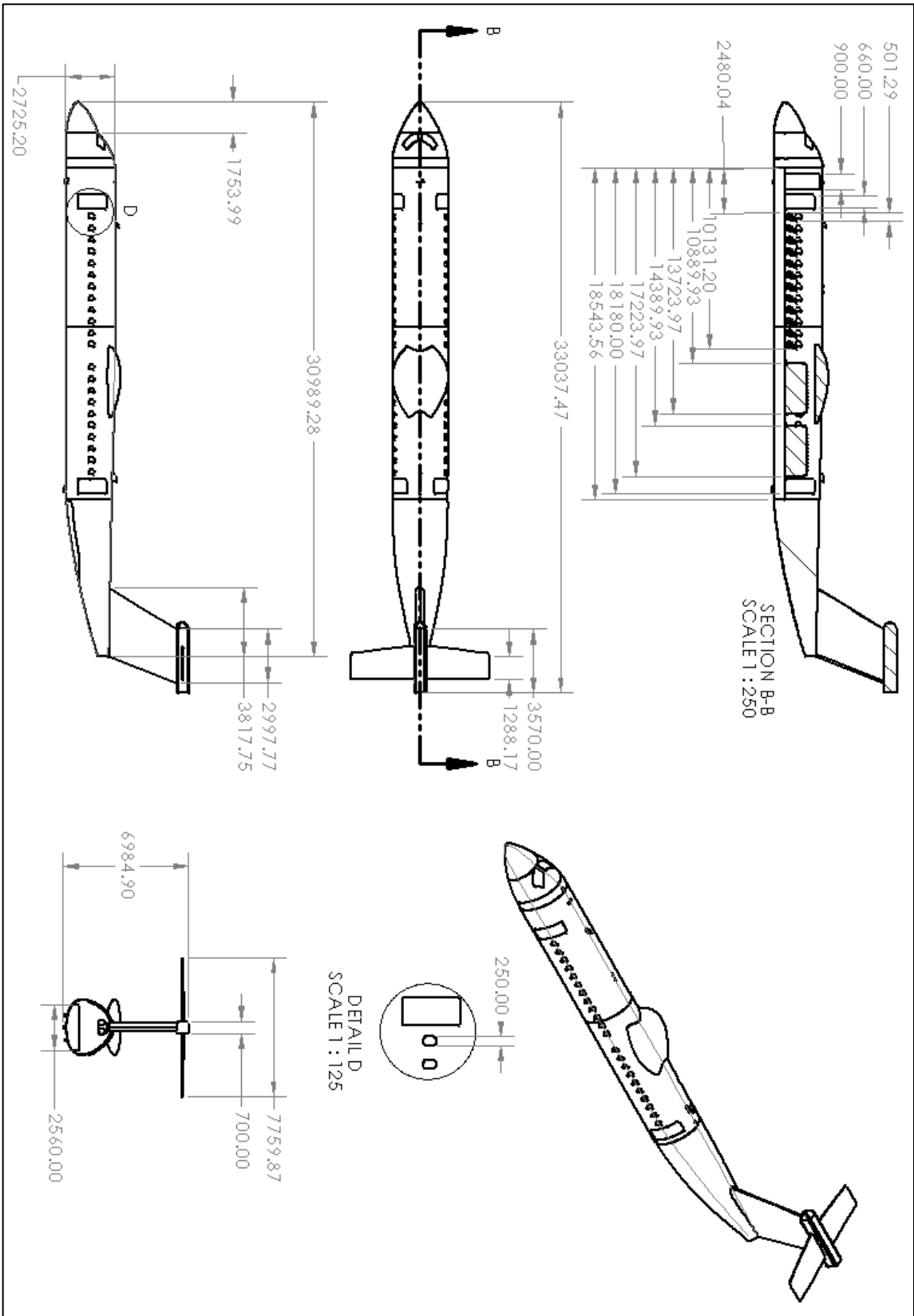
% Labels and title
xlabel('MTOW (Kg)', 'FontSize', 20);
ylabel('Maximum Range (Km)', 'FontSize', 20);
title('MTOW vs Maximum Range', 'FontSize', 20);
grid on;
legend('Other Aircraft', 'FR25', 'Confidence Band', 'Trendline', 'Location', 'best');

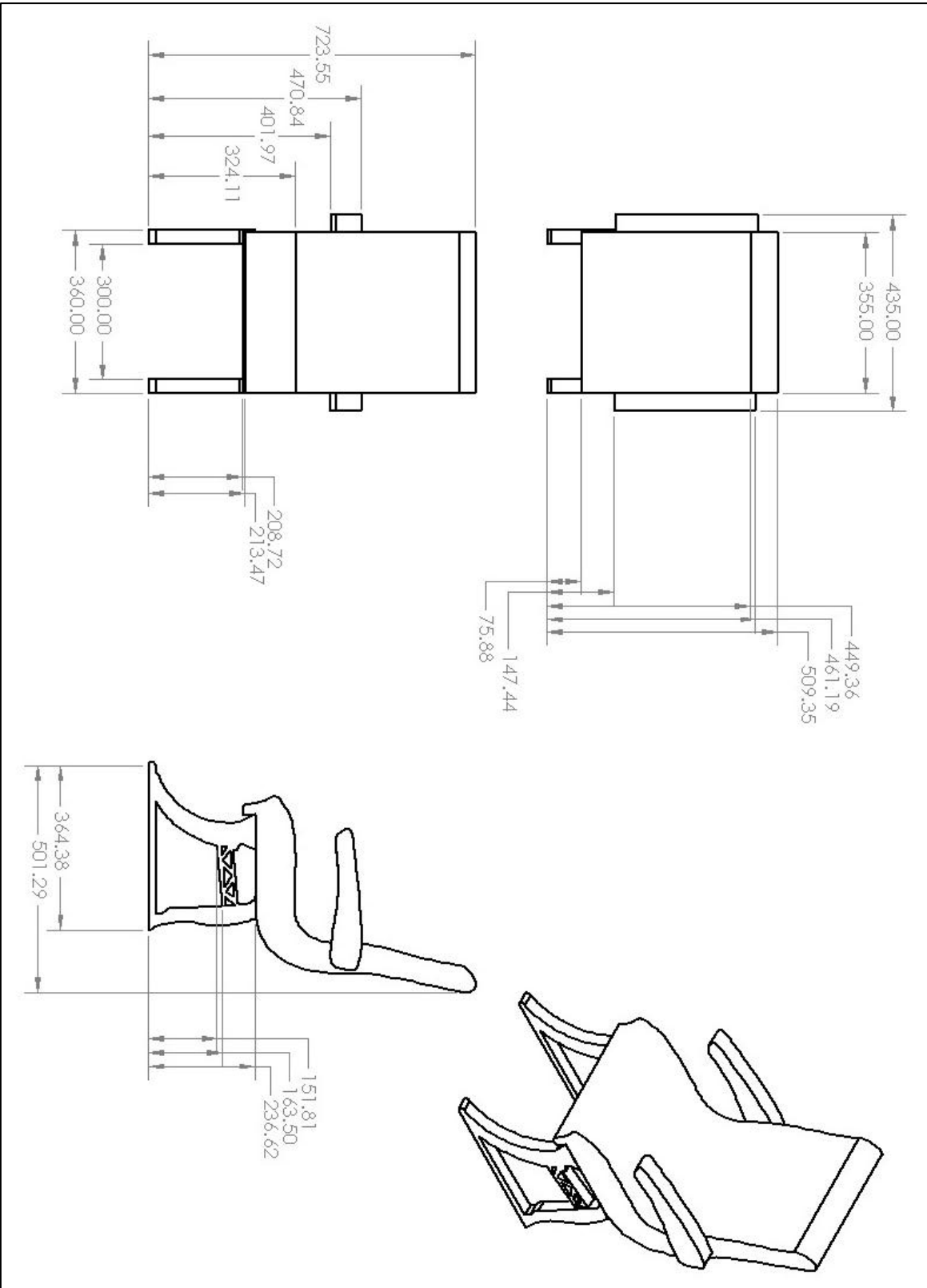
% Optional: tighten view with some padding to avoid clipped labels
xlim([min(MTOW)-1000, max(MTOW)+4000]);
ylim([min(Max_Range)-2000, max(Max_Range)+6000]);

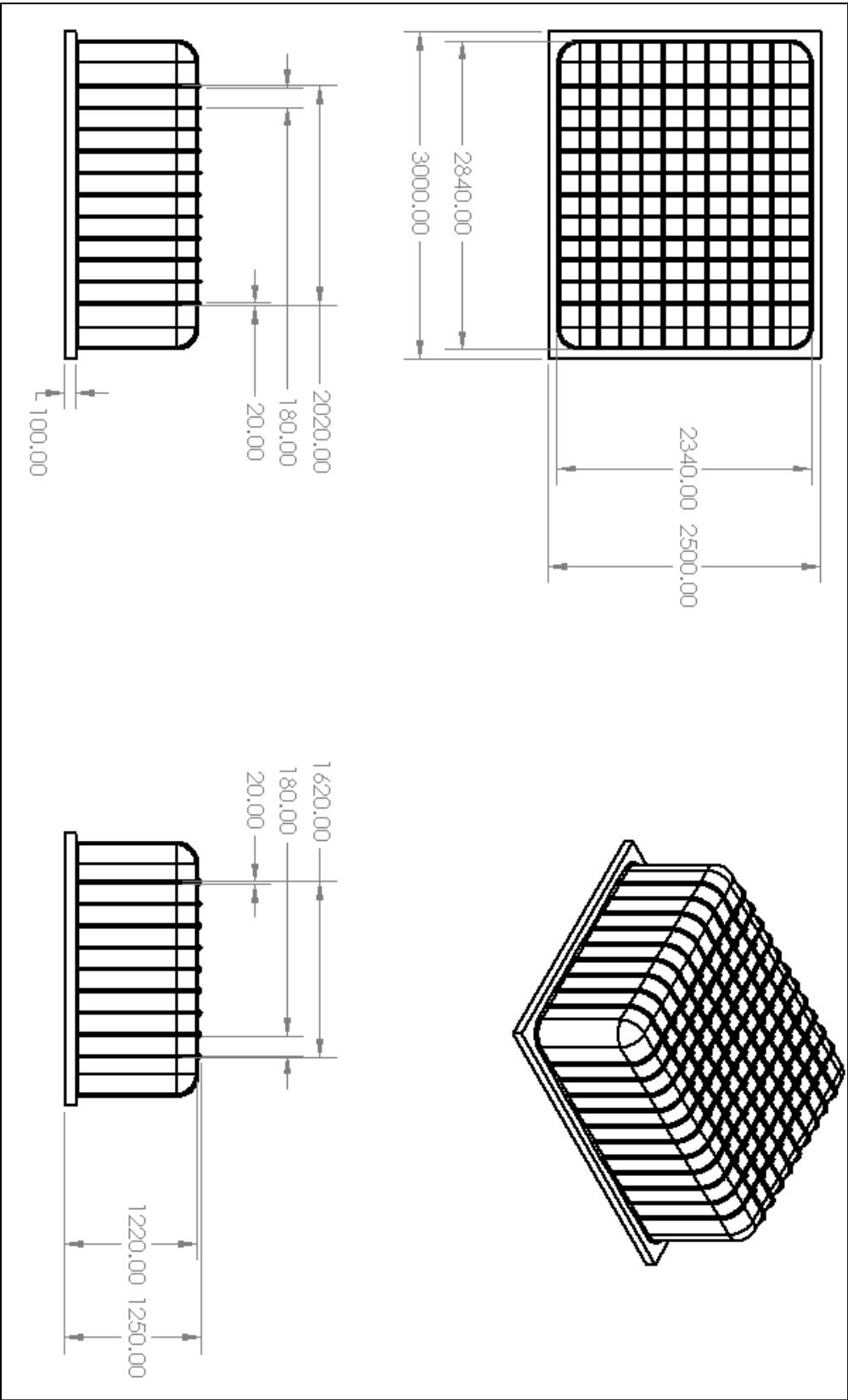
% Notes:
% - If pdist/squareform are unavailable, replace with a simple nearest-neighbor
  check.
% - Adjust baseOffset, altOffset, labelXShift, and crowdThresh percentile as needed
  for your data scale.

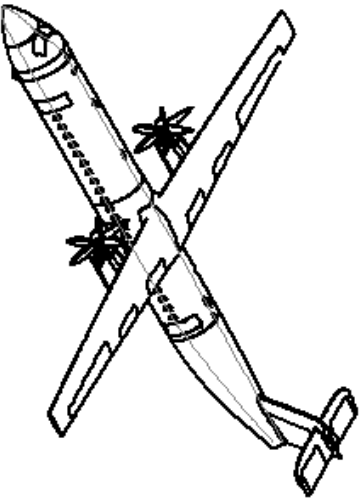
```











Annex D: Mission Profile Plot MATLAB Code

```
% Mission Profile Plot

% Define phases
phases = {'Taxi Out','Take-off','Climb','Cruise','Descent','Approach & Landing','Taxi In'};
n = numel(phases);

% Duration (min) per phase
duration = [15 2 15 116.5 15 5 10];

% Fuel consumption per phase (%)
fuel_consumed = [1.05 2.34 14.03 45.75 1.75 0.59 0.70];
fuel_remaining = 100 - cumsum(fuel_consumed);

% Altitude profile (m) per phase
altitude_m = [0 0 7630 7630 0 0 0];

%% Plot 1: Duration + Fuel Consumption/Remaining
figure;
yyaxis left
b = bar(duration,'FaceColor',[0.6 0.8 1]); % save bar handle
ylabel('Duration (minutes)');
ylim([0, 150]);

yyaxis right
hold on;
p1 = plot(1:n,fuel_consumed,'--og','LineWidth',1.5,'MarkerSize',6);
p2 = plot(1:n,fuel_remaining,'-or','LineWidth',1.5,'MarkerSize',6);
ylabel('Fuel (%)');
ylim([0 110]);

xticks(1:n);
xticklabels(phases);
xtickangle(25);
title('Mission Profile: Duration, Fuel Consumption, and Remaining Fuel');

legend([b p1 p2],'Duration','Fuel Consumed','Fuel Remaining','Location','best');

% --- Annotations ---

% Duration labels (on left axis)
yyaxis left;
leftYL = ylim;
offsetDur = 0.03 * (leftYL(2) - leftYL(1)); % 3% of axis height

if isprop(b,'XEndPoints') && isprop(b,'YEndPoints')
    xPos = b.XEndPoints;
    yPos = b.YEndPoints;
else
```

```

        xPos = 1:n;
        yPos = duration;
    end

    for i = 1:n
        text(xPos(i), yPos(i) + offsetDur, sprintf('%.1f min', duration(i)), ...
            'HorizontalAlignment','center', 'VerticalAlignment','bottom', 'FontSize',9);
    end

    % Fuel remaining labels (on right axis)
    yyaxis right;
    rightYL = ylim;
    offsetFuel = 0.03 * (rightYL(2) - rightYL(1)); % 3% offset on right axis
    for i = 1:n
        text(i, fuel_remaining(i) + offsetFuel, sprintf('%.1f%', fuel_remaining(i)), ...
            'Color','r','FontSize',8, 'HorizontalAlignment','center', 'VerticalAlignment','bottom');
    end

    %% Plot 2: Altitude Profile (in meters)
    figure;
    plot(1:n, altitude_m, '-s', 'Color',[1 0.5 0], 'LineWidth',2, 'MarkerFaceColor','y');
    ylabel('Altitude (meters)');
    ylim([0 9000]);

    xticks(1:n);
    xticklabels(phases);
    xtickangle(25);
    title('Mission Profile: Altitude per Phase (meters)');

    % Add altitude labels
    for i = 1:n
        text(i, altitude_m(i) + 300, sprintf('%d m', altitude_m(i)), ...
            'HorizontalAlignment','center', 'Color',[1 0.5 0]);
    end
end

```

Annex E: Total Parasite Drag MATLAB Code

```
% FR25 DRAG ANALYSIS - UPDATED FOR ACTUAL MTOW
% WITH OPERATIONAL CRUISE SPEED COMPARISON
clear; clc;

% Input parameters for FR25
S_ref = 72;           % Wing reference area (m^2)
MTOW_kg = 30693.07;   % Maximum Take-Off Weight (kg)
g = 9.81;             % Gravitational acceleration (m/s^2)
W = MTOW_kg * g;      % Weight in Newtons (N)
rho = 0.549;          % Air density (kg/m^3) at service ceiling (~6,000m)
CD0 = 0.0191;         % Parasite drag coefficient
AR = 11.4;            % Aspect ratio
e = 0.716;            % Oswald efficiency factor

% Speed range (m/s)
V = linspace(30, 250, 150);

% Calculate aerodynamic parameters for each speed
CL = 2 * W ./ (rho * V.^2 * S_ref); % Lift coefficient
CDi = CL.^2 ./ (pi * AR * e);        % Induced drag coefficient
CD = CD0 + CDi;                      % Total drag coefficient

% Drag forces (N)
DP = 0.5 * rho * V.^2 * S_ref * CD0; % Parasite drag
DI = 0.5 * rho * V.^2 .* S_ref .* CDi; % Induced drag
DT = DP + DI;                        % Total drag

% Efficiency metrics
LD = CL ./ CD;                      % Lift-to-drag ratio

% =====
% FIND KEY OPERATIONAL POINTS
% =====

% 1. Find minimum drag point (L/D maximum)
[DT_min, idx_min] = min(DT);
V_min = V(idx_min);
LD_max = LD(idx_min);
DP_min = DP(idx_min);
DI_min = DI(idx_min);
CL_min = CL(idx_min);
CD_min = CD(idx_min);

% 2. Find cruise speed point (185 m/s)
V_cruise = 185; % m/s
[~, idx_cruise] = min(abs(V - V_cruise));
V_cruise_actual = V(idx_cruise);
DT_cruise = DT(idx_cruise);
DP_cruise = DP(idx_cruise);
DI_cruise = DI(idx_cruise);
CL_cruise = CL(idx_cruise);
CD_cruise = CD(idx_cruise);
```

```

LD_cruise = LD(idx_cruise);

% =====
% CREATE PLOT WITH MULTIPLE MARKERS
% =====

figure; hold on; box on;

% Plot drag components
plot(V, DT, 'b-', 'LineWidth', 2.5);      % Total Drag
plot(V, DP, 'k-.', 'LineWidth', 2);       % Parasite Drag
plot(V, DI, 'k:', 'LineWidth', 2);        % Induced Drag

% Mark L/D Maximum Point (minimum drag)
plot(V_min, DT_min, 'ro', 'MarkerSize', 24, 'MarkerFaceColor', 'r', 'LineWidth',
2.5);
plot([V_min V_min], [0 DT_min+20000], 'r--', 'LineWidth', 2);
text(V_min-5, DT_min+20000, 'L/D_{max}', 'FontSize', 24, 'FontWeight', 'bold', ...
'BackgroundColor', 'yellow', 'EdgeColor', 'red', 'LineWidth', 1.5);

% Mark Cruise Speed Point (185 m/s) - FIXED: using 'green' instead of 'lightgreen'
plot(V_cruise_actual, DT_cruise, 'gs', 'MarkerSize', 24, 'MarkerFaceColor', 'g',
'LineWidth', 2.5);
plot([V_cruise_actual V_cruise_actual], [0 DT_cruise+20000], 'g--', 'LineWidth', 2);
text(V_cruise_actual-5, DT_cruise+20000, 'Cruise (185 m/s)', 'FontSize', 24,
'FontWeight', 'bold', ...
'BackgroundColor', 'yellow', 'EdgeColor', 'green', 'LineWidth', 1.5);

% Labels and formatting
xlabel('Airspeed (m/s)', 'FontSize', 24, 'FontWeight', 'bold');
ylabel('Drag (N)', 'FontSize', 24, 'FontWeight', 'bold');
title('FR25 Drag Analysis: L/D Maximum vs. Operational Cruise Speed', 'FontSize', 24,
'FontWeight', 'bold');
legend('Total Drag', 'Parasite Drag', 'Induced Drag', 'L/D_{max} Point', 'Cruise
Speed (185 m/s)', ...
'Location', 'northwest', 'FontSize', 24, 'FontWeight', 'bold');
grid on;
set(gca, 'FontSize', 24);
hold off;

% =====
% CONSOLE OUTPUT - DETAILED COMPARISON
% =====

fprintf('\n');
fprintf('=====
||\n');
fprintf('||          FR25 DRAG ANALYSIS - OPERATIONAL COMPARISON (MTOW = %.0f kg)
||\n', MTOW_kg);
fprintf('=====
||\n\n');

fprintf('POINT 1: AERODYNAMIC OPTIMUM (L/D Maximum)\n');
fprintf('-----
--\n');

```

```

fprintf(' Speed:                                %.2f m/s (%.0f km/hr / %.0f knots)\n',
V_min, V_min*3.6, V_min*1.94384);
fprintf(' Total Drag:                            %.0f N\n', DT_min);
fprintf(' Parasite Drag:                             %.0f N\n', DP_min);
fprintf(' Induced Drag:                               %.0f N\n', DI_min);
fprintf(' Drag Coefficient (CD):                       %.4f\n', CD_min);
fprintf(' Lift Coefficient (CL):                       %.4f\n', CL_min);
fprintf(' L/D Ratio:                                   %.2f\n', LD_max);
fprintf(' Characteristic:                             BEST RANGE (Maximum distance on fuel)\n\n');

fprintf('POINT 2: OPERATIONAL CRUISE SPEED (FR25 Design)\n');
fprintf('-----\n\n');
fprintf(' Speed:                                %.2f m/s (%.0f km/hr / %.0f knots)\n',
V_cruise_actual, V_cruise_actual*3.6, V_cruise_actual*1.94384);
fprintf(' Total Drag:                            %.0f N\n', DT_cruise);
fprintf(' Parasite Drag:                             %.0f N\n', DP_cruise);
fprintf(' Induced Drag:                               %.0f N\n', DI_cruise);
fprintf(' Drag Coefficient (CD):                       %.4f\n', CD_cruise);
fprintf(' Lift Coefficient (CL):                       %.4f\n', CL_cruise);
fprintf(' L/D Ratio:                                   %.2f\n', LD_cruise);
fprintf(' Characteristic:                             COMMERCIAL OPTIMUM (Speed + Efficiency\nbalance)\n\n');

fprintf('COMPARATIVE ANALYSIS\n');
fprintf('-----\n\n');
fprintf(' Speed difference:                         %.0f m/s (%.0f%% faster than L/D max)\n',
V_cruise_actual - V_min, (V_cruise_actual - V_min)/V_min*100);
fprintf(' Drag increase:                           %.0f N (%.1f%% higher than minimum)\n',
DT_cruise - DT_min, (DT_cruise - DT_min)/DT_min*100);
fprintf(' L/D ratio decrease:                       %.2f (%.1f%% of maximum)\n', LD_max -
LD_cruise, (LD_cruise/LD_max)*100);
fprintf(' Efficiency retained:                     %.1f%%\n', (LD_cruise/LD_max)*100);
fprintf('\n ENGINEERING TRADE-OFF:\n');
fprintf(' Sacrifice %.1f%% efficiency for:\n', 100 - (LD_cruise/LD_max)*100);
fprintf(' - 5-6 commercial flights per day (vs 2-3 at L/D max)\n');
fprintf(' - Competitive market speed (top 15%% of turboprops)\n');
fprintf(' - 65%% power margin for safety and contingencies\n\n');

fprintf('AIRCRAFT CAPABILITY SUMMARY\n');
fprintf('-----\n\n');
fprintf(' Best Range Speed:                         %.0f m/s (%.0f km/hr)\n', V_min, V_min*3.6);
fprintf(' Recommended Cruise:                       %.0f m/s (%.0f km/hr)\n', V_cruise_actual,
V_cruise_actual*3.6);
fprintf(' Design Philosophy:                         Optimized for commercial regional\noperations\n');
fprintf('=====

```

Annex F: Wing Loading Plot MATLAB Code

```
% Load data from Excel
filename = 'Aircraft Data.xlsx';
data = readtable(filename);

% Extract relevant columns
MTOW = data("MTOW_Kg_");
Wingloading = data("WingLoading_Kg_m_2_");
AircraftNames = data("AircraftModel");

% Identify FR25 index
fr25_idx = strcmp(AircraftNames, 'FR25');

% Create scatter plot
figure;
scatter(MTOW(~fr25_idx), Wingloading(~fr25_idx), 80, 'filled', 'MarkerFaceColor',
'b'); % other aircraft
hold on;
scatter(MTOW(fr25_idx), Wingloading(fr25_idx), 120, '^', 'filled', 'MarkerFaceColor',
'r'); % FR25

% Improved label placement: staggered vertical offsets + leader lines for close
points
baseOffset = 5; % base vertical offset in Wingloading units (tune for
your data scale)
altOffset = baseOffset * 0.6; % alternate offset for staggering
labelXShift = range(MTOW) * 0.01; % shift labels ~1% of MTOW range to the right

% Compute pairwise distances to detect crowded points (2D Euclidean)
pts = [MTOW(:), Wingloading(:)];
try
    D = squareform(pdist(pts)); % requires Statistics and Machine
Learning Toolbox
    nz = D(D>0);
    if isempty(nz)
        crowdThresh = inf;
    else
        crowdThresh = prctile(nz, 10); % 10th percentile of non-zero distances
as crowd threshold
    end
catch
    % Fallback: simple nearest-neighbor distance without toolboxes
    n = size(pts,1);
    D = inf(n);
    for i = 1:n
        for j = i+1:n
            dij = hypot(pts(i,1)-pts(j,1), pts(i,2)-pts(j,2));
            D(i,j) = dij; D(j,i) = dij;
        end
    end
    nz = D(D>0 & isfinite(D));
    if isempty(nz)
        crowdThresh = inf;
    end
end
```

```

else
    crowdThresh = prctile(nz, 10);
end
end

for i = 1:numel(MTOW)
    % Determine offset strategy
    if fr25_idx(i)
        vOffset = 1.5 * baseOffset; % make FR25 stand out
    else
        % simple stagger pattern to reduce collisions
        if mod(i,3)==0
            vOffset = -altOffset;
        elseif mod(i,2)==0
            vOffset = altOffset;
        else
            vOffset = 0;
        end
    end

    % If point is crowded, increase offset and draw a leader line
    isCrowded = any(D(i, :) > 0 & D(i, :) < crowdThresh);
    if isCrowded
        vOffset = vOffset + baseOffset; % push label further away
        % Leader line from point to label position
        plot([MTOW(i), MTOW(i) + labelXShift], ...
            [Wingloading(i), Wingloading(i) + vOffset], ...
            '-', 'Color', [0.6 0.6 0.6], 'LineWidth', 0.5);
    end

    % Place label
    text(MTOW(i) + labelXShift, Wingloading(i) + vOffset, AircraftNames{i}, ...
        'FontSize', 15, 'HorizontalAlignment', 'left', 'VerticalAlignment',
        'middle');
end

% Linear fit and confidence band
[p,S,mu] = polyfit(MTOW, Wingloading, 1);
xFit = linspace(min(MTOW), max(MTOW), 200);
[yFit, delta] = polyval(p, xFit, S, mu);

% Slightly wider confidence band
scale_factor = 1.2; % 20% wider band
fill([xFit fliplr(xFit)], [yFit + scale_factor*delta fliplr(yFit -
scale_factor*delta)], ...
    [1 0.8 0.8], 'EdgeColor', 'none', 'FaceAlpha', 0.5);

% Regression line
plot(xFit, yFit, 'r-', 'LineWidth', 2);

% Axes formatting
ax = gca;
ax.FontSize = 18; % larger tick labels

% Labels and title

```

```

xlabel('MTOW (Kg)', 'FontSize', 20);
ylabel('Wing Loading (Kg/m^2)', 'FontSize', 20);
title('MTOW vs Wing Loading', 'FontSize', 20);
grid on;
legend('Other Aircraft', 'FR25', 'Confidence Band', 'Trendline', 'Location', 'best');

% Optional: add padding so labels are not clipped
xpad = range(MTOW) * 0.05;
ypad = range(Wingloading) * 0.10;
if xpad == 0, xpad = 1000; end
if ypad == 0, ypad = 5; end
xlim([min(MTOW)-xpad, max(MTOW)+xpad]);
ylim([min(Wingloading)-ypad, max(Wingloading)+ypad]);

```

Annex G: Hybrid Aircraft L/D Plot MATLAB Code

```
% Hybrid Aircraft L/D and Maximum Range Calculation

% Constants
g = 9.81;    % gravity [m/s^2]

% Aircraft and aerodynamic parameters
S_ref = 72;    % Wing reference area [m^2]
W_OE_kg = 19933; % Operational Empty Weight [kg]
W_PL_kg = 3000; % Payload weight [kg]
W_OE = W_OE_kg * g; % OEW in Newtons
W_PL = W_PL_kg * g; % Payload in Newtons
rho = 0.549; % Air density [kg/m^3], e.g. cruise altitude
CD0 = 0.0191; % Parasite drag coefficient
AR = 11.4; % Wing aspect ratio
e = 0.716; % Oswald efficiency

% Battery and fuel parameters
battery_mass_kg = 400; % Battery mass
battery_energy_density_Wh_per_kg = 250; % Battery specific energy [Wh/kg]
E0_bat = battery_mass_kg * battery_energy_density_Wh_per_kg * 3600; % Battery energy [J]

fuel_mass_start_kg = 2000; % Initial fuel mass (kg)
fuel_mass_end_kg = 300; % Fuel reserve after flight (kg)
ef = 43e6; % Jet fuel energy density [J/kg]
ebat = battery_energy_density_Wh_per_kg * 3600; % Battery energy density [J/kg]

Ef_start = fuel_mass_start_kg * ef; % Initial fuel energy [J]
Ef_end = fuel_mass_end_kg * ef; % Fuel reserve energy [J]

% System efficiencies
eta1 = 0.85; % Transmission/prop efficiency (fuel path)
eta2 = 0.92; % Electric motor efficiency (battery path)
eta3 = 0.97; % Overall hybrid system efficiency

% Hybridization ratio (fraction of propulsion electric)
Phi = 0.3;

% Speed range
V = linspace(20, 200, 100); % Speed range [m/s]

% Preallocate arrays
CL = zeros(size(V));
CD = zeros(size(V));
LD = zeros(size(V));

for i = 1:length(V)
    CL(i) = 2*(W_OE + W_PL) / (rho * V(i)^2 * S_ref);
    CDi = CL(i)^2 / (pi * AR * e);
    CD(i) = CD0 + CDi;
    LD(i) = CL(i) / CD(i);
end
```

```

% Find maximum L/D and corresponding speed index
[LD_max, idx_max] = max(LD);

% Plot L/D vs Speed
V_knots = V * 1.94384;
figure; hold on; box on;
plot(V_knots, LD, 'b-', 'LineWidth', 2);
plot(V_knots(idx_max), LD_max, 'ro', 'MarkerFaceColor', 'r');
text(V_knots(idx_max), LD_max + 1, 'L/D_{max}', 'FontSize', 12,
'BackgroundColor', 'cyan');
xlabel('Airspeed (knots)');
ylabel('L/D');
title('Lift-to-Drag Ratio vs Airspeed');
grid on;
set(gca, 'FontSize', 12);
hold off;

% Calculate max range with hybrid equation
numerator = W_OE + W_PL + (g / ebat) * E0_bat + (g / ef) * Ef_start;
denominator = W_OE + W_PL + (g / ebat) * E0_bat + (g / ef) * Ef_end;

hybrid_efficiency = eta1 + eta2 * (Phi / (1 - Phi));

R_meters = eta3 * ef / g * LD_max * hybrid_efficiency * log(numerator / denominator);
R_nm = R_meters / 1852; % Convert meters to nautical miles

fprintf('Hybrid aircraft estimated max range: %.1f nautical miles\n', R_nm);

```

Annex G: Constraint analysis Plot MATLAB Code

```
% FR25 Constraint Analysis - Regulatory Constraints (Corrected, Classic Cruise
Constraint)
% --- Aircraft and Engine Data ---
rho = 1.225;           % Sea level air density, kg/m^3
g = 9.81;              % Gravity, m/s^2
S = 72;                % Wing area, m^2
W_TO = 30693.07 * g;   % Max takeoff weight, Newtons (TBC)
% --- Use landing weight for design W/S! ---
WL = 0.85 * W_TO;      % Landing weight, Newtons (typical post-cruise fuel burn)
WS_design = WL / S;    % Design wing loading for landing (N/m^2)
WS_range = linspace(0.5*WS_design, 1.5*WS_design, 100);

% --- Performance Parameters ---
V_TO = 64;             % Takeoff speed (m/s)
V_L = 55.6;            % Approach/landing speed (m/s)
ROC = 5;               % Rate of climb, m/s
V_C = 185;             % Cruise speed (m/s)
V_climb = 80;          % Climbing speed (m/s)
mu = 0.1;              % Ground friction coefficient

% --- Aerodynamic Coefficients and Factors ---
CD0 = 0.019;           % Zero-lift drag coefficient
CL_max_TO = 2.2;       % Max lift coefficient for takeoff
CL_max_L = 2.7;        % Max lift coefficient for landing
CL_cruise = 0.45;      % Typical cruise CL
CL_climb = 3.0;        % Lift coefficient at climb speed
CD_CL_v2 = 0.03;       % Drag-to-lift ratio at V2
K_cruise = 0.035;      % Induced drag factor for cruise

% --- Engine Output ---
P_engine = 3382000;    % Shaft Power per engine, W
n_engines = 2;         % Number of engines
P_total = P_engine * n_engines;
T_actual = P_total / V_TO; % Thrust at takeoff, N
TW_actual = T_actual / W_TO; % Thrust-to-weight ratio

% --- Regulatory Engine-out Takeoff Constraint (Raymer Eqn 9.20) ---
b_w = 28.7;            % Wingspan (m)
L_TO = 1700;           % Takeoff field length (m)
N_eng = n_engines;
WTO_per_bw2 = (WS_range * S) / b_w^2;
takeoff_reg = 0.78 * ((N_eng ./ (0.89*N_eng - 1)) .* ...
    (WTO_per_bw2 ./ (L_TO * rho * g)).^0.5 + ...
    42 ./ L_TO .* (N_eng ./ (0.89*N_eng - 1)));
TW_TOreg_at_design = 0.78 * ((N_eng / (0.89*N_eng - 1)) * ...
    ((WS_design * S) / b_w^2) / (L_TO * rho * g))^0.5 + ...
    42 / L_TO * (N_eng / (0.89*N_eng - 1));

% --- Regulatory Landing Constraint (vertical line) ---
WS_landing_limit = 0.335 * rho * V_L^2 * CL_max_L;
% --- Regulatory Engine-out Climb Constraint (Eq 9.6) ---
```

```

gamma_v2 = 0.024; % Minimum climb gradient (2.4%)
TW_climb = (gamma_v2 + CD_CL_v2) / (1 - 1/n_engines) * ones(size(WS_range));
TW_climb_at_design = (gamma_v2 + CD_CL_v2) / (1 - 1/n_engines);
% --- Maneuver Constraint ---
n_turn = 2.5;
V_turn = 100;
k = 0.04;
q_turn = 0.5 * rho * V_turn^2;
CL_turn = (WS_range * n_turn) ./ q_turn;
CL_turn(CL_turn <= 0) = nan;
TW_turn = CD0 ./ CL_turn * n_turn + ...
    k * n_turn^3 * WS_range ./ (q_turn * CL_turn.^3);
CL_turn_design = (WS_design * n_turn)/q_turn;
if CL_turn_design > 0
    TW_turn_at_design = CD0/CL_turn_design*n_turn + ...
        k*n_turn^3*WS_design/(q_turn*CL_turn_design^3);
else
    TW_turn_at_design = inf;
end
% --- Cruise Constraint (CLASSIC FORMULA, VISIBLE ON PLOT) ---
rho_cruise = 1.225; % kg/m^3, for sea-level cruise (or use altitude value)
V_cruise = V_C;
CD0_cruise = CD0; % Use same as drag table, can adjust if needed
K_cruise = 0.035;
CL_cruise_vec = WS_range ./ (0.5 * rho_cruise * V_cruise^2); % For each W/S
TW_cruise_curve = (CD0_cruise ./ CL_cruise_vec) + (K_cruise .* CL_cruise_vec);
TW_cruise_at_design = (CD0_cruise / (WS_design / (0.5 * rho_cruise * V_cruise^2))) +
...
    K_cruise * (WS_design / (0.5 * rho_cruise * V_cruise^2));
disp('Cruise constraint range:');
disp([min(TW_cruise_curve), max(TW_cruise_curve)]);

% --- Interpolate required T/W at design point for each constraint ---
TW_required_takeoff = interp1(WS_range, takeoff_reg, WS_design);
TW_required_climb = TW_climb_at_design;
TW_required_turn = interp1(WS_range, TW_turn, WS_design);
TW_required_cruise = TW_cruise_at_design;

% --- Feasibility ---
is_left_of_landing = WS_design < WS_landing_limit;
TW_req_at_design = max([TW_required_takeoff, TW_required_climb, TW_required_turn,
    TW_required_cruise]);
isFeasible = (TW_actual > TW_req_at_design) && is_left_of_landing;

% --- Plotting ---
figure; hold on; grid on;
plot(WS_range, takeoff_reg, 'm-', 'LineWidth', 2, 'DisplayName', 'Takeoff
(Regulatory, Engine-out)');
plot(WS_range, TW_climb, 'g-', 'LineWidth', 2, 'DisplayName', 'Climb (Regulatory,
Engine-out)');
plot(WS_range, TW_turn, 'c-', 'LineWidth', 2, 'DisplayName', '2.5g Turn');
plot(WS_range, TW_cruise_curve, 'k--', 'LineWidth', 2, 'DisplayName', 'Cruise');
xline(WS_landing_limit, 'b--', 'LineWidth', 2, 'DisplayName', 'Landing
(Regulatory)');

```

```

% --- Design Point and Lines ---
% xline(WS_design, 'r-', 'LineWidth', 2.5, 'DisplayName', 'Design W/S');
% yline(TW_actual, 'r-.', 'LineWidth', 2, 'DisplayName', 'Actual T/W');
plot(WS_design, TW_actual, 'ro', 'MarkerSize', 14, 'LineWidth', 3, 'DisplayName',
'Design Point');

% --- Margin indicator line ---
% plot([WS_design, WS_design], [TW_req_at_design, TW_actual], 'r:', 'LineWidth', 2.5,
'DisplayName', 'Design Margin');

xlabel('Wing Loading W/S (N/m^2)', 'FontSize', 24);
ylabel('Thrust-to-Weight Ratio T/W', 'FontSize', 24);
title('Aircraft Constraint Diagram for FR25 Turboprop Hybrid (Regulatory
Constraints)', 'FontSize', 24, 'FontWeight', 'bold');
legend('Location', 'best', 'FontSize', 18, 'FontWeight', 'bold');
set(gca, 'FontSize', 24);
xlim([0 max(WS_range)*1.1]);
ylim([0 0.5]); % Adjusted for realistic T/W range
hold off;

% --- Feasibility Analysis Report ---
fprintf('\n');
fprintf('===== \n');
fprintf('    DESIGN FEASIBILITY ANALYSIS - FR25 \n');
fprintf('===== \n \n');

fprintf('DESIGN PARAMETERS: \n');
fprintf('  Design Wing Loading (W/S):    %.2f N/m^2 \n', WS_design);
fprintf('  Actual Thrust-to-Weight (T/W): %.4f \n \n', TW_actual);

fprintf('CONSTRAINT ANALYSIS AT DESIGN POINT: \n');
fprintf('  Takeoff:   Required T/W = %.4f   |   Margin = %+.4f   [%s] \n', ...
    TW_required_takeoff, TW_actual - TW_required_takeoff, ...
    iif(TW_actual >= TW_required_takeoff, '✓ PASS', 'X FAIL'));
fprintf('  Climb:     Required T/W = %.4f   |   Margin = %+.4f   [%s] \n', ...
    TW_required_climb, TW_actual - TW_required_climb, ...
    iif(TW_actual >= TW_required_climb, '✓ PASS', 'X FAIL'));
fprintf('  Turn:      Required T/W = %.4f   |   Margin = %+.4f   [%s] \n', ...
    TW_required_turn, TW_actual - TW_required_turn, ...
    iif(TW_actual >= TW_required_turn, '✓ PASS', 'X FAIL'));
fprintf('  Cruise:    Required T/W = %.4f   |   Margin = %+.4f   [%s] \n \n', ...
    TW_required_cruise, TW_actual - TW_required_cruise, ...
    iif(TW_actual >= TW_required_cruise, '✓ PASS', 'X FAIL'));

fprintf('LANDING CONSTRAINT: \n');
fprintf('  Maximum Allowable W/S:    %.2f N/m^2 \n', WS_landing_limit);
fprintf('  Design W/S:               %.2f N/m^2 \n', WS_design);
fprintf('  Margin:                   %.2f N/m^2   [%s] \n \n', ...
    WS_landing_limit - WS_design, iif(is_left_of_landing, '✓ PASS', 'X FAIL'));

fprintf('OVERALL FEASIBILITY: \n');
fprintf('  Maximum Required T/W:     %.4f \n', TW_req_at_design);
fprintf('  Available T/W:            %.4f \n', TW_actual);
fprintf('  Total Margin:              %.4f (%.1f%) \n \n', ...
    TW_actual - TW_req_at_design, (TW_actual - TW_req_at_design) / TW_req_at_design);

```

```

    TW_actual - TW_req_at_design, 100 * (TW_actual - TW_req_at_design) /
    TW_req_at_design);

if isFeasible
    fprintf('>>> ✓✓✓ DESIGN IS FEASIBLE ✓✓✓ <<<\n\n');
    [~, idx_critical] = max([TW_required_takeoff, TW_required_climb,
    TW_required_turn, TW_required_cruise]);
    constraints_names = {'Takeoff', 'Climb', 'Turn', 'Cruise'};
    fprintf('Most Critical Constraint: %s\n\n', constraints_names{idx_critical});
else
    fprintf('>>> XXX DESIGN IS NOT FEASIBLE XXX <<<\n\n');
    if ~is_left_of_landing
        fprintf('REASON: Wing loading exceeds landing limit!\n\n');
    else
        fprintf('REASON: Insufficient thrust-to-weight ratio!\n\n');
        [~, idx_critical] = max([TW_required_takeoff, TW_required_climb,
    TW_required_turn, TW_required_cruise]);
        constraints_names = {'Takeoff', 'Climb', 'Turn', 'Cruise'};
        fprintf('Critical Constraint: %s\n\n', constraints_names{idx_critical});
    end
end

fprintf('=====\n\n');

% Helper function for conditional text
function result = iif(condition, true_val, false_val)
    if condition
        result = true_val;
    else
        result = false_val;
    end
end
end

```

Annex H: Blended Wing Analysis Plot MATLAB Code

```
% FR25 BLENDED WING AIRFOIL ANALYSIS
% Simplified approach with proper aerodynamic modeling

clear; clc; close all;

% =====
% WING GEOMETRY & AIRFOIL DATA
% =====

% Airfoil data (from XFOIL)
airfoils = struct();

% NACA 4415 (Root) - 44.4%
airfoils(1).name = 'NACA 4415';
airfoils(1).weight = 0.444;
airfoils(1).alpha_0 = -5.0;
airfoils(1).CL_0 = 0.45;
airfoils(1).CL_5 = 1.00;

% NACA 2412 (Mid) - 33.3%
airfoils(2).name = 'NACA 2412';
airfoils(2).weight = 0.333;
airfoils(2).alpha_0 = -2.5;
airfoils(2).CL_0 = 0.25;
airfoils(2).CL_5 = 0.75;

% NACA 0012 (Tip) - 22.2%
airfoils(3).name = 'NACA 0012';
airfoils(3).weight = 0.222;
airfoils(3).alpha_0 = 0.0;
airfoils(3).CL_0 = 0.0;
airfoils(3).CL_5 = 0.5;

% FR25 parameters
S_ref = 72;           % Wing area (m^2)
W = 301099;           % Weight (N)
AR = 11.4;             % Aspect ratio
e = 0.716;             % Oswald efficiency
CD0 = 0.0191;          % Parasite drag
rho = 0.549;           % Air density at cruise

% Angle of attack range
alpha = linspace(-15, 20, 300);

% =====
% CALCULATE BLENDED AIRFOIL CHARACTERISTICS
% =====

CL_blend = zeros(size(alpha));
CD_blend = zeros(size(alpha));

for i = 1:length(alpha)
```

```

a = alpha(i);
CL_temp = 0;
CD_temp = 0;

% Blend across three airfoils
for j = 1:3
    af = airfoils(j);
    w = af.weight;

    % Calculate CL using linear lift curve slope
    CL_alpha = (af.CL_5 - af.CL_0) / 5.0;
    CL_j = CL_alpha * (a - af.alpha_0);

    % Apply gentle stall curve (smooth transition, not abrupt)
    if a > 14
        % Smooth stall reduction after 14 degrees
        stall_factor = 1.0 - 0.15 * ((a - 14)/2)^1.5;
        CL_j = CL_j * max(stall_factor, 0.7);
    elseif a < -12
        % Negative stall
        stall_factor = 1.0 - 0.15 * ((a + 12)/2)^1.5;
        CL_j = CL_j * max(stall_factor, 0.5);
    end

    % Calculate CD (parasite + induced)
    k = 1 / (pi * AR * e);
    CD_j = CD0 + k * CL_j^2;

    % Add stall drag smoothly
    if a > 14 || a < -12
        stall_angle_excess = max(abs(a) - 14, 0);
        CD_j = CD_j + 0.02 * (stall_angle_excess)^1.2;
    end

    % Blend
    CL_temp = CL_temp + w * CL_j;
    CD_temp = CD_temp + w * CD_j;
end

CL_blend(i) = CL_temp;
CD_blend(i) = CD_temp;
end

% Prevent negative CD
CD_blend = max(CD_blend, 0.01);

% Calculate L/D
LD_blend = CL_blend ./ CD_blend;

% =====
% FIND CRUISE POINT
% =====

CL_cruise_target = 0.445; % At 185 m/s
[~, idx_cruise] = min(abs(CL_blend - CL_cruise_target));

```

```

alpha_cruise = alpha(idx_cruise);
CL_cruise = CL_blend(idx_cruise);
CD_cruise = CD_blend(idx_cruise);
LD_cruise = LD_blend(idx_cruise);

% =====
% CREATE PLOTS
% =====

figure('Position', [100 100 1600 600]);

% Plot 1: CL vs Alpha
subplot(2,3,1); hold on; grid on; box on;
for j = 1:3
    af = airfoils(j);
    CL_line = ((af.CL_5 - af.CL_0) / 5.0) * (alpha - af.alpha_0);
    plot(alpha, CL_line, '--', 'LineWidth', 1.5, 'DisplayName', af.name);
end
plot(alpha, CL_blend, 'k-', 'LineWidth', 3, 'DisplayName', 'Blended Wing');
plot(alpha_cruise, CL_cruise, 'ro', 'MarkerSize', 12, 'MarkerFaceColor', 'r',
'LineWidth', 2.5, 'DisplayName', 'Cruise');
xlabel('Angle of Attack (degrees)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Lift Coefficient (CL)', 'FontSize', 11, 'FontWeight', 'bold');
title('CL vs Angle of Attack', 'FontSize', 12, 'FontWeight', 'bold');
legend('Location', 'northwest', 'FontSize', 9);
xlim([-15 20]); ylim([-1.2 1.8]);
grid on;
hold off;

% Plot 2: CD vs Alpha
subplot(2,3,2); hold on; grid on; box on;
plot(alpha, CD_blend, 'b-', 'LineWidth', 2.5);
plot(alpha_cruise, CD_cruise, 'ro', 'MarkerSize', 12, 'MarkerFaceColor', 'r',
'LineWidth', 2.5);
xlabel('Angle of Attack (degrees)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Drag Coefficient (CD)', 'FontSize', 11, 'FontWeight', 'bold');
title('CD vs Angle of Attack', 'FontSize', 12, 'FontWeight', 'bold');
xlim([-15 20]); ylim([0 0.30]);
grid on;
hold off;

% Plot 3: L/D vs Alpha
subplot(2,3,3); hold on; grid on; box on;
plot(alpha, LD_blend, 'g-', 'LineWidth', 2.5);
plot(alpha_cruise, LD_cruise, 'ro', 'MarkerSize', 12, 'MarkerFaceColor', 'r',
'LineWidth', 2.5);
xlabel('Angle of Attack (degrees)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('L/D Ratio', 'FontSize', 11, 'FontWeight', 'bold');
title('Aerodynamic Efficiency (L/D)', 'FontSize', 12, 'FontWeight', 'bold');
xlim([-15 20]); ylim([0 25]);
grid on;
hold off;

% Plot 4: Drag Polar
subplot(2,3,4); hold on; grid on; box on;

```

```

plot(CD_blend, CL_blend, 'b-', 'LineWidth', 2.5);
plot(CD_cruise, CL_cruise, 'ro', 'MarkerSize', 12, 'MarkerFaceColor', 'r',
'LineWidth', 2.5);
xlabel('Drag Coefficient (CD)', 'FontSize', 11, 'FontWeight', 'bold');
ylabel('Lift Coefficient (CL)', 'FontSize', 11, 'FontWeight', 'bold');
title('Drag Polar', 'FontSize', 12, 'FontWeight', 'bold');
grid on;
xlim([0 0.35]);
hold off;

% =====
% CONSOLE OUTPUT
% =====

fprintf('\n');
fprintf('FR25 BLENDED WING AERODYNAMIC ANALYSIS\n');
fprintf('\n\n');

fprintf('CRUISE FLIGHT CONDITIONS (V = 185 m/s):\n');
fprintf('-----\n');
fprintf('  Angle of Attack:          %.2f degrees\n', alpha_cruise);
fprintf('  Lift Coefficient (CL):    %.4f\n', CL_cruise);
fprintf('  Drag Coefficient (CD):    %.6f\n', CD_cruise);
fprintf('  L/D Ratio:                %.2f\n', LD_cruise);
fprintf('  Cruise CD (parasite):     %.6f\n', CD0);
fprintf('\nSAFETY MARGINS:\n');
fprintf('  Distance to stall (+):    %.1f degrees\n', 14 - alpha_cruise);
fprintf('  Distance to stall (-):    %.1f degrees\n', alpha_cruise - (-12));
fprintf('  Status:                   ✓ SAFE OPERATION\n');

fprintf('\nAIRFOIL BLEND WEIGHTS:\n');
for j = 1:3
    fprintf('  %s:                      %.1f%%\n', airfoils(j).name,
airfoils(j).weight*100);
end

fprintf('\n');

```